

FIRST AID

Complete Study Notes

Complete, Chapter-wise Study Notes for the Whole Syllabus

Prepared strictly on the prescribed syllabus — based on St. John Ambulance, Indian Red Cross Society, AHA/ERC 2020–25 resuscitation guidelines and standard textbooks.

WHAT THIS BOOK GIVES YOU

- **Every syllabus topic covered in depth** — nothing left to a second book: outline of first aid, structure and functions of the body, dressings and bandages, CPR, wounds, haemorrhage, shock, electric shock, artificial respiration, asphyxia, fractures and dislocation, unconsciousness and fainting, epilepsy and hysteria, poisons and food poisoning, common conditions, transport of the injured and common medicines.
- **Exam-oriented presentation** — definitions, classifications, signs and symptoms, step-by-step management, do's and don'ts, normal values, and comparison tables in the exact form in which questions are set.
- **Colour-coded learning aids** — blue Key Point boxes, red Exam Focus boxes, purple Mnemonic boxes, green Definition boxes and amber Caution boxes.
- **One-line recap** at the end of every chapter for last-minute revision, plus quick-reference appendices of normal values, first-aid records, antidotes and abbreviations.

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PART I • FOUNDATIONS OF FIRST AID

CHAPTER 1

Outline of First Aid

Meaning, aims, principles, scope, golden rules, duties of a first aider, triage, the first-aid kit, medico-legal aspects and emergency helplines.

SYLLABUS COVERED IN THIS CHAPTER

- Outline of the First-Aid — definition, aims, objects, scope and limitations of first aid.
- Principles, golden rules, qualities and responsibilities of a first aider; DRABC approach, primary and secondary survey, triage.
- First-aid kit, personal protection, medico-legal aspects, consent, Good Samaritan law, emergency numbers and records.

1.1 Meaning and Definition of First Aid

The term *First Aid* is made of two words — **First**, meaning the earliest or initial help, and **Aid**, meaning help or assistance. It therefore means the **first help given to a sick or injured person before the arrival of qualified medical assistance**.

STANDARD DEFINITIONS (LEARN AT LEAST ONE WORD-PERFECT)

- **First aid** is the *immediate, temporary and skilled* assistance given to a person who has suddenly fallen ill or met with an accident, using the material available at hand, until the services of a doctor are obtained or during transport to hospital.
- **St. John Ambulance definition:** "First aid is the skilled application of accepted principles of treatment on the occurrence of an accident or in the case of sudden illness, using facilities or materials available at the time."
- **First aider:** a person who has received a certificate from an authorised/recognised training body (St. John Ambulance, Indian Red Cross Society, etc.) certifying competence to give first aid.
- **Key words in every definition:** *immediate – temporary – skilled – on the spot – with available material – till medical help arrives.*

First aid is therefore **not** a substitute for medical treatment; it is the vital bridge between the moment of injury and definitive medical care. Deaths in the first few minutes after an accident are usually due to three correctable causes — **airway obstruction, severe bleeding and shock** — all of which lie within the power of a trained first aider.

Related terms you must be able to distinguish

Term	What it means
First aid	Immediate, temporary, skilled help with available material; ends when qualified help takes over.
Home nursing	Continued care of a sick person at home after the doctor's advice; long-term, not an emergency skill.
Emergency care / EMS	Organised pre-hospital system (ambulance, paramedics, control room) that continues what the first aider began.
Triage	Sorting of many casualties according to severity so that the most urgent are treated first.
Golden hour	The first 60 minutes after serious injury — definitive treatment within this hour gives the best chance of survival.
Platinum ten minutes	The first 10 minutes of the golden hour, in which the casualty should be assessed, stabilised and moved.
Bystander / Good Samaritan	An untrained onlooker who helps; protected by law in India (see 1.14).

1.2 Aims and Objects of First Aid

The aims are classically remembered as the **THREE P's**. Every examination asks these, usually as 'the first aim of first aid is...' — the answer is always **to preserve/save life**.

The 3 P's	Aim	How it is achieved in practice
1. Preserve life	To save life — the first and foremost aim (of the casualty, of bystanders and of the first aider himself).	Open the airway, give rescue breaths and chest compressions, arrest severe bleeding, treat shock.
2. Prevent worsening	To prevent the condition from becoming worse (prevent further injury and complications).	Immobilise fractures, cover wounds, do not move unnecessarily, prevent infection, keep the casualty warm and reassured.

The 3 P's	Aim	How it is achieved in practice
3. Promote recovery	To promote recovery and reduce the period of illness or disability.	Relieve pain and anxiety, handle gently, position comfortably, arrange early and correct transport to hospital.

Expanded list of objects (St. John Ambulance)

1. To **sustain life** — maintain airway, breathing and circulation.
2. To **prevent deterioration** of the casualty's condition.
3. To **relieve pain** and suffering, and to give reassurance.
4. To **prevent infection** of wounds.
5. To **prevent shock**, or to treat it if already present.
6. To **arrange removal** of the casualty to shelter, home or hospital by the most suitable method.
7. To **hand over** the casualty to a doctor/hospital with a clear report of what was found and what was done.
8. To **stay with** the casualty until responsibility is properly handed over.

1.3 Scope and Limitations of First Aid

Scope (what a first aider does)

- Assesses the scene for danger and makes it safe.
- Assesses the casualty (primary and secondary survey) and diagnoses the condition as far as possible from history, signs and symptoms.
- Gives immediate, appropriate treatment on the spot with whatever material is available (improvisation is a core first-aid skill).
- Decides priorities when there are several casualties (triage).
- Arranges safe transport and hands over the casualty with a report.
- Prevents infection and cross-infection; protects himself/herself.
- Gives psychological support — reassurance is a genuine treatment.

Limitations (what a first aider must NOT do)

LIMITS OF FIRST AID — VERY COMMONLY ASKED

- First aid is **temporary**, not definitive treatment; it ends the moment qualified medical help takes charge.
- A first aider must **not diagnose disease, prescribe medicines or give injections**.
- Must not attempt to reduce a fracture or dislocation, or push back protruding organs/bone ends.
- Must not give anything by mouth to an unconscious, semi-conscious, convulsing or abdominal-injury casualty.
- Must not exceed training, and must not delay professional help by over-treating.
- Must not discuss the casualty's condition with bystanders/press (confidentiality), nor accept a fee for first aid.

1.4 General Principles of First Aid

These principles govern every emergency, whatever the cause:

1. **Be calm, quick and methodical** — do not panic; panic spreads.
2. **Make the area safe** before touching the casualty (electricity, gas, traffic, fire, water, collapsing structures). *Safety order: self — scene — casualty.*
3. **Remove the casualty from danger, or the danger from the casualty** — whichever is quicker and safer.
4. **Treat the most urgent condition first** — the order of priority is **Airway — Breathing — Circulation (bleeding) — Shock — fractures — minor injuries** ('treat the worst first').
5. **Handle gently**; do not move the casualty more than necessary, and never move a suspected spine injury without support.
6. **Reassure** the casualty and the relatives; never say the injury looks bad, never allow the casualty to see his own wound if avoidable.
7. **Loosen tight clothing** at neck, chest and waist; remove only as much clothing as necessary (tear along the seam).
8. **Do not allow crowding**; disperse onlookers and use them for errands (calling ambulance, fetching blankets).
9. **Guard against and treat shock** in every serious injury — keep the casualty warm, lying down, head low.
10. **Nothing by mouth** if unconscious, if surgery/anaesthesia is likely, or in abdominal injury.
11. **Arrange transport** by the most suitable means, in the correct position, without haste and without jolting.
12. **Record and report** — note the time, findings, treatment given and hand over to the doctor.

13. Do not attempt too much; do only what is necessary and stop when help arrives.

THE FOUR VITAL FIRST-AID ACTIONS (THE 'BIG FOUR' THAT SAVE LIVES)

- Restore **breathing** (open the airway, artificial respiration).
- Restore **circulation** (CPR / chest compressions).
- **Arrest severe bleeding**.
- **Treat shock** and poisoning promptly.

1.5 Golden Rules, Do's and Don'ts

DO (Golden Rules)	DO NOT
Keep yourself safe first; wear gloves if available.	Do not rush in blindly — a dead rescuer helps nobody.
Shout for help and telephone 112/108 early.	Do not waste time looking for perfect equipment — improvise.
Check response — airway — breathing — bleeding, in that order.	Do not treat a minor wound while the airway is blocked.
Stop severe bleeding at once by direct pressure and elevation.	Do not apply a tourniquet routinely.
Lay the casualty down; loosen tight clothes; cover to keep warm.	Do not give alcohol, and do not apply hot-water bottles directly.
Turn an unconscious, breathing casualty into the recovery position.	Do not give food or drink to an unconscious casualty.
Immobilise fractures before moving the casualty.	Do not try to straighten (reduce) a fracture or dislocation.
Cover wounds with a clean sterile dressing.	Do not touch the wound, wash it with unclean water, or breathe over it.
Reassure constantly; stay with the casualty.	Do not allow the casualty to walk or to be crowded around.
Hand over with a clear verbal and written report.	Do not exceed your training or delay medical help.

1.6 Qualities of a Good First Aider

A frequently asked list. Remember the memory aid below.

MNEMONIC — 'A GOOD FIRST AIDER IS COOL PRO'

- **C** – Calm, confident and composed (does not panic).
- **O** – Observant — notices signs, hazards and clues at the scene.
- **O** – Organised and methodical; sets priorities.
- **L** – Leadership — takes charge, directs bystanders.
- **P** – Perceptive, Prompt and Practical (improvises with what is at hand).
- **R** – Resourceful, Responsible and Reassuring (tactful, sympathetic, gentle).
- **O** – Objective — keeps within limits, knows when to stop, and maintains **confidentiality**.

- Additional qualities regularly listed in textbooks: *gentle, tactful, sympathetic, cheerful, dexterous (skilful hands), explicit speech, good memory, personal hygiene, up-to-date knowledge and regular refresher training*.
- A first aider should **retrain/refresh every 2–3 years**, because guidelines (especially CPR) change.

1.7 Duties and Responsibilities of a First Aider

1. **Assess the situation** quickly and calmly — what happened, how many casualties, what are the dangers?
2. **Protect** the casualty and others from further danger.
3. **Prevent infection** — wash hands, wear gloves, do not cough over wounds.
4. **Assess the casualty** — primary survey (DRABC) then secondary survey (head-to-toe) and history.
5. **Give immediate treatment** according to priority.
6. **Arrange for medical aid / ambulance** — or send a bystander with a clear message.
7. **Arrange removal** to hospital, home or shelter in the correct position.
8. **Remain with the casualty** until handed over to a responsible person.
9. **Report** all findings and treatment; make a written record; **maintain confidentiality**.
10. **Reassure** relatives and protect the casualty's belongings (hand over valuables to police/relatives with witnesses).

WHAT INFORMATION MUST BE GIVEN WHEN CALLING AN AMBULANCE?

- Your **name and telephone number** (so the control room can call back).
- **Exact location** with landmarks, and the best route/approach.

- **What happened** (type of accident/illness) and **when** it happened.
- **Number, sex and approximate age** of casualties.
- **Nature of injuries** and the condition of each casualty (conscious? breathing? bleeding? trapped?).
- **Hazards still present** — fire, gas, live wires, chemicals, traffic.
- **Never hang up first** — let the control room end the call, and repeat the address before ringing off.

1.8 Approaching an Emergency: the DRABC / DRSABCD Action Plan

The action plan is the single most important sequence in first aid. Two forms are taught; both are correct and examinable.

Letter	Stands for	What you actually do
D	Danger	Check for danger to yourself, bystanders and casualty. Remove danger or casualty.
R	Response	Shake gently and shout — 'Are you all right? Open your eyes!' Use the AVPU scale (Alert, Voice, Pain, Unresponsive).
S	Send/Shout for help	Shout for help; ask a bystander to dial 112/108 and fetch an AED.
A	Airway	Open the airway — head tilt + chin lift (jaw thrust if spinal injury suspected). Clear visible obstruction; remove loose dentures.
B	Breathing	Look, listen and feel for normal breathing for up to 10 seconds . Gasping (agonal breathing) is NOT normal breathing.
C	Circulation / CPR	If not breathing normally — start CPR 30:2. If breathing — check and control severe bleeding, then recovery position.
D	Defibrillation / Disability	Attach an AED as soon as available; otherwise assess Disability (pupils, limb movement) and Deformity .
E	Exposure / Examine	Expose and examine the whole body for hidden injuries, then cover to prevent hypothermia.

Primary survey versus secondary survey

	Primary survey	Secondary survey
Purpose	Detect and treat immediately life-threatening conditions.	Detect all other injuries and illnesses once life-threats are controlled.
Sequence	D-R-A-B-C (+D, E)	History — Signs & Symptoms — head-to-toe examination — vital signs
Time taken	Seconds to 1–2 minutes.	3–5 minutes.
Repeat	Immediately if the casualty deteriorates.	Vital signs every 10 minutes (every 5 if serious).

History taking — the AMPLE and SAMPLE rules

SAMPLE (symptoms & history)	AMPLE (pre-hospital hand-over)
S – Signs and symptoms	A – Allergies
A – Allergies	M – Medicines being taken
M – Medication	P – Past illness / pregnancy
P – Past medical history	L – Last meal (time of last food/drink)
L – Last meal	E – Events leading to the injury
E – Events leading to the incident	—

Information is gathered from **three sources**: the casualty (*subjective* = symptoms), your own examination (*objective* = signs) and the bystanders/scene (*history*).

Signs and symptoms — know the difference

Symptoms (subjective)	Signs (objective)
Felt and described by the casualty — pain, nausea, giddiness, thirst, numbness, weakness, cold feeling.	Found by the first aider using sight, hearing, touch and smell — pallor, bleeding, swelling, deformity, sweating, rapid pulse, vomiting, smell of alcohol/petrol on breath.

Normal vital signs (to be memorised — asked every year)

Vital sign	Normal adult	Child (1–12 yr)	Infant (<1 yr)
Pulse rate	60–100 beats/min (average 72)	80–120/min	100–160/min
Respiratory rate	12–20 breaths/min (average 16–18)	20–30/min	30–60/min
Blood pressure	120/80 mm Hg	~100/65 mm Hg	~80/50 mm Hg
Body temperature	98.4°F / 37°C (range 97–99°F)	Same	Same
SpO ₂ (oxygen saturation)	95–100 %	95–100 %	95–100 %
Capillary refill	< 2 seconds	< 2 seconds	< 2 seconds
Pupils	Equal, round, react to light (3–5 mm)	Same	Same

1.9 Triage — Sorting Multiple Casualties

Triage (French **trier** = to sort) is the classification of casualties according to the urgency of their need for treatment when resources are limited. The rule is: **do the most for the most**.

Priority	Colour tag	Category	Examples
Priority 1 (P1)	RED	Immediate / critical — treat first	Airway obstruction, severe external bleeding, tension pneumothorax, shock, unconscious casualty who is breathing, chest injuries
Priority 2 (P2)	YELLOW	Urgent — treatment can wait 1–2 hours	Closed fractures of long bones, moderate burns without airway involvement, abdominal injury with stable vitals, back injuries
Priority 3 (P3)	GREEN	Delayed / minor — 'walking wounded'	Minor cuts, sprains, small burns, minor fractures of hand/foot, anxiety states
Priority 4 (P4)	BLACK / WHITE	Expectant or dead	No breathing even after the airway is opened; injuries incompatible with life (massive head injury)

- **START triage** = *Simple Triage And Rapid Treatment* — assesses **RPM**: **R**espiration (>30/min = red), **P**erfusion (capillary refill >2 s or absent radial pulse = red) and **M**ental status (cannot obey commands = red). Each casualty should be sorted in **under 60 seconds**.
- First step of START: ask everyone who can walk to move to a marked area — they are automatically **GREEN**.
- Triage must be **repeated** (dynamic), because a casualty's category can change.
- In a mass-casualty situation, the first aider does only two things while triaging: **open the airway** and **stop major bleeding**.

1.10 The First-Aid Box / Kit

A first-aid box should be made of metal or strong plastic, be dust- and damp-proof, be clearly marked with a **white cross on a green background** (the international first-aid sign), be kept in a known, easily reachable place, and be checked and restocked regularly by a named person.

Group	Contents	Chief uses
Dressings	Sterile gauze pieces, non-adherent (paraffin) dressings, cotton wool, adhesive dressings (Band-Aid), eye pads, sterile pads	Cover wounds, absorb blood and discharge, control bleeding
Bandages	Roller (cotton) bandages 2.5/5/7.5/10 cm, triangular bandages, crepe (elastic) bandage, adhesive plaster tape, safety pins, clips	Hold dressings, support and immobilise, make slings
Antiseptics	Povidone-iodine (Betadine) solution/ointment, chlorhexidine, 70 % alcohol/spirit swabs, hydrogen peroxide, potassium permanganate, savlon/dettol	Clean skin and wounds, prevent infection
Instruments	Blunt-nosed scissors, tweezers/forceps, safety pins, disposable gloves, torch, thermometer, tourniquet (for medical use only), resuscitation face-shield/pocket mask, kidney tray	Cutting, removal of splinters, protection, examination, safe rescue breathing
Medicines (basic)	Paracetamol, ORS sachets, antiseptic cream, oral antihistamine (cetirizine/chlorpheniramine), antacid, soda bicarb, glucose, smelling salts (sal volatile), calamine lotion, silver-sulphadiazine cream	Fever and pain, dehydration, allergy, acidity, burns, faint, stings
Records & misc.	Notebook and pencil, first-aid manual, list of emergency numbers, cotton sling, plastic bags, blanket/space blanket, drinking water, hand sanitiser	Record keeping, warmth, waste disposal, hygiene

KIT RULES — EXAM POINTS

- International first-aid symbol: **white cross on green background** (green = safety). The **red cross on white** is the protective emblem of the Red Cross and must not be used on ordinary kits.
- The box must contain **no loose tablets**, no expired items, and must be **locked away from children but never locked to the user**.
- Contents must be checked **after every use** and at least once a month; a checklist is kept inside the lid.
- Factories Act, 1948 (Sec. 45): a first-aid box for **every 150 workers**, in charge of a trained person, available during working hours; an ambulance room is required where **more than 500 workers** are employed.

1.11 Personal Protection and Infection Control for the First Aider

- **Universal (standard) precautions:** treat every casualty's blood and body fluid as potentially infectious (HIV, hepatitis B and C).
- **Hand hygiene:** wash with soap and running water for 40–60 seconds before and after every casualty; alcohol rub (20–30 s) if hands are not visibly soiled.
- **Barriers:** disposable gloves, plastic apron, eye protection, face-shield or pocket mask for rescue breathing.
- **Avoid** direct mouth-to-mouth contact if a barrier is available; compression-only CPR is acceptable for an untrained/unwilling rescuer.
- **Cover your own cuts** with a waterproof dressing before giving first aid.
- **Never re-cap or re-use needles;** dispose of sharps in a puncture-proof container; soiled dressings go into a sealed plastic bag (yellow bag → incineration).
- **Needle-stick / splash injury:** wash immediately with soap and water (do not squeeze or scrub, do not use bleach on the wound), irrigate eyes with water, report at once and seek **post-exposure prophylaxis (PEP) within 2 hours** (not later than 72 hours).

1.12 The Chain of Survival and the Golden Hour

Link	Action	Why it matters
1. Early recognition and call for help	Recognise cardiac arrest / serious injury and dial 112 or 108.	Nothing else can start until help is summoned.
2. Early CPR	Immediate bystander chest compressions.	Doubles or triples survival; buys time for the brain (irreversible damage after 3–4 minutes without oxygen).
3. Early defibrillation	Use an AED within 3–5 minutes.	Survival falls by 7–10 % for every minute of delay in defibrillation.
4. Early advanced care	Paramedics, drugs, airway, hospital ICU.	Stabilises the heart rhythm and treats the cause.
5. Post-resuscitation care / rehabilitation	Targeted temperature management, intensive care, rehabilitation.	Preserves brain function and quality of life.

TIME FACTS THAT ARE ASKED AGAIN AND AGAIN

- **Golden hour** — first **60 minutes** after major trauma; definitive care within it markedly improves survival.
- **Platinum 10 minutes** — ideal on-scene time for a critical trauma casualty.
- **Brain damage** begins after **3–4 minutes** of complete oxygen deprivation; it is usually irreversible after **8–10 minutes**.
- **Breathing check** during DRABC: not more than **10 seconds**.
- **Clinical death** = stopped breathing and circulation (reversible); **biological/brain death** = irreversible death of brain cells.
- Survival from cardiac arrest without CPR falls by about **7–10 % per minute**.

1.13 Medico-Legal Aspects of First Aid

Consent

Type of consent	Meaning / when used
Expressed (informed) consent	The conscious, mentally sound adult casualty says or nods that he accepts help. Always introduce yourself, say you are trained, and ask permission.
Implied consent	Assumed when the casualty is unconscious, confused, seriously injured or a minor with no guardian present — the law presumes a reasonable person would want life-saving help (doctrine of emergency).

Type of consent	Meaning / when used
Consent for a minor	Taken from the parent/guardian; if absent, implied consent applies. In India, valid consent age for medical treatment is 18 years (12 years for examination).
Refusal of consent	A competent adult may refuse first aid. Do not force treatment; record the refusal, ask a witness to sign, and call the ambulance anyway.

Duties, negligence and protection in law

Duty of care — A legal/ethical obligation to act; it applies to a designated workplace first aider on duty. An ordinary bystander in India has a moral duty and is protected when he helps.

Standard of care — You are judged by what a **reasonable person with the same training** would have done in the same circumstances — not by the outcome.

Negligence — Failure to give the expected standard of care, causing harm. Four elements: **duty, breach, causation, damage**.

Abandonment — Leaving a casualty before handing over to someone of equal or greater ability — a punishable act.

Confidentiality — Information about the casualty must not be revealed to the press, employer or onlookers; give it only to the doctor, ambulance staff or police.

Good Samaritan protection (India) — Supreme Court guidelines (2016) and **Section 134A of the Motor Vehicles (Amendment) Act, 2019** protect a bystander who helps a road accident victim: he **cannot** be held civilly or criminally liable, cannot be forced to reveal his identity, and cannot be detained or asked to pay for treatment.

Duty of the hospital — Every hospital (government or private) must give immediate emergency treatment to an accident victim — *Parmanand Katara v. Union of India* (1989): **saving life takes precedence over all formalities and police procedures**.

Preserving evidence — At a crime scene, disturb as little as possible; note the original position of the casualty and objects; hand over removed clothing to the police.

Records and reporting

- Keep a **first-aid register/accident report** noting: date and time of incident, name/age/sex, what happened, findings (signs and symptoms), vital signs with times, treatment given, time of calling help, to whom the casualty was handed over, and your name and signature.
- Records are needed for insurance, workmen's compensation, police enquiry and for later medical care. Write facts only — no opinions.

1.14 Emergency Telephone Numbers in India

Number	Service
112	All-in-one national emergency helpline (ERSS) — police, fire, ambulance, disaster
108	Free emergency ambulance (EMRI) — accident/medical emergency
102	Free ambulance for pregnant women, mothers and infants (JSSK)
100 / 112	Police
101	Fire brigade
104	Health advice / medical helpline
1098	CHILDLINE — children in distress
1091	Women's helpline; 181 women in distress
1073 / 1033	Road accident / national highway helpline
14567	Senior citizen (Elderline) helpline
1800-11-6117	National Poisons Information Centre, AIIMS New Delhi (24 × 7)
14416 / 1800-891-4416	Tele-MANAS mental-health helpline; 9152987821 iCALL
1070 / 1077	State / district disaster management control room
1962	Animal ambulance / veterinary helpline

1.15 History and Organisations of First Aid (General Awareness)

Fact	Detail
Who coined the term 'first aid'	Johannes Friedrich August von Esmarch (German surgeon) — <i>erste hilfe</i> ; he also introduced the Esmarch bandage (triangular bandage) and is called the <i>father of first aid</i> .
Esmarch triangular bandage	A triangular cloth bandage; the standard first-aid bandage even today.

Fact	Detail
Red Cross founded	1863 , Geneva, by Jean Henri Dunant (first Nobel Peace Prize, 1901). Emblem: red cross on white — the reverse of the Swiss flag.
First Geneva Convention	1864 — protection of the wounded in war.
St. John Ambulance Association	Founded 1877 in England; the pioneer of organised first-aid training and certification.
Indian Red Cross Society	Established 1920 (Indian Red Cross Society Act XV of 1920); President of India is its President.
World First Aid Day	Second Saturday of September every year (started 2000 by IFRC).
World Red Cross Day	8 May — Henri Dunant's birthday.
World Health Day / WHO	7 April ; WHO founded 1948, headquarters Geneva.
Ambulance	From Latin <i>ambulare</i> (to walk). Modern ambulance service traces to the Napoleonic surgeon Dominique-Jean Larrey , who introduced field ambulances and battlefield triage.
Modern CPR	Mouth-to-mouth ventilation revived by Peter Safar and James Elam (1950s); closed-chest compression by Kouwenhoven, Jude and Knickerbocker (1960) ; Safar is called the <i>father of CPR</i> .
Florence Nightingale	Founder of modern nursing (Crimean War, 'Lady with the Lamp'); International Nurses Day 12 May .

1.16 Chapter Recap — One-Line Revision

- First aid = **immediate, temporary, skilled** help with **available material** until medical aid arrives.
- Aims = **3 P's**: Preserve life (first aim), Prevent worsening, Promote recovery.
- Action plan = **D-R-S-A-B-C-D**; treatment priority = **airway** → **breathing** → **bleeding** → **shock**.
- Breathing is checked for a maximum of **10 seconds**; brain damage starts in **3–4 minutes**.
- Triage colours = **RED (immediate)** – **YELLOW (urgent)** – **GREEN (delayed)** – **BLACK (dead/expectant)**.
- First-aid symbol = **white cross on green**; Red Cross emblem = **red cross on white**.
- Ambulance = **108**; universal emergency = **112**; poison information = **1800-11-6117**.
- Term 'first aid' coined by **Esmarch**; Red Cross founded by **Henri Dunant (1863)**; St. John Ambulance **1877**; Indian Red Cross **1920**.
- Never give **anything by mouth** to an unconscious casualty; never **reduce** a fracture or dislocation; never **exceed your training**.

CHAPTER 2

Structure and Functions of the Body

Cells, tissues and systems — skeletal, muscular, circulatory, respiratory, nervous, digestive, urinary, endocrine, skin, sense organs and reproductive systems, with all the normal values a first aider must know.

SYLLABUS COVERED IN THIS CHAPTER

- Structure and functions of the body — organisation of the body, all major systems with their parts, functions and normal values.
- Anatomical terms, body cavities and regions used while describing injuries; landmarks used in first aid (pressure points, CPR site).

2.1 Levels of Organisation of the Human Body

The body is built up in a definite order. Anatomy is the study of the *structure* of the body; physiology is the study of its *functions*.

Level	Description	Example
1. Cell	Smallest living, structural and functional unit of the body (discovered by Robert Hooke, 1665).	Red blood cell, nerve cell
2. Tissue	A group of similar cells performing the same function.	Muscle tissue, nervous tissue
3. Organ	Two or more tissues combining to perform a specific function.	Heart, lung, stomach
4. System	A group of organs working together for one major function.	Circulatory system, digestive system
5. Organism	All systems working together — the complete human being.	Human body

NUMBERS TO REMEMBER

- The adult body has about **75–100 trillion cells**, **4 basic tissue types**, and roughly **11 organ systems**.
- Body water = **about 60 %** of body weight in an adult male (55 % female, 75 % in a newborn). Two-thirds of it is **intracellular**.
- Average adult blood volume = **5–6 litres (about 8 % of body weight; 70 ml/kg)**.
- **Homeostasis** = maintenance of a constant internal environment (temperature, pH 7.35–7.45, glucose, fluids). Failure of homeostasis = disease.

2.2 The Cell and the Four Basic Tissues

Chief parts of a cell

Part	Function
Cell (plasma) membrane	Selectively permeable outer covering; controls entry and exit of substances.
Cytoplasm	Jelly-like fluid holding the organelles; site of metabolic reactions.
Nucleus	Control centre; contains DNA/chromosomes (46 in 23 pairs); directs cell activity and division.
Mitochondria	Powerhouse of the cell — produces energy as ATP by aerobic respiration.
Ribosomes	Protein synthesis ('protein factory').
Endoplasmic reticulum	Transport channel; rough ER = protein, smooth ER = fat/lipid synthesis.
Golgi apparatus	Packaging and secretion of proteins ('post office').
Lysosome	Contains digestive enzymes — 'suicidal bag' of the cell; destroys worn-out parts and bacteria.
Centrosome	Takes part in cell division.

The four primary tissues

Tissue	Where found	Function
Epithelial	Skin surface, lining of mouth, gut, air passages, glands, blood vessels	Protection, absorption, secretion, filtration
Connective	Bone, cartilage, tendon, ligament, blood, fat (adipose), lymph	Support, binding, transport, storage, defence — the most abundant tissue type

Tissue	Where found	Function
Muscular	Skeletal muscle, heart, walls of gut and vessels	Movement, maintenance of posture, propulsion, heat production
Nervous	Brain, spinal cord, nerves	Reception of stimuli, conduction of impulses, control and co-ordination

2.3 Anatomical Terms, Planes, Cavities and Regions

These words are used in every description of injury — learn them once and they save confusion for ever.

Term	Meaning	Term	Meaning
Anatomical position	Standing erect, eyes forward, arms at sides, palms facing forward	Superior / Inferior	Above / below
Anterior (ventral)	Front of the body	Posterior (dorsal)	Back of the body
Medial / Lateral	Nearer to / away from the midline	Proximal / Distal	Nearer to / farther from the trunk (root of a limb)
Superficial / Deep	Near the surface / away from the surface	Supine / Prone	Lying face up / lying face down
Flexion / Extension	Bending / straightening a joint	Abduction / Adduction	Moving away from / towards the midline
Palmar / Plantar	Palm of hand / sole of foot	Unilateral / Bilateral	One side / both sides

Planes of the body

- **Sagittal (median) plane** — vertical, divides the body into right and left halves.
- **Coronal (frontal) plane** — vertical, divides into front and back.
- **Transverse (horizontal/axial) plane** — divides into upper and lower parts.

Body cavities and their contents

Cavity	Contents
Cranial cavity	Brain (protected by the skull and meninges)
Vertebral (spinal) canal	Spinal cord
Thoracic cavity	Heart in the pericardium, lungs in the pleurae, oesophagus, trachea, great vessels, thymus (mediastinum = the space between the lungs)
Abdominal cavity	Stomach, small and large intestine, liver, gall bladder, pancreas, spleen, kidneys, suprarenal glands
Pelvic cavity	Urinary bladder, rectum, reproductive organs, sigmoid colon

The **diaphragm** separates the thoracic from the abdominal cavity. The abdomen is divided for description into **nine regions** (right hypochondrium, epigastrium, left hypochondrium; right lumbar, umbilical, left lumbar; right iliac/inguinal, hypogastrium/suprapubic, left iliac/inguinal) or more simply into **four quadrants** by lines crossing at the umbilicus.

WHY THIS MATTERS IN FIRST AID

- Pain in the **right hypochondrium** — liver/gall bladder; **right iliac fossa** — appendix; **epigastrium** — stomach/heart (a heart attack is often mistaken for 'gas').
- Injury below the **5th rib on the left** can tear the **spleen**; on the right, the **liver** — both cause severe internal bleeding.
- Always describe an injury using the casualty's **own right and left**, not yours.

2.4 The Skeletal System

The adult human skeleton has **206 bones** (a newborn has about **270–300**, which fuse with growth). The skeleton is divided into the **axial skeleton (80 bones)** and the **appendicular skeleton (126 bones)**.

Functions of the skeleton (mnemonic: S-P-M-M-S)

- **Shape**, support and framework of the body.
- **Protection** of vital organs (skull — brain, ribs — heart and lungs, vertebrae — spinal cord, pelvis — bladder and reproductive organs).
- **Movement** — bones act as levers for muscles; joints allow motion.
- **Mineral storage** — 99 % of body **calcium** and much phosphorus is stored in bone.

- Site of blood-cell formation (**haematopoiesis**) in **red bone marrow**; yellow marrow stores fat.

Count of bones — the most examined table in anatomy

Region	Bones	Number
AXIAL SKELETON		80
Skull – cranium	Frontal 1, Parietal 2, Temporal 2, Occipital 1, Sphenoid 1, Ethmoid 1	8
Skull – face	Maxilla 2, Zygomatic 2, Nasal 2, Lacrimal 2, Palatine 2, Inferior nasal concha 2, Vomer 1, Mandible 1	14
Ear ossicles	Malleus, incus, stapes — 3 in each ear	6
Hyoid	The only bone that has no joint with another bone	1
Vertebral column	Cervical 7, Thoracic 12, Lumbar 5, Sacrum 1 (5 fused), Coccyx 1 (4 fused)	26
Thoracic cage	Ribs 24 (12 pairs) + Sternum 1	25
APPENDICULAR SKELETON		126
Shoulder girdle	Clavicle 2 + Scapula 2	4
Upper limbs	Humerus 2, Radius 2, Ulna 2, Carpals 16, Metacarpals 10, Phalanges 28	60
Pelvic girdle	Hip bone (ilium + ischium + pubis fused) — 2	2
Lower limbs	Femur 2, Patella 2, Tibia 2, Fibula 2, Tarsals 14, Metatarsals 10, Phalanges 28	60
TOTAL		206

- Bones in one **hand** = **27**; in one **foot** = **26**; one upper limb (including girdle) = 32; one lower limb (including hip bone) = 31.
- Vertebral column in a child = **33 vertebrae**; in an adult = **26** (because 5 sacral fuse into 1 and 4 coccygeal fuse into 1). It has **4 curves** and about **100 joints**.
- Ribs: **12 pairs** — **true ribs 1–7** (joined directly to the sternum by their own cartilage), **false ribs 8–10** (joined indirectly) and **floating ribs 11–12** (not joined in front).
- The **sternum** has 3 parts — manubrium, body and xiphoid process; chest compressions are given on the **lower half of the sternum**.

Types of bones

Type	Example
Long	Femur, humerus, tibia, radius, ulna, metacarpals
Short	Carpals (wrist) and tarsals (ankle)
Flat	Skull bones, sternum, ribs, scapula
Irregular	Vertebrae, facial bones, hip bone
Sesamoid	Patella (kneecap) — the largest sesamoid bone
Pneumatic (air-filled)	Maxilla, frontal, sphenoid, ethmoid (contain sinuses)

Record facts about bones

Superlative	Bone
Longest, heaviest and strongest bone	Femur (thigh bone) — about ¼ of the body height
Smallest bone	Stapes (stirrup) in the middle ear
Largest / broadest flat bone	Hip bone; the largest flat bone of the skull is the parietal
Strongest, hardest substance in the body	Enamel of the tooth (bone is the hardest <i>tissue</i>)
Only bone with no articulation	Hyoid bone in the neck
Most commonly fractured bone	Clavicle (collar bone)
Longest bone of the upper limb	Humerus
Bone that protects the brain	Skull (cranium); the soft spot in a baby's skull is the fontanelle
Movable bone of the skull	Mandible (lower jaw) — also the strongest facial bone

Structure of a long bone

- Periosteum** — tough outer membrane carrying blood vessels; **compact bone** — dense outer layer; **spongy (cancellous) bone** — inner honeycomb; **medullary cavity** — contains marrow.
- Epiphysis** (end), **diaphysis** (shaft), **epiphyseal plate** (growth plate — injury to it in children can stunt growth).

- Bone cells: **osteoblasts** (bone-forming), **osteocytes** (mature) and **osteoclasts** (bone-resorbing). Vitamin **D** and calcium are needed for bone formation; deficiency causes **rickets** in children and **osteomalacia/osteoporosis** in adults.

Joints (articulations)

Class	Movement	Examples
Fibrous (synarthrosis)	Immovable / fixed	Sutures of the skull, teeth in sockets
Cartilaginous (amphiarthrosis)	Slightly movable	Between vertebral bodies, symphysis pubis, first rib–sternum
Synovial (diarthrosis)	Freely movable — have a capsule, synovial membrane and fluid	Shoulder, hip, knee, elbow, wrist

Type of synovial joint	Movement	Example
Ball and socket	All directions (most mobile)	Shoulder (most mobile and most commonly dislocated), hip
Hinge	Flexion and extension only	Elbow, knee, ankle, interphalangeal joints
Pivot (trochoid)	Rotation	Atlanto-axial joint (turning the head 'no'), radio-ulnar
Gliding (plane)	Sliding	Between carpals, tarsals, acromio-clavicular
Saddle	Two directions + limited rotation	Carpometacarpal joint of the thumb
Condylloid (ellipsoid)	Flexion, extension, abduction, adduction	Wrist, metacarpophalangeal (knuckle)

- **Ligament** = tough fibrous band joining **bone to bone** (stabilises a joint). **Tendon** = cord joining **muscle to bone**. **Cartilage** = smooth elastic tissue covering joint surfaces.
- **Synovial fluid** lubricates and nourishes the joint cartilage.
- The **atlas** (1st cervical vertebra) supports the skull; the **axis** (2nd) allows rotation. **C7** is the vertebra prominens felt at the base of the neck.

2.5 The Muscular System

There are about **600–650 skeletal muscles**, forming **40–45 % of body weight**. Muscles work by contraction, always in **antagonistic pairs** (one flexes, the other extends).

Feature	Skeletal (striated, voluntary)	Smooth (non-striated, involuntary)	Cardiac
Where	Attached to bones of limbs, trunk, face	Walls of stomach, intestine, blood vessels, bladder, uterus, iris	Only in the wall (myocardium) of the heart
Control	Voluntary — under conscious control	Involuntary — autonomic nerves and hormones	Involuntary , but striated; has its own pacemaker
Appearance	Striated, long cylindrical, multinucleate	Spindle-shaped, single central nucleus, no striations	Striated, branched, intercalated discs
Fatigue	Fatigues quickly	Slow, sustained, does not fatigue easily	Never fatigues (works lifelong)
Function	Movement, posture, heat production (shivering)	Peristalsis, vasoconstriction, emptying of bladder	Pumping of blood — about 70 beats/min for life

Important muscle facts and major muscles

Muscle / fact	Detail
Largest / strongest muscle	Gluteus maximus (buttock) — largest; masseter (jaw) — strongest by force; the heart is the hardest-working muscle
Smallest muscle	Stapedius (in the middle ear)
Longest muscle	Sartorius (front of thigh)
Chief muscle of respiration	Diaphragm (dome-shaped; supplied by the phrenic nerve , C3–C5) assisted by the intercostals
Muscles of the shoulder/arm	Deltoid (abducts arm), biceps brachii (flexes elbow), triceps (extends elbow), pectoralis major (chest), trapezius and latissimus dorsi (back)
Muscles of the leg	Quadriceps femoris (extends knee), hamstrings (flex knee), gastrocnemius and soleus (calf — Achilles tendon), tibialis anterior
Abdominal muscles	Rectus abdominis, external and internal oblique, transversus abdominis — protect viscera and help in expiration, coughing, defaecation
Strongest tendon	Achilles (calcaneal) tendon — attaches calf muscles to the heel bone
Energy for contraction	ATP ; fuel is glucose and oxygen. Anaerobic work produces lactic acid — muscle cramp and fatigue

Muscle / fact	Detail
Muscle tone	Continuous partial contraction that maintains posture; lost in unconsciousness — this is why the tongue falls back and blocks the airway

2.6 The Circulatory (Cardio-Vascular) System

Consists of the **heart** (pump), the **blood vessels** (pipes) and the **blood** (transport medium). Discovery of circulation — **William Harvey**.

The heart

- Cone-shaped muscular organ the size of a closed fist; weight **250–350 g**; lies in the **mediastinum**, behind the sternum, **two-thirds to the left** of the midline; apex points down and to the left at the level of the 5th intercostal space.
- Covered by a double membrane, the **pericardium**; its wall has three layers — **endocardium** (inner), **myocardium** (thick muscle) and **epicardium/pericardium** (outer).
- **Four chambers** — right atrium, right ventricle, left atrium, left ventricle. The **left ventricle has the thickest wall** because it pumps blood to the whole body.
- **Valves** prevent backflow: **tricuspid** (right atrium → right ventricle), **bicuspid/mitral** (left atrium → left ventricle), **pulmonary** (right ventricle → pulmonary artery) and **aortic** (left ventricle → aorta). Heart sounds '**lub**' (**S1**) = closure of tricuspid and mitral, '**dub**' (**S2**) = closure of aortic and pulmonary valves.
- **Pacemaker = SA (sino-atrial) node** in the right atrium → AV node → bundle of His → Purkinje fibres. The heart's own blood supply comes from the **coronary arteries** (first branches of the aorta); their blockage causes **heart attack (myocardial infarction)**.
- **Cardiac output** = stroke volume (≈ 70 ml) $\times$ heart rate (≈ 72 /min) = about **5 litres/minute**. The heart beats about **100,000 times a day**.

Circulation — the two circuits

Circuit	Path	Note
Pulmonary circulation (heart → lungs → heart)	Right atrium → right ventricle → pulmonary artery → lungs → pulmonary veins → left atrium	The pulmonary artery is the only artery carrying deoxygenated blood ; the pulmonary vein is the only vein carrying oxygenated blood
Systemic circulation (heart → body → heart)	Left atrium → left ventricle → aorta → arteries → capillaries of the body → veins → superior and inferior vena cava → right atrium	Carries oxygen and nutrients to every cell and removes CO ₂ and wastes
Portal circulation	Veins of the stomach, intestine, spleen and pancreas → hepatic portal vein → liver → hepatic vein → inferior vena cava	Allows the liver to process absorbed food and detoxify
Coronary circulation	Aorta → right and left coronary arteries → heart muscle	Supplies the myocardium itself

Blood vessels

Feature	Artery	Vein	Capillary
Direction	Carries blood away from the heart	Carries blood towards the heart	Connects arterioles to venules
Blood	Oxygenated (except pulmonary artery)	Deoxygenated (except pulmonary vein)	Exchange of gases, nutrients and wastes with tissues
Wall	Thick, muscular, elastic	Thin, less elastic	One cell thick (endothelium only)
Valves	Absent	Present (prevent backflow against gravity)	Absent
Pressure / flow	High pressure, pulsating, rapid	Low pressure, steady, slow	Very slow — allows exchange
Bleeding pattern	Bright red, spurts in jets with each beat	Dark red, steady flow / oozes	Oozes slowly from the whole surface
Position	Deep (protected)	Mostly superficial	In every tissue

- **Largest artery = aorta**; **largest veins = superior and inferior vena cava**; **largest lymphatic vessel = thoracic duct**.
- Total length of all blood vessels $\approx$ **96,000 km**; capillaries are the most numerous (~ 5 – 10 μ m wide).
- **Blood pressure** = the pressure of blood on arterial walls. Normal adult = **120/80 mm Hg** (systolic/diastolic); measured with a **sphygmomanometer** over the **brachial artery**. Hypertension $\geq 140/90$; hypotension $< 90/60$.

Blood — composition and functions

Component	Normal value	Function
Plasma (55 %)	91 % water + proteins (albumin, globulin, fibrinogen), salts, glucose, hormones, wastes	Transport medium; maintains blood volume, pressure and osmotic balance
RBC (erythrocytes)	Male 4.5–5.5 million/mm ³ ; female 4.0–5.0 million/mm ³ ; lifespan 120 days	Carry oxygen as oxyhaemoglobin; no nucleus; made in red bone marrow, destroyed in the spleen ('graveyard of RBCs')
Haemoglobin	Male 13–18 g/dl , female 12–16 g/dl	Iron-containing red pigment that binds oxygen; low level = anaemia
WBC (leucocytes)	4,000–11,000/mm³	Defence against infection; neutrophils (most numerous, 60–70 %) phagocytose bacteria; lymphocytes make antibodies; eosinophils in allergy
Platelets (thrombocytes)	1.5–4.5 lakh/mm³ (150,000–450,000)	Clotting of blood; lifespan 8–10 days
Clotting time / bleeding time	Clotting 3–8 min; bleeding 1–4 min	Clotting needs vitamin K , calcium, prothrombin and fibrinogen: fibrinogen → fibrin mesh → clot

- **Functions of blood (mnemonic: T-R-P-D)** — Transport (O₂, CO₂, food, hormones, wastes), Regulation (temperature, pH, water balance), Protection (WBC, antibodies, clotting) and Defence against infection.
- **pH of blood = 7.35–7.45** (slightly alkaline). Colour is due to haemoglobin; blood is **8 % of body weight**.
- **Blood groups (Landsteiner)** — A, B, AB, O. **O negative = universal donor; AB positive = universal recipient**. Rh factor present = Rh positive. Group **O** is commonest in India, **AB** the rarest.
- **Anticoagulants:** heparin (natural, from liver/mast cells), citrate and EDTA (in blood bags); stored blood keeps for **35–42 days** at 2–6 °C.

Pulse — where to feel it

Pulse point	Artery / site	Use in first aid
Radial	Thumb-side of the front of the wrist	Routine pulse counting in a conscious casualty
Carotid	Side of the neck, beside the windpipe	Pulse check in a collapsed casualty (trained rescuers only, ≤ 10 s)
Brachial	Inner side of the arm above the elbow	Blood pressure; pulse check in infants ; pressure point for arm bleeding
Femoral	Groin, midway along the fold	Pressure point for severe leg bleeding
Temporal	In front of the ear	Bleeding from the scalp
Facial	Lower border of the jaw	Bleeding from the face
Popliteal / Posterior tibial / Dorsalis pedis	Behind the knee / behind the inner ankle / top of the foot	To confirm circulation beyond a splint or bandage

Normal pulse = **60–100 beats/min** (average 72). **Tachycardia** > 100/min; **bradycardia** < 60/min. Note rate, **rhythm** (regular or irregular) and **volume** (strong, weak, thready). A **rapid, weak, thready pulse is the classical sign of shock**.

The lymphatic system (part of circulation and defence)

- Consists of **lymph**, lymph capillaries and vessels, **lymph nodes**, **spleen**, **thymus**, tonsils and Peyer's patches.
- Functions — drains excess tissue fluid back to the blood, absorbs **fats** from the intestine (through lacteals), filters out bacteria in the nodes, and produces **lymphocytes** and antibodies.
- **Spleen** — largest lymphatic organ, in the left hypochondrium; stores blood, destroys old RBCs; easily ruptured in left-sided abdominal injury → severe internal bleeding.
- Lymph flows in **one direction only** (towards the heart) and is moved by muscle contraction and breathing; the **thoracic duct** empties into the left subclavian vein.

2.7 The Respiratory System

Respiration supplies **oxygen** to the cells and removes **carbon dioxide**. Breathing (ventilation) is only its mechanical part.

Part	Points to remember
Nose	Air is warmed, moistened and filtered ; lined by ciliated mucous membrane. The nose is the correct route of breathing.
Pharynx (throat)	Common passage for air and food; divided into naso-, oro- and laryngo-pharynx.
Larynx (voice box)	Contains the vocal cords; the epiglottis closes it during swallowing so that food goes into the oesophagus, not the windpipe.

Part	Points to remember
Trachea (windpipe)	About 10–12 cm long, held open by 16–20 C-shaped cartilage rings ; divides at the level of T4–T5 into two bronchi.
Bronchi and bronchioles	Right bronchus is shorter, wider and more vertical — therefore inhaled foreign bodies usually enter the right lung.
Alveoli	About 300 million thin-walled air sacs; total surface area 70–80 m² ; site of gas exchange by diffusion across a membrane one cell thick.
Lungs	Cone-shaped, spongy; right lung has 3 lobes, left has 2 (to make room for the heart — cardiac notch). Covered by the pleura with pleural fluid between its two layers.
Diaphragm	Main muscle of breathing; contracts and flattens during inspiration.

Mechanism of breathing

	Inspiration (active)	Expiration (passive at rest)
Muscles	Diaphragm contracts and descends; external intercostals contract and raise the ribs	Diaphragm and intercostals relax; elastic recoil of the lungs
Chest cavity	Increases in all three diameters	Decreases
Pressure inside	Falls below atmospheric — air rushes in	Rises above atmospheric — air is pushed out
Duration	About 2 seconds	About 3 seconds

Respiratory values and air composition

Value	Normal figure
Respiratory rate — adult	12–20/min (average 16–18); child 20–30; infant 30–60; newborn 40–60
Tidal volume	500 ml (about 6–8 ml/kg) per quiet breath
Vital capacity	4,000–5,000 ml (about 4.8 litres in men)
Total lung capacity	About 6,000 ml
Residual volume	1,200 ml — air that can never be breathed out
Dead space	About 150 ml of air that does not take part in exchange
Ratio of respiration to pulse	About 1 : 4
Control centre	Medulla oblongata (respiratory centre), stimulated chiefly by a rise in CO₂ in the blood

Gas	Inspired (atmospheric) air	Expired air	Alveolar air
Oxygen	21 %	16 %	14 %
Carbon dioxide	0.04 %	4 %	5.5 %
Nitrogen	79 %	79 %	80 %
Water vapour	Variable	Saturated	Saturated
Temperature	Atmospheric	Body temperature	Body temperature

FIRST-AID SIGNIFICANCE

- Expired air still contains **16 % oxygen** — that is why **mouth-to-mouth respiration works**.
- The brain uses **20 % of the body's oxygen**; it cannot store oxygen, hence damage begins within 3–4 minutes of stopped breathing.
- **Difficulty in breathing = dyspnoea**; absent breathing = **apnoea**; blue discoloration from lack of oxygen = **cyanosis** (seen first in lips, tongue, nail beds and ear lobes); rapid breathing = tachypnoea.
- In an unconscious casualty the commonest cause of airway obstruction is the **tongue falling back** — hence head tilt and chin lift.

2.8 The Nervous System

The control and communication system of the body. Structural unit = the **neuron** (nerve cell), which cannot regenerate once destroyed.

Division	Parts	Function
Central nervous system (CNS)	Brain + spinal cord	Receives, interprets and stores information; initiates responses

Division	Parts	Function
Peripheral nervous system (PNS)	12 pairs of cranial nerves + 31 pairs of spinal nerves	Carries sensory impulses to and motor impulses from the CNS
Autonomic nervous system (ANS)	Sympathetic and parasympathetic divisions	Involuntary control of heart, glands, gut, vessels and pupils

Parts of the brain and their functions

Part	Function
Cerebrum (largest part; 2 hemispheres, grey outer cortex)	Higher functions — intelligence, memory, reasoning, speech, emotion, will, and the centres for voluntary movement and sensation. The left hemisphere controls the right side of the body (crossed control).
Cerebellum ('little brain', behind and below)	Balance, posture, co-ordination of voluntary movement and muscle tone. Damage → staggering gait, tremors.
Mid-brain	Relay for visual and auditory reflexes; pupil reflex.
Pons	Bridge between the parts of the brain; helps control breathing rhythm.
Medulla oblongata (lowest part, continues as the spinal cord)	Vital centres — respiratory, cardiac (heart rate) and vasomotor (blood pressure) centres, plus reflexes of swallowing, coughing, sneezing and vomiting. Injury here is rapidly fatal.
Hypothalamus	Temperature regulation , hunger, thirst, sleep, emotion; controls the pituitary gland.
Thalamus	Relay station for sensations, especially pain.
Brain stem	Mid-brain + pons + medulla; controls consciousness and vital functions.

- Brain weight ≈ **1,300–1,400 g**; it contains about **86–100 billion neurons** and receives **15–20 % of the cardiac output**.
- Meninges** = three protective membranes — **dura mater** (outer, tough), **arachnoid mater** (middle) and **pia mater** (inner, delicate). Inflammation = **meningitis**.
- Cerebrospinal fluid (CSF)** — about **130–150 ml**, circulates in the ventricles and around the cord, acting as a **shock absorber** and nutrient medium. Clear watery fluid leaking from the ear or nose after a head injury is CSF and indicates a **fractured base of the skull**.
- Spinal cord** — about **45 cm** long, extends from the medulla to the level of **L1–L2**; gives off **31 pairs** of spinal nerves (**cervical 8, thoracic 12, lumbar 5, sacral 5, coccygeal 1**).
- Reflex action** — an involuntary, rapid, protective response (knee jerk, blinking, withdrawal from heat). Reflex arc = receptor → sensory neuron → **spinal cord** → motor neuron → effector muscle. It does **not** involve the brain, which is why it is fast.

The twelve cranial nerves (mnemonic and functions)

MNEMONIC FOR ORDER

- On Old Olympus' Towering Top, A Finn And German Viewed Some Hops** — Olfactory, Optic, Oculomotor, Trochlear, Trigeminal, Abducent, Facial, Auditory (vestibulocochlear), Glossopharyngeal, Vagus, Spinal accessory, Hypoglossal.
- For type: **Some Say Marry Money, But My Brother Says Big Brains Matter More** (S = sensory, M = motor, B = both).

No.	Nerve	Type	Main function
I	Olfactory	Sensory	Smell
II	Optic	Sensory	Vision
III	Oculomotor	Motor	Movement of the eyeball; constriction of the pupil; raising the eyelid
IV	Trochlear	Motor	Downward and outward eye movement (superior oblique)
V	Trigeminal	Both	Sensation of the face; muscles of chewing — the largest cranial nerve
VI	Abducent	Motor	Outward movement of the eye (lateral rectus)
VII	Facial	Both	Muscles of facial expression; taste from the front of the tongue; tears and saliva
VIII	Vestibulocochlear (auditory)	Sensory	Hearing and balance
IX	Glossopharyngeal	Both	Taste from the back of the tongue; swallowing; gag reflex
X	Vagus	Both	Longest cranial nerve — supplies heart, lungs and gut; slows the heart; parasympathetic
XI	Spinal accessory	Motor	Movement of the neck and shoulder muscles

No.	Nerve	Type	Main function
XII	Hypoglossal	Motor	Movement of the tongue

Sympathetic versus parasympathetic (autonomic) actions

Organ	Sympathetic ('fight or flight')	Parasympathetic ('rest and digest')
Heart	Rate and force increase	Rate decreases
Bronchi	Dilate (more air)	Constrict
Pupils	Dilate	Constrict
Skin vessels	Constrict → pallor, cold clammy skin	Dilate
Sweat glands	Increased sweating	Little effect
Gut / digestion	Slowed, mouth becomes dry	Peristalsis and secretion increase
Adrenal gland	Secretes adrenaline	—
Chief transmitter	Noradrenaline / adrenaline	Acetylcholine

All the classical signs of **shock and fear** — pale cold clammy skin, rapid pulse, dilated pupils, dry mouth, rapid breathing — are sympathetic (adrenaline) effects. This is why they appear together.

2.9 The Digestive System

The alimentary canal is a muscular tube about **9 metres (30 feet)** long from mouth to anus, with accessory organs opening into it.

Part	Length / size	Function and points to remember
Mouth	—	Mastication (chewing) and mixing with saliva; salivary amylase (ptyalin) begins starch digestion. Three pairs of salivary glands — parotid (largest), submandibular, sublingual; 1–1.5 litres of saliva daily.
Teeth	32 permanent, 20 milk (deciduous) teeth	Incisors (cut), canines (tear), premolars and molars (grind); hardest substance = enamel .
Pharynx & oesophagus	Oesophagus 25 cm	Swallowing (deglutition); food is pushed by peristalsis . The epiglottis guards the airway.
Stomach	Capacity 1–1.5 litres	Stores and churns food; secretes HCl (kills bacteria, pH 1.5–3.5), pepsin (protein), rennin and mucus; forms chyme . Intrinsic factor for vitamin B ₁₂ absorption.
Small intestine	6–7 m — duodenum 25 cm, jejunum, ileum	Main site of digestion and absorption ; villi enormously increase surface area; receives bile and pancreatic juice in the duodenum.
Large intestine	1.5 m — caecum, appendix, colon, rectum, anal canal	Absorption of water , salts and some vitamins; formation and storage of faeces; bacterial flora make vitamin K and B.
Liver	Largest gland/internal organ, 1.2–1.5 kg , right hypochondrium	Secretes bile (emulsifies fat), stores glycogen, iron and vitamins, detoxifies poisons and alcohol, forms urea and plasma proteins, produces heat, destroys old RBCs. Only organ that can regenerate .
Gall bladder	Stores 30–50 ml	Stores and concentrates bile.
Pancreas	12–15 cm, behind the stomach	Dual gland — exocrine: pancreatic juice (amylase, lipase, trypsin); endocrine: insulin and glucagon from the islets of Langerhans.

Digestive enzymes at a glance

Enzyme	Source	Acts on	Converts to
Salivary amylase (ptyalin)	Saliva	Starch	Maltose
Pepsin	Stomach (with HCl)	Proteins	Peptones
Rennin	Stomach (infants)	Milk casein	Curd
Trypsin	Pancreas	Proteins/peptones	Amino acids
Pancreatic amylase	Pancreas	Starch	Maltose/glucose
Lipase (+ bile salts)	Pancreas	Fats	Fatty acids + glycerol
Maltase, sucrase, lactase	Small intestine	Disaccharides	Glucose (monosaccharides)
Bile (not an enzyme)	Liver	Fat	Emulsified fat droplets

End products absorbed: carbohydrates as **glucose**, proteins as **amino acids**, fats as **fatty acids and glycerol** (through lacteals into the lymph). Balanced diet = carbohydrate, protein, fat, vitamins, minerals, water and fibre. **1 g carbohydrate/protein = 4 kcal; 1 g fat = 9 kcal.**

2.10 The Urinary (Excretory) System

Organ	Facts
Kidneys (2)	Bean-shaped, 11 × 6 × 3 cm , ~150 g each, at the back of the abdomen at T12–L3; the right kidney is slightly lower (liver above it). Functional unit = nephron (about 1 million per kidney).
Ureters (2)	25–30 cm muscular tubes carrying urine from kidney to bladder by peristalsis.
Urinary bladder	Muscular reservoir in the pelvis; capacity 400–600 ml ; desire to pass urine at about 200–300 ml.
Urethra	Male 18–20 cm , female 4 cm (hence urinary infection is commoner in women).

- **Functions of the kidney:** excretion of urea, uric acid, creatinine and drugs; regulation of **water and electrolyte balance**, **blood pressure** (renin) and **acid-base balance**; secretion of **erythropoietin** (stimulates RBC formation) and activation of **vitamin D**.
- Urine formation = **glomerular filtration** – **selective reabsorption** – **tubular secretion**. About **180 litres** are filtered daily, of which **1–1.5 litres** are excreted as urine.
- Normal urine — pale amber, aromatic, pH ~6, specific gravity 1.010–1.030, 96 % water + urea (main solid waste), uric acid, creatinine, salts. **Sugar, protein, blood, ketones and pus are abnormal**.
- Normal output **≥ 30 ml/hour (0.5 ml/kg/h)**. Reduced output (**oliguria**) or none (**anuria**) is an important sign of **shock, dehydration or severe burns**. Excessive urine = polyuria; painful = dysuria; blood in urine = haematuria.
- Other excretory organs: **skin** (water, salt, urea in sweat), **lungs** (CO₂ and water vapour), **liver** (bile pigments) and **large intestine** (faeces).

2.11 The Endocrine System

Ductless glands that pour **hormones** ('chemical messengers') directly into the blood. Action is slower but longer-lasting than nervous control.

Gland	Chief hormone(s)	Action / deficiency-excess
Pituitary ('master gland', pea-sized, base of brain)	Growth hormone, TSH, ACTH, FSH/LH, prolactin, ADH (vasopressin) and oxytocin from the posterior lobe	Controls other glands and growth. Excess GH → gigantism/acromegaly; deficiency → dwarfism; lack of ADH → diabetes insipidus
Thyroid (largest endocrine gland, in the neck)	Thyroxine (T4) , T3 and calcitonin	Controls metabolic rate . Deficiency → cretinism (child), myxoedema (adult), goitre with iodine lack; excess → thyrotoxicosis with bulging eyes
Parathyroid (4, behind the thyroid)	Parathormone	Regulates calcium ; deficiency → tetany (muscle spasm, cramps)
Adrenal (suprarenal) , above each kidney	Medulla: adrenaline (epinephrine) and noradrenaline. Cortex: cortisol, aldosterone, sex hormones	Adrenaline = the emergency hormone (fight or flight): raises heart rate, BP and blood sugar, dilates pupils and bronchi — used as the drug of choice in anaphylaxis . Cortisol deficiency → Addison's disease
Pancreas — islets of Langerhans	Insulin (beta cells) and glucagon (alpha cells)	Insulin lowers blood glucose; glucagon raises it. Lack of insulin → diabetes mellitus ; excess insulin or missed meal → hypoglycaemia
Ovaries / Testes	Oestrogen and progesterone / testosterone	Secondary sexual characters, reproduction, menstrual cycle
Thymus	Thymosin	Maturation of T-lymphocytes (immunity); largest in childhood
Pineal	Melatonin	Sleep-wake (circadian) rhythm

Normal **fasting blood glucose** = **70–110 mg/dl**; post-prandial < 140 mg/dl. Below **70 mg/dl** = **hypoglycaemia** (a first-aid emergency — give sugar); above 200 mg/dl random suggests diabetes.

2.12 The Skin (Integumentary System) and Temperature Regulation

The skin is the **largest organ of the body** — area about **1.5–2 m²**, weight about 4 kg (16 % of body weight).

Layer	Details
Epidermis (outer)	No blood vessels; several layers of epithelium; outermost dead keratin layer; contains melanin (colour, protects from UV). Renews every 4 weeks.
Dermis (true skin)	Contains blood vessels , nerve endings , sweat glands , sebaceous (oil) glands , hair follicles and collagen/elastic fibres. Pain of a burn comes from these nerve endings.
Subcutaneous layer	Adipose (fatty) tissue — insulation, energy store, shock absorption.

Functions of the skin (mnemonic: S-H-A-P-E-S)

- **Sensation** — touch, pain, pressure, heat and cold receptors.
- **Heat regulation** — sweating and dilatation of vessels lose heat; constriction and shivering conserve it.
- **Absorption** of some drugs and ointments, and **excretion** of water, salt and urea as sweat (500–700 ml daily).
- **Protection** — barrier against injury, micro-organisms, chemicals and ultraviolet light (the body's **first line of defence**).
- Elaboration of **vitamin D** from sunlight.
- Storage of fat and water; **cosmetic/identity** (fingerprints).

TEMPERATURE FACTS

- Normal body temperature **98.4 °F (37 °C)**; oral, axillary (0.5° lower) and rectal (0.5° higher) sites. °C = (°F – 32) × 5/9.
- **Fever (pyrexia)** > 99 °F; **hyperpyrexia** > 105 °F; **hypothermia** < 95 °F (35 °C).
- Heat is produced by metabolism, muscle activity and shivering; lost by **radiation, conduction, convection and evaporation** (sweating) — and also through breathing, urine and faeces.
- The **hypothalamus** is the body's thermostat.
- Loss of skin (as in burns) causes loss of fluid, heat and the barrier against infection — the three dangers of a major burn.

2.13 The Sense Organs

Organ	Structure	Function / first-aid relevance
Eye	Sclera (white), cornea (transparent front), iris with pupil , lens, aqueous and vitreous humour, retina with rods (dim light) and cones (colour), optic nerve, conjunctiva, lacrimal (tear) gland	Vision. Pupil signs are vital: equal and reacting = normal; unequal = head injury/stroke; pinpoint = morphine/opium poisoning or organophosphate; widely dilated and fixed = cardiac arrest, deep unconsciousness, atropine/dhatura poisoning
Ear	Outer — pinna and canal up to the eardrum (tympanic membrane); middle — 3 ossicles (malleus, incus, stapes) and the Eustachian tube to the throat; inner — cochlea (hearing) and semicircular canals (balance)	Hearing and equilibrium . Bleeding or clear fluid from the ear after a head injury suggests a fractured base of skull — never plug the ear
Nose	Nasal cavity with ciliated mucosa, turbinates, olfactory nerve endings; rich blood supply in the septum	Smell; filters, warms and moistens air. The rich vessels explain epistaxis (nose bleed)
Tongue	Papillae with taste buds; muscular organ	Taste (sweet, salt, sour, bitter, umami), speech, chewing and swallowing. Falls back and blocks the airway in unconsciousness
Skin	Nerve endings in the dermis	Touch, pressure, pain, heat and cold

2.14 The Reproductive System (in brief)

- **Male**: two testes (in the scrotum) producing **sperm** and **testosterone**; epididymis, vas deferens, seminal vesicles, prostate gland, urethra and penis.
- **Female**: two **ovaries** producing **ova** and the hormones oestrogen and progesterone; fallopian tubes (site of fertilisation), **uterus** (for pregnancy), cervix, vagina and external genitalia; **breasts** for lactation.
- **Menstrual cycle** — about **28 days**; ovulation on about day 14. **Pregnancy (gestation)** = **280 days / 40 weeks / 9 calendar months**, divided into three trimesters.
- First-aid relevance: pregnancy alters first aid — a pregnant casualty is turned onto her **left side** (to prevent the uterus pressing on the inferior vena cava), CPR compressions are given slightly higher on the sternum, and bleeding in pregnancy is always an emergency.

2.15 Master Table of Normal Values (learn by heart)

Parameter	Normal value
Body temperature	98.4 °F / 37 °C
Pulse (adult)	60–100 beats/min (avg 72)
Respiration (adult)	12–20 breaths/min (avg 16–18)
Blood pressure (adult)	120/80 mm Hg
Blood volume	5–6 litres (8 % of body weight)

Parameter	Normal value
Haemoglobin	M 13–18 g/dl, F 12–16 g/dl
RBC count	4.5–5.5 million/mm ³
WBC count	4,000–11,000/mm ³
Platelet count	1.5–4.5 lakh/mm ³
Blood pH	7.35–7.45
Fasting blood sugar	70–110 mg/dl
Urine output	1–1.5 litre/day (≥ 30 ml/hour)
Tidal volume	500 ml
Cardiac output	5 litres/min
SpO ₂	95–100 %
CSF volume	130–150 ml
Number of bones (adult)	206
Number of muscles	About 600–650
Chromosomes	46 (23 pairs)
Normal sleep (adult)	7–8 hours

2.16 Chapter Recap — One-Line Revision

- Cell → tissue → organ → system → organism; 4 tissue types; connective tissue is the most abundant.
- 206 bones** = axial 80 + appendicular 126; 27 in a hand, 26 in a foot; **femur** largest, **stapes** smallest, **clavicle** most often fractured.
- Vertebrae **7-12-5-5-4**; ribs **12 pairs** (7 true, 3 false, 2 floating).
- Heart has 4 chambers and 4 valves; **left ventricle** is thickest; pacemaker = **SA node**; pulmonary artery carries deoxygenated blood.
- Blood = plasma 55 % + cells 45 %; RBC lifespan **120 days**; **O– universal donor, AB+ universal recipient**.
- Right lung **3 lobes**, left **2**; inhaled foreign bodies go to the **right** lung; expired air has **16 % O₂** and **4 % CO₂**.
- Vital centres (breathing, heart, BP) are in the **medulla oblongata**; balance in the **cerebellum**; thermostat in the **hypothalamus**.
- 12 cranial nerves, 31 pairs of spinal nerves**; vagus is the longest, trigeminal the largest.
- Liver** is the largest gland; **thyroid** the largest endocrine gland; **pituitary** the master gland; **skin** the largest organ.
- Adrenaline explains every sign of shock — pale, cold, clammy skin with a rapid weak pulse and dilated pupils.

PART II • LIFE-SAVING SKILLS

CHAPTER 3

Dressings and Bandages

Types of dressing; the triangular bandage and all its folds and slings; the cotton roller bandage and its turns; the rubber (elastic) bandage; special bandages, rules, dangers and improvisation.

SYLLABUS COVERED IN THIS CHAPTER

- Dressing and Bandages — use of **triangular bandages** and **cotton roller bandage**, **rubber bandage** and **different types of dressing**.
- General rules of bandaging, slings, signs of a tight bandage, improvisation and care of the casualty.

3.1 Definitions and Purpose

DEFINITIONS

- **Dressing** — a **protective covering applied directly over a wound or burn** to arrest bleeding, prevent infection and absorb discharge. It must be **sterile (or at least clean)**, **absorbent**, **porous** and **larger than the wound** on all sides.
- **Bandage** — a **piece of cloth or material used to hold a dressing in position**, to apply pressure, to support or immobilise a part, or to prevent/reduce swelling. A bandage **never** touches the wound directly.
- **Pad** — thick soft material (cotton wool, folded cloth) placed over a dressing to apply pressure or to fill a hollow.
- **Compress** — a folded cloth applied wet: **cold compress** (sprain, bruise, nose bleed) or **hot compress** (boil, pain relief).

Purpose of a DRESSING	Purpose of a BANDAGE
Controls bleeding by direct pressure.	Holds the dressing or splint in position.
Prevents entry of germs (prevents infection).	Applies pressure to control bleeding.
Absorbs blood and discharge.	Gives support to an injured limb or joint.
Protects the wound from further injury and from air/dust.	Immobilises a fracture, dislocation or wounded part.
Relieves pain by covering exposed nerve endings.	Restricts swelling (compression) and helps reduce it.
Promotes healing by keeping the wound moist and clean.	Helps in lifting and carrying the casualty.

3.2 Different Types of Dressing

Type of dressing	Description	Chief uses
Dry sterile (gauze) dressing	Sterile gauze squares in a sealed packet, covered with a pad of cotton wool and held by a bandage; the standard first-aid dressing	Most open wounds; absorbs blood and allows the wound to breathe
Adhesive dressing (Band-Aid, sticking plaster)	Small gauze pad with an adhesive backing, often waterproof	Small clean cuts, abrasions; no separate bandage needed
Sterile (standard/first field) dressing	A sterile pad already attached to a roller bandage in a sealed packet; used by the armed forces and ambulance services	Bleeding wounds — quickest of all dressings; one hand can apply it
Non-adherent / paraffin (tulle gras) dressing	Gauze impregnated with paraffin or silicone so that it does not stick	Burns , scalds, grazes and raw surfaces — painless to remove
Medicated dressing	Gauze soaked in an antiseptic or medicament (povidone-iodine, silver sulphadiazine, honey, framycetin)	Infected wounds, ulcers, burns — on medical advice only
Wet (moist) dressing / compress	Cloth wrung out in cold (or warm) water or saline	Cold: sprains, bruises, stings; warm: to relieve pain and soften crusts
Pressure dressing	Dressing plus a thick pad, firmly bandaged over the wound	Severe bleeding
Occlusive (air-tight) dressing	Plastic sheet, cling film or a sterile pad taped on three sides ('flutter valve')	Sucking chest wound (open pneumothorax), exposed abdominal organs
Ring (doughnut) pad	A ring made by winding a narrow bandage round the fingers, then binding it	Embedded foreign body , protruding bone end, fractured skull — pressure is put round, not on, the injury

Type of dressing	Description	Chief uses
Eye pad	Sterile pad with a bandage or shield	Eye injuries; the uninjured eye may also be covered to stop movement
Improvised dressing	Clean handkerchief, freshly laundered linen, towel, sanitary pad, clean cloth; in the last resort the casualty's own clean clothing	When no sterile dressing is at hand — remember: any clean cover is better than none
Cotton wool	Absorbent cotton — never placed directly on a wound because its fibres stick to it	Only as padding over a gauze dressing

Rules for applying a dressing

1. **Wash your hands** and wear disposable gloves if available.
2. Do not touch the wound or the surface of the dressing that will touch the wound; **hold a dressing by its edges/corners**.
3. Do not cough, sneeze or breathe over the wound or dressing; **do not talk** over it.
4. The dressing must **extend at least 2.5 cm (1 inch) beyond the wound** on all sides.
5. Place the dressing **directly on the wound — do not slide it** into position.
6. Add a pad of cotton wool over the gauze and hold it with a bandage or adhesive strapping.
7. If blood soaks through, **do not remove the dressing** — apply another pad and bandage over it and press firmly.
8. Do not apply cotton wool, fluffy material or antiseptic powder directly to a wound; do not use anything sticky on a burn.
9. Never remove a dressing that has stuck to a wound — leave that to the doctor.
10. Change dressings only when soaked, or as advised; check circulation beyond the dressing.

3.3 General Rules of Bandaging

1. **Explain and reassure** — tell the casualty what you are going to do.
2. Make the casualty **sit or lie comfortably** in a well-supported position; support the injured part while bandaging.
3. Stand **in front of the casualty** (and on the injured side) so that you can watch his face for signs of pain.
4. Bandage the injured part in the **position in which it is to remain** (joints slightly flexed, not fully straight).
5. Apply from the **inner to the outer side** and from **below upwards (distal to proximal)** — towards the heart — so that blood is not trapped in the limb.
6. Hold the **head (roll) of a roller bandage uppermost/outermost** and unroll only a little at a time, keeping the roll in the right hand for a right limb.
7. Each turn should **overlap the previous one by about two-thirds ($\frac{1}{2}$ – $\frac{2}{3}$)** so that no skin shows and the pressure is even.
8. Bandage **firmly enough to control bleeding or to support, but never so tightly as to stop circulation**; avoid uneven or slack turns.
9. **Pad hollows and bony points** (armpit, elbow, knee, ankle, between fingers) with cotton wool to prevent chafing and to spread pressure.
10. **Leave the finger tips and toes exposed** whenever possible so that circulation can be checked.
11. Finish off by **tucking in the end, or with a reef knot, safety pin, clip or adhesive tape**; the **knot must not lie over a wound, a bony point, or the back of a part on which the casualty will lie**.
12. Use a **reef knot** — it is flat, secure and easy to untie.
13. **Check circulation** immediately after applying and again every **10 minutes**; loosen at once if there are signs of tightness.
14. Avoid bandaging **over a fresh wound with unequal pressure**; never apply a wet bandage that will shrink on drying.
15. In fractures, bandages must be applied **above and below the fracture site — never over it**.
16. Remove or replace a bandage that becomes loose, soiled or wet.

SIGNS THAT A BANDAGE IS TOO TIGHT — LOOSEN AT ONCE (MNEMONIC: THE 5 P'S + 1)

- **Pain** — throbbing or increasing pain beyond the bandage.
- **Pallor** — the part becomes pale, then **blue (cyanosed)**.
- **Pulselessness** — no pulse beyond the bandage.
- **Paraesthesia** — tingling, 'pins and needles', numbness.
- **Paralysis** — inability to move the fingers or toes.
- **Coldness and swelling** of the part; **capillary refill of the nail bed slower than 2 seconds** (press the nail:

colour should return at once).

3.4 Types of Bandage — Overview

Bandage	Material / form	Chief use
Triangular bandage	Cotton cloth 1 m × 1 m square cut diagonally into two triangles	Slings, holding dressings, immobilising, improvised pads and tourniquets — the most versatile first-aid bandage
Roller bandage (cotton/open-weave)	A long strip of cotton, gauze or flannel rolled up; widths 2.5–15 cm	Holding dressings, support, pressure over a dressing, immobilisation
Rubber / elastic (crepe) bandage	Woven cotton with rubber or elastic threads; stretches	Firm support and compression for sprains, strains, varicose veins, swelling
Adhesive bandage / strapping	Zinc-oxide plaster, micropore tape, adhesive strips	Fixing dressings, strapping ribs/joints, closing small wounds
Tubular gauze / net bandage	Seamless tube applied with an applicator	Fingers, toes, head, limbs
Special (shaped) bandages	T-bandage, four-tailed, many-tailed (Sculptetus), suspensory, capeline	Perineum, jaw, abdomen, scrotum, scalp
Plaster of Paris	Bandage impregnated with gypsum, applied wet	Hospital immobilisation of fractures — not a first-aid measure

3.5 The Triangular Bandage

Introduced by **Esmarch**, the triangular bandage is the first aider's most useful single article. It is made by cutting a **square metre of cloth (about 38–40 inches) diagonally**, giving two bandages; the two short sides measure about **1 m** each and the long side (base) about **1.3–1.4 m**.

Parts of a triangular bandage

- **Point** — the corner opposite the base (the apex).
- **Base** — the longest side.
- **Ends (extremities)** — the two corners at either end of the base.
- **Sides** — the two shorter borders.

The three forms (folds) of a triangular bandage

Form	How it is made	Uses
Open (whole) bandage	Used unfolded, as a triangle	Slings ; large dressings of the scalp, chest, back, hand, foot, hip/buttock and stump
Broad-fold bandage	Bring the point down to the base , then fold once more in the same direction — gives a band about 15–20 cm wide	Immobilising and supporting — securing splints, tying legs together, fractures of the thigh/leg, chest and rib injuries
Narrow-fold bandage	Fold the broad bandage once again lengthwise — a band about 7–10 cm wide	Figure-of-eight bandage of the ankle/foot , fixing dressings, improvised tourniquet , tying feet together, making a ring pad , collar-and-cuff sling

- A **cravat** is another name for the narrow-fold (or broad-fold) bandage.
- To store a triangular bandage, fold the ends to the centre repeatedly and then fold in half — it becomes a small packet for the kit.
- Always finish with a **reef knot**; place a pad under the knot if it presses on a bony point.

Uses of the triangular bandage — complete list

Application	Method in brief
Arm (large/St John) sling	For injuries of the forearm, wrist and hand , and for fractured ribs. Support the forearm across the chest, hand slightly higher than the elbow . Place the open bandage between the chest and forearm, point towards the elbow of the injured side; bring the lower end up over the shoulder and tie the two ends in the hollow above the collar bone on the injured side ; tuck or pin the point in front of the elbow. Finger tips must be visible.

Application	Method in brief
Elevation sling	For hand injuries with bleeding, fractured collar bone, crushed hand, injured shoulder — keeps the hand well raised to reduce bleeding and swelling. Place the casualty's forearm diagonally across the chest, fingers towards the opposite shoulder ; lay the bandage over the forearm with one end over the sound shoulder; tuck the base under the hand and forearm; bring the lower part under the elbow and across the back; tie at the hollow of the neck; twist and tuck the point in at the elbow.
Collar-and-cuff (clove-hitch) sling	A narrow-fold bandage tied as a clove hitch round the wrist and the ends taken round the neck — used for a fractured humerus/upper arm where the weight of the arm should hang, and when a full sling cannot be applied.
Triangular (improvised) sling	Made with the casualty's coat, jacket, shirt front, belt, tie, scarf, dupatta or by pinning the sleeve to the clothing.
Scalp / head dressing (capelline)	Fold a hem along the base; place the base on the forehead just above the eyebrows with the point hanging down the back; cross the ends behind the head over the point, bring them round the forehead and tie in front; then draw the point down and pin it up on top.
Chest / back	Place the point over the shoulder on the injured side, base across the chest; tie the two ends at the back, leaving one end long, and tie that end to the point.
Hand	Lay the hand palm-down on the open bandage with the wrist on the base and fingers towards the point; fold the point over the back of the hand; cross the ends over the wrist and tie.
Foot	Place the foot on the centre of the bandage with the heel towards the base; bring the point over the instep, fold the sides over and tie the ends round the ankle.
Knee / elbow	Point placed on the thigh (or upper arm), base below the joint; cross the ends behind the joint, bring them up and tie; bring the point down and pin.
Hip / buttock	Tie a narrow-fold bandage round the waist; apply the open bandage with the point tucked under the waist band and the base round the thigh.
Shoulder	Point up the side of the neck, base folded over the upper arm; tie the ends on the outer side of the arm and fix the point under a sling.
Stump (amputation)	Cover the stump with the point brought over the end; cross the ends round the stump and tie; pin the point.
Other uses	As a broad bandage to tie splints or legs together; as a narrow bandage for a figure-of-eight ankle bandage; as a ring pad , a cold compress , an eye pad cover , an improvised tourniquet , a sling for carrying, a filter/mask, and to tie a casualty to a stretcher.

SLINGS — THE MOST-ASKED COMPARISON

- **Arm sling** — supports the forearm **horizontally/slightly raised**; for forearm, wrist and hand injuries and fractured ribs; hand is **slightly above the level of the elbow**.
- **Elevation sling** — the hand is placed near the **opposite shoulder (well above the heart)**; for **bleeding hand injuries, fractured collar bone and injured shoulder**.
- **Collar-and-cuff sling** — supports only the **wrist**, letting the arm hang; for a **fractured upper arm (humerus)**.
- In every sling the knot is tied in the **hollow above the collar bone on the injured side** and the **finger tips are left exposed**.

3.6 The Cotton Roller Bandage

A roller bandage is a strip of cotton, open-weave gauze, flannel or calico rolled into a cylinder. It is the commonest bandage for holding dressings and giving support.

Parts of a roller bandage

- **Head (roll)** — the rolled-up portion held in the hand.
- **Tail (free end)** — the loose end applied first.
- **Body** — the portion being unrolled.
- A **single-headed** bandage is rolled from one end; a **double-headed** bandage is rolled from both ends towards the middle (used for the head and jaw).

Standard widths — which size for which part

Part of the body	Width of bandage
Finger	2.5 cm (1 inch)
Hand	5 cm (2 inches)
Arm / forearm	5–6 cm (2–2½ inches)
Leg (lower limb)	7.5–9 cm (3–3½ inches)
Trunk, chest, abdomen, thigh	10–15 cm (4–6 inches)

Part of the body	Width of bandage
Head / scalp	5–7.5 cm
Usual length	5–6 metres (bandages are supplied 3–10 m long)

Basic turns of a roller bandage — learn the table

Turn	How it is done	Where it is used
Simple / circular turn	Turns laid exactly one over the other, at right angles to the limb	To begin and end every bandage; for wrist, neck and finger and for holding a dressing on a uniform part
Spiral turn	Turns run obliquely up the limb, each overlapping the previous by two-thirds	Parts of uniform thickness — fingers, upper arm, trunk
Reverse spiral (spiral reverse)	Each spiral turn is twisted (reversed) on itself with the thumb to make it lie flat	Tapering (cone-shaped) parts — forearm and leg
Figure-of-eight turn	Ascending and descending oblique turns crossing each other in the form of the figure 8	Joints — elbow, knee, ankle , wrist; gives firm support
Spica	A modified figure-of-eight in which the turns overlap like the husk of an ear of corn	Shoulder, hip, groin and thumb ('thumb spica')
Divergent spica	Turns pass alternately above and below the joint, diverging from the centre	Over a flexed elbow or knee
Recurrent turn	Turns are carried backwards and forwards over the end of the part and then fixed by circular turns	Finger tip, stump of an amputated limb, head
Locking/anchoring turn	First turn placed obliquely, its corner turned down and covered by the next turn	To fix the beginning of any bandage

Method of applying a roller bandage

1. Choose the **correct width** for the part; use a **firmly rolled** bandage.
2. Face the casualty; support the limb in the position it is to remain.
3. Hold the **roll upwards in the right hand** for a right limb and place the **outer surface of the tail on the part, below the injury**.
4. Make **two firm circular (anchoring) turns** to fix the tail.
5. Bandage **upwards and from within outwards**, each turn covering **two-thirds** of the previous one.
6. Use the turn suited to the part (spiral, reverse spiral, figure-of-eight, spica or recurrent).
7. Finish with **one or two circular turns** and fasten with a safety pin, clip, adhesive tape, or by splitting the end and tying a reef knot.
8. **Check the circulation** of the fingers/toes at once and again after 10 minutes.

Particular applications

Finger — Anchor at the wrist, carry to the finger, apply spiral turns, finish with a figure-of-eight round the wrist (so it cannot slip off).

Hand and wrist — Anchor at the wrist, cross the palm, figure-of-eight turns round the hand, leaving the thumb free; finish at the wrist.

Elbow / knee — Flex the joint slightly; apply a figure-of-eight or divergent spica with the crossing over the front of the elbow (back of the knee is padded).

Ankle and foot — Begin with a circular turn round the instep, carry the bandage in **figure-of-eight turns round the ankle** and finish above the ankle.

Leg / forearm — Reverse spiral turns from below upwards.

Scalp / head — Circular turns round the forehead and occiput, or a double-headed bandage; secure at the side.

Stump — Recurrent turns over the end, then circular turns.

3.7 The Rubber (Elastic / Crepe) Bandage

A rubber bandage is made of cotton woven with **rubber or elastic threads**, so that it **stretches and grips** the part and exerts continuous, even pressure. The historical **Esmarch's rubber bandage** was used to squeeze blood out of a limb before surgery and as a tourniquet.

Uses of a rubber/elastic bandage

- **Firm support and compression** of a **sprain, strain or bruise** (the 'C' of the **RICE** treatment — Rest, Ice, Compression, Elevation).
- To **reduce and prevent swelling (oedema)** after injury, and to prevent re-accumulation.

- Support for **varicose veins**, for **thrombosis prophylaxis** and for **venous ulcers** (elastic/crepe compression bandage).
- Support to a **joint or muscle during and after activity**, and over a **plaster-free mild fracture** while awaiting medical help.
- Firm pressure over a dressing** in bleeding, and to hold splints and dressings on rounded or moving parts (the stretch keeps it in place).
- Application over a **snake-bitten limb** as a *pressure immobilisation bandage* (firm, as for a sprained ankle, plus a splint) — in current practice done with a crepe/elastic bandage, **never a tourniquet**.
- As an **emergency tourniquet or Esmarch bandage by medical personnel only**.

Advantages, precautions and dangers

Advantages	Precautions / dangers
Stretches, so it fits rounded and moving parts and does not slip.	It is easy to apply too tightly — the commonest cause of a constricted limb; never stretch it to its full length.
Gives continuous, even and adjustable compression.	Must be applied evenly from the toes/fingers upwards , never leaving a tight ring.
Washable and reusable.	Never use on an open, dirty or infected wound as the only cover, and never directly over a burn.
Comfortable and allows some movement.	Do not leave on overnight or during sleep; remove and re-apply every few hours; check circulation every 10–15 minutes at first.
Can be used over dressings and splints.	Contra-indicated where circulation is already poor (arterial disease, diabetic limb) and in an unsplinted fracture .

CREPE / RUBBER BANDAGE VS COTTON ROLLER BANDAGE

- Cotton roller** — non-elastic; holds dressings and gives moderate support; cheap; absorbs discharge; can loosen with movement.
- Crepe / rubber** — elastic; gives **compression and firm support**; does not absorb; stays in place on joints; but carries a **greater risk of over-tightness**.
- For a **sprained ankle** the bandage of choice is a **crepe bandage in figure-of-eight turns**.

3.8 Special and Improvised Bandages

Bandage	Shape and use
T-bandage (single/double T)	A horizontal band with one or two vertical tails — holds dressings on the perineum, groin and anus
Four-tailed bandage	A strip split at both ends — for the chin/jaw, nose, back of the head, elbow, knee
Many-tailed (Scultetus) bandage	Overlapping tails stitched to a central band — for the abdomen and chest , where re-application must be easy
Capeline bandage	A double-headed bandage for the scalp/whole head
Suspensory bandage	Supports the scrotum
Tubular gauze	Seamless tube pushed on with an applicator — fingers, toes, limbs, head
Adhesive strapping	Fixes dressings, supports sprains, strapping of fractured ribs (rarely used now)
Improvised bandages	Handkerchief, dupatta, scarf, necktie, belt, stockings, torn bed sheet, towel, shawl, rope (padded), bark, clothing — the first aider improvises rather than delays
Improvised splints	Umbrella, walking stick, broom, cricket bat, rolled newspaper/magazine, cardboard, wooden plank, pillow, folded blanket; the casualty's own sound limb or the trunk can also serve as a splint

3.9 Chapter Recap — One-Line Revision

- Dressing touches the wound; bandage holds the dressing.** A dressing must be sterile, absorbent and larger than the wound by 2.5 cm.
- Triangular bandage = **1 m square cut diagonally**; parts = point, base, ends, sides; forms = **open, broad-fold, narrow-fold**.
- Arm sling** → forearm/wrist/ribs; **elevation sling** → bleeding hand, fractured collar bone; **collar-and-cuff** → fractured humerus.
- Roller bandage widths: finger **2.5 cm**, hand **5 cm**, arm **5–6 cm**, leg **7.5–9 cm**, trunk **10–15 cm**.
- Turns: circular (start/finish), spiral (uniform part), **reverse spiral (forearm/leg)**, **figure-of-eight (joints/ankle)**, spica (shoulder/hip/thumb), recurrent (stump/finger tip).

- Bandage **from below upwards, inner to outer**, overlapping **two-thirds**; tie with a **reef knot**; keep finger tips visible.
- Tight bandage = **pain, pallor, pulselessness, paraesthesia, paralysis, coldness and blueness** — **loosen immediately**.
- **Rubber/crepe bandage** = compression and support (sprains, varicose veins, RICE); greatest danger is over-tightness.
- **Cotton wool must never be placed directly on a wound**; never remove a blood-soaked dressing — add another over it.

CHAPTER 4

Cardio-Pulmonary Resuscitation (CPR)

Basic Life Support: recognition of cardiac arrest, high-quality chest compressions, rescue breathing, the AED, CPR in children and infants, choking, the recovery position and special situations.

SYLLABUS COVERED IN THIS CHAPTER

- Cardio-pulmonary resuscitation — meaning, indications, complete step-by-step technique in adults, children and infants; one- and two-rescuer CPR; compression-only CPR.
- Automated External Defibrillator, recovery position, choking (Heimlich manoeuvre), complications, when to start and when to stop, and CPR in special situations.

4.1 Meaning and Definitions

DEFINITIONS

- **CPR (cardio-pulmonary resuscitation)** — an emergency life-saving procedure combining **chest compressions** (to circulate blood artificially) with **rescue breathing** (to oxygenate the blood), performed when the heart and breathing have stopped. *Cardio* = heart, *pulmonary* = lungs, *resuscitation* = revival.
- **Cardiac arrest** — sudden and complete cessation of effective pumping by the heart; the casualty is **unresponsive, not breathing normally and has no pulse**.
- **Respiratory arrest** — breathing has stopped but the heart is still beating (it will stop within minutes if untreated).
- **Basic Life Support (BLS)** — the whole set of skills that maintain airway, breathing and circulation without drugs or advanced equipment: recognition, calling for help, CPR and AED use.
- **Advanced Cardiac Life Support (ACLS)** — BLS plus drugs, advanced airways, manual defibrillation and monitoring, by trained medical teams.
- **Defibrillation** — delivery of a controlled electric shock to restore a normal heart rhythm in ventricular fibrillation.

CPR does **not usually restart the heart by itself**; it keeps oxygenated blood flowing to the **brain and heart muscle** until a defibrillator and advanced care arrive. Good CPR provides only about **25–33 % of the normal cardiac output** — which is why it must be of high quality and started at once.

4.2 Causes and Recognition of Cardiac Arrest

Cardiac causes	Non-cardiac causes
Heart attack (myocardial infarction) — the commonest cause in adults	Asphyxia — choking, drowning, strangulation, smoke inhalation
Ventricular fibrillation and other arrhythmias	Electrocution and lightning strike
Heart failure, cardiomyopathy, valve disease	Severe haemorrhage and shock
Blunt injury over the chest (commotio cordis)	Poisoning and drug overdose
Congenital heart disease (in the young)	Head injury , stroke, severe allergy (anaphylaxis) , hypothermia, severe hypoglycaemia , electrolyte disturbance

- In **adults** cardiac arrest is usually **cardiac (a rhythm problem)** — therefore **compressions and early defibrillation** come first.
- In **children and infants** it is usually **respiratory (asphyxial)** — therefore **breaths matter as much as compressions**, and CPR begins with **5 rescue breaths** in the European/Indian sequence.

Signs of cardiac arrest — how to recognise it in seconds

Sign	How to check
Unresponsiveness	Shake the shoulders gently and shout 'Are you all right?' — no reply, no movement
Absent or abnormal breathing	Look for chest movement, listen and feel at the mouth for not more than 10 seconds . Agonal gasping (occasional noisy gasps) is <i>not</i> normal breathing — treat as cardiac arrest
Absent pulse	Carotid pulse in adults/children, brachial or femoral in infants — checked only by trained rescuers, for ≤ 10 seconds . Laypersons should not delay CPR to feel for a pulse
Other signs	Deathly pale or blue-grey (cyanosed) skin; widely dilated pupils not reacting to light; no heart sounds; sometimes a brief seizure at the onset

4.3 The Chain of Survival and When to Start CPR

CPR is begun in **any** unresponsive casualty who is **not breathing normally**. Do not wait for a pulse check, and do not wait for the ambulance.

Start CPR when	Do NOT start / stop CPR when
Casualty is unresponsive and not breathing normally (or only gasping).	There are obvious signs of irreversible death — rigor mortis, decomposition, decapitation, injuries incompatible with life, dependent lividity.
After a drowning, electrocution, choking, poisoning or lightning strike with no breathing.	The casualty starts to breathe normally, coughs or moves — turn into the recovery position and monitor.
Cardiac arrest in a hypothermic casualty — continue for longer ('nobody is dead until warm and dead').	Qualified help takes over , or a valid 'do not resuscitate' order exists.
Cardiac arrest in pregnancy, in trauma, in a child — always.	The rescuer is exhausted or the scene becomes unsafe .

CRITICAL TIME FACTS

- Chances of survival fall by **7–10 % for every minute** without CPR and defibrillation.
- **Brain damage** starts within **3–4 minutes** of arrest and is usually irreversible by **8–10 minutes**.
- Immediate bystander CPR **doubles to triples** survival.
- The target is **CPR within 4 minutes** and **defibrillation within 8 minutes** (ideally 3–5).
- Interruptions in compressions must be kept to **less than 10 seconds**; 'chest compression fraction' should exceed **60 %** of the time.

4.4 Adult CPR — Step-by-Step Procedure

The modern sequence is **C-A-B (Compressions – Airway – Breathing)**, adopted in 2010 in place of the older A-B-C, so that compressions are not delayed.

1. **D — Danger.** Make sure it is safe for you (traffic, live wires, gas, water). Wear gloves if available.
2. **R — Response.** Tap the shoulders and shout. If there is no response, the casualty is unconscious.
3. **S — Shout / Send for help.** Shout for a bystander; get someone to **dial 112 / 108** and **fetch an AED**. If alone with a phone, put it on **speaker** and call while starting CPR. For an adult, **call first**; for a child or a drowning casualty, **give 1 minute of CPR first, then call** ('phone fast').
4. **Position the casualty** — flat on the **back on a firm, hard surface** (the floor, not a bed), arms by the side; kneel beside the chest. Loosen tight clothing and expose the chest.
5. **Check breathing** — open the airway with **head tilt–chin lift** and look, listen and feel for **≤ 10 seconds**. Gasping = no breathing.
6. **C — Start 30 chest compressions.** Place the **heel of one hand on the centre of the chest (lower half of the sternum)**, the other hand on top with fingers interlocked; keep **arms straight, elbows locked, shoulders directly above the hands**, and compress using the weight of your upper body from the hips.
7. **A — Open the Airway** again by head tilt–chin lift (jaw thrust if spinal injury is suspected); remove any visible obstruction, loose dentures or vomit.
8. **B — Give 2 rescue breaths.** Pinch the nose, seal your mouth over the casualty's mouth (or use a pocket mask/face shield) and blow steadily for about **1 second** until the chest rises. Allow the chest to fall, then give the second breath. Each breath ≈ 500–600 ml — just enough for a **visible chest rise**.
9. **Continue cycles of 30 : 2** without interruption. Change rescuers **every 2 minutes (about 5 cycles)** to avoid fatigue, taking less than 10 seconds to change over.
10. **Attach the AED as soon as it arrives** and follow its voice prompts.
11. **Continue until** the casualty breathes normally, qualified help takes over, or you are physically exhausted.

The five components of high-quality CPR (the numbers to memorise)

Component	Adult standard
Rate of compressions	100–120 per minute (≈ 2 per second)
Depth of compressions	5–6 cm (2–2.4 inches) — at least $\frac{1}{3}$ of the chest depth
Compression : ventilation ratio	30 : 2 (one or two rescuers in an adult)
Recoil	Allow complete chest recoil after each compression; do not lean on the chest
Interruptions	Minimise — no pause longer than 10 seconds

Component	Adult standard
Hand position	Heel of the hand on the lower half of the sternum (centre of the chest, between the nipples)
Rescue breath volume/time	Chest just rises; 1 second per breath; avoid over-inflation
Cycles per 2 minutes	About 5 cycles of 30 : 2
Rescuer change	Every 2 minutes , in < 10 seconds

COMMON MISTAKES THAT MAKE CPR USELESS

- Compressing **too slowly, too shallowly** or in the **wrong place** (over the xiphoid or the ribs).
- **Leaning** on the chest so that it does not recoil — the heart cannot refill.
- **Bent elbows** — wastes energy and reduces depth.
- Long pauses for pulse checks, moving the casualty, or fumbling with the AED.
- **Over-ventilation** (too many, too forceful breaths) — causes stomach inflation, vomiting and raises pressure in the chest, reducing blood flow.
- CPR on a **soft surface (bed, sofa)** or with the casualty sitting.

4.5 Compression-Only (Hands-Only) CPR

- If a rescuer is **untrained, unwilling or unable to give rescue breaths**, he should give **continuous chest compressions at 100–120/min without pauses**.
- This is as effective as standard CPR in the **first few minutes** of an adult cardiac arrest witnessed by a bystander.
- Hands-only CPR is **not suitable** for: **children and infants, drowning, choking, drug overdose**, and arrests that are **not witnessed** — all of these are asphyxial and need breaths.
- Telephone-guided (dispatcher-assisted) CPR is recommended — the ambulance control room will guide a bystander.

4.6 CPR in Children and Infants

A **newborn** is up to 1 month, an **infant** under 1 year, a **child** from 1 year to puberty; from puberty onwards adult technique is used.

Point	Adult	Child (1 yr – puberty)	Infant (< 1 yr)
Hands used	Two hands (heel of one, other on top)	One or two hands (one hand is enough in a small child)	Two fingers (one rescuer) or two thumbs encircling (two rescuers)
Site	Lower half of the sternum, centre of the chest	Lower half of the sternum	Just below the nipple line , on the lower sternum
Depth	5–6 cm	About 5 cm / 1/3 of chest depth	About 4 cm / 1/3 of chest depth
Rate	100–120/min	100–120/min	100–120/min
Ratio (single rescuer)	30 : 2	30 : 2	30 : 2
Ratio (two healthcare rescuers)	30 : 2	15 : 2	15 : 2
Initial breaths	None (start compressions)	5 rescue breaths first (ERC/Indian practice)	5 rescue breaths first ; puffs from the cheeks only
Airway opening	Head tilt + chin lift	Head tilt + chin lift	Neutral position — do not over-extend the neck
Rescue breath technique	Mouth-to-mouth, nose pinched	Mouth-to-mouth	Mouth-to-mouth-and-nose (cover both) or mouth-to-nose
Pulse check point	Carotid	Carotid or femoral	Brachial (inner upper arm) or femoral
When to call help if alone	Call first , then CPR	CPR for 1 minute (5 cycles), then call	CPR for 1 minute, then call
AED	Adult pads and energy	Paediatric pads/attenuator if available (1–8 yr)	Manual defibrillator preferred; AED with paediatric pads if needed

PAEDIATRIC POINTS THAT ARE ASKED

- In children, cardiac arrest is usually due to **respiratory causes** — so open the airway and give **5 initial rescue breaths**.
- Compress with **two fingers** in an infant (single rescuer) and with the **two-thumb encircling technique** when two rescuers are present.

- Never use full adult force on a child; depth is **one-third of the chest** in both children and infants.
- **Newborn resuscitation** uses a ratio of **3 : 1** with a rate of 120 events/min, and the **head is kept in a neutral position**.

4.7 One-Rescuer versus Two-Rescuer CPR

Feature	One rescuer	Two rescuers
Ratio (adult)	30 : 2	30 : 2
Ratio (child/infant, healthcare providers)	30 : 2	15 : 2
Roles	Same rescuer compresses and ventilates	One compresses, the other manages the airway and breaths and watches for chest rise
Change over	Rest is impossible — fatigue sets in within 1–2 minutes	Change every 2 minutes , taking less than 10 seconds
Advantage	Can be started immediately by anyone	Better quality, less fatigue, faster AED use and better monitoring

4.8 Rescue Breathing (Ventilation) Technique

- **Mouth-to-mouth** — the standard: head tilt–chin lift, pinch the nose, wide seal over the mouth, blow for **1 second** watching the chest rise, release and let the chest fall.
- **Mouth-to-nose** — used when the mouth is injured, jaw is clenched (trismus), the mouth cannot be sealed, or in rescue from water.
- **Mouth-to-mouth-and-nose** — in **infants**.
- **Mouth-to-stoma** — in a casualty with a tracheostomy/laryngectomy opening in the neck; seal the mouth and nose.
- **Mouth-to-mask / face shield** — preferred for protection against infection; the one-way valve prevents contact with vomit and blood.
- **Bag-valve-mask (Ambu bag)** — by trained rescuers; two-person technique gives a better seal; can deliver **100 % oxygen**.
- If the chest does **not** rise: **re-check the head tilt, re-seal the mouth, look for an obstruction** — but do **not** give more than **2 attempts** before resuming compressions.
- **Gastric inflation** from forceful breaths causes vomiting and aspiration — if vomiting occurs, turn the head to the side, clear the mouth and continue.

4.9 The Automated External Defibrillator (AED)

An AED is a portable device that analyses the heart rhythm and, if a **shockable rhythm (ventricular fibrillation or pulseless ventricular tachycardia)** is present, delivers an electric shock. It is designed to be used by laypersons and **will not shock a casualty who does not need it**.

1. Switch the AED on as soon as it arrives; **continue CPR while the pads are being attached**.
2. Bare and dry the chest. Attach the pads: one **below the right collar bone**, the other on the **left side of the chest below the armpit (mid-axillary line)** — the *anterolateral* position.
3. Ensure nobody is touching the casualty; say **'Stand clear!'** while the machine analyses.
4. If a shock is advised, press the **shock button** (or it may be automatic), then **resume compressions immediately** for 2 minutes before the next analysis.
5. If **no shock is advised**, resume CPR at once.
6. Leave the pads attached and the AED on until help arrives.

Situation	AED precaution
Wet chest / casualty in water	Drag clear of water and dry the chest before applying pads
Hairy chest	Shave or wipe quickly, or press the pads firmly; a spare set of pads may be needed
Medication patch on the chest	Remove the patch and wipe the skin
Pacemaker or implanted defibrillator (a lump under the skin)	Place the pad at least 8–12 cm (3–5 in) away from the device
Metal jewellery / metal surface	Remove necklaces near the pads; move the casualty off a metal sheet; do not touch the casualty during the shock
Child 1–8 years	Use paediatric pads/attenuator ; if not available, adult pads may be used

Situation	AED precaution
Infant < 1 year	A manual defibrillator is preferred; an AED may be used if nothing else is available
Pregnancy / trauma	AED is used normally — it is safe
Oxygen flowing	Move the oxygen mask/tubing at least 1 m away before the shock (fire risk)

- The most common rhythm in a witnessed adult cardiac arrest is **ventricular fibrillation**, and the only effective treatment for it is **defibrillation**.
- **Non-shockable rhythms** — asystole and pulseless electrical activity (PEA); these need CPR and drugs, not a shock.
- A **precordial thump** is no longer recommended in first aid.

4.10 The Recovery Position

Used for an **unconscious casualty who is breathing normally** and has no injury preventing it. It keeps the airway open by letting the tongue fall forward and allows vomit, blood and saliva to drain out of the mouth.

1. Kneel beside the casualty; remove spectacles and bulky objects from the pockets.
2. Straighten the legs; place the **arm nearer to you at right angles to the body, elbow bent, palm upwards**.
3. Bring the **far arm across the chest** and hold the back of the hand against the near cheek.
4. With the other hand, **grasp the far leg above the knee and pull it up**, keeping the foot on the ground.
5. Pull on the far leg and **roll the casualty towards you** onto the side.
6. Adjust the upper leg so that the **hip and knee are bent at right angles**.
7. **Tilt the head back** gently to keep the airway open, and make sure the **mouth points slightly downwards**.
8. Cover with a blanket, **monitor breathing continuously**, and **change sides every 30 minutes** to prevent pressure injury.

POSITION OF THE CASUALTY — A FAVOURITE EXAM TABLE

- **Unconscious but breathing** → recovery (lateral) position.
- **Cardiac arrest** → flat on the **back on a hard surface**.
- **Shock** → lying flat, **legs raised 20–30 cm**, head low and turned to one side.
- **Fainting** → lie down and raise the legs; or sit with the head between the knees.
- **Breathlessness, heart attack, asthma, chest injury** → **sitting up / half-sitting (Fowler's)**, supported, leaning slightly forward.
- **Head injury / suspected spinal injury** → **do not move**; keep flat and support the head in line with the body.
- **Abdominal injury / after abdominal pain** → lying on the back with **knees drawn up** and supported.
- **Snake bite / bleeding limb** → keep the affected part **below heart level** for a snake bite, and **raised** for bleeding.
- **Pregnant casualty** → **left lateral position**.

4.11 Choking (Foreign-Body Airway Obstruction)

Choking is the mechanical obstruction of the airway by a foreign body — the commonest cause of accidental death in small children (sweets, peanuts, coins, balloons, small toys) and a frequent cause of adult death at meals (meat, bones, dentures) and with alcohol.

Mild (partial) obstruction	Severe (complete) obstruction
Casualty can speak, cough and breathe	Cannot speak, cough, cry or breathe
Coughing is effective and forceful	Cough is silent or absent; wheezy, gasping attempts at breathing
Casualty is distressed but not blue	Clutches the throat (universal sign of choking), face becomes red then blue-grey (cyanosed) , eyes bulging, then loses consciousness
Treatment: encourage him to cough ; do nothing else; stay with him and watch	Treatment: 5 back blows → 5 abdominal thrusts (Heimlich) → repeat; if unconscious, start CPR

Treatment of severe choking — conscious adult or child (over 1 yr)

1. Stand at the side and slightly behind; support the chest with one hand and **lean the casualty well forward**.
2. Give up to **5 sharp back blows** between the shoulder blades with the heel of your hand. Check after each blow.
3. If unsuccessful, give up to **5 abdominal thrusts (Heimlich manoeuvre)**: stand behind, put both arms round the upper abdomen, make a fist and place the thumb side **midway between the navel and the lower end of the breast bone**, grasp it with the other hand and pull **sharply inwards and upwards**.

4. Continue alternating **5 back blows and 5 abdominal thrusts**; recheck the mouth after each cycle and remove any visible object with the fingers (**never blind finger-sweeps**).
5. If the casualty becomes **unconscious**, lower him carefully to the ground, call 112/108 and **begin CPR starting with chest compressions** — compressions may expel the object.
6. **Every casualty who has received abdominal thrusts must be seen by a doctor**, because of the risk of internal injury.

Special group	Modified technique
Infant under 1 year	NO abdominal thrusts. Lay the baby face down along your forearm with the head low, supporting the jaw — 5 back blows between the shoulder blades — turn face up — 5 chest thrusts with two fingers on the lower sternum. Repeat; call for help; start CPR if unresponsive.
Pregnant woman or very obese adult	Use chest thrusts (hands on the lower half of the sternum) instead of abdominal thrusts.
Casualty in a wheelchair or unable to stand	Apply abdominal thrusts from behind the chair, or lay him down and use the heel of the hand on the upper abdomen.
Self-treatment (alone)	Give yourself abdominal thrusts with your own fist, or press the upper abdomen sharply against a hard edge — a chair back, table or railing .
Unconscious casualty	Chest compressions (CPR) generate higher airway pressure than abdominal thrusts — start CPR .

REMEMBER

- The **Heimlich manoeuvre = abdominal thrusts**, described by **Dr. Henry Heimlich (1974)**.
- **Never** slap the back of a person who is coughing effectively while sitting upright, and never give abdominal thrusts to an **infant**.
- **Never do a blind finger sweep** — it may push the object deeper.
- A choking child should never be held upside down and shaken.

4.12 CPR in Special Situations

Situation	Modification
Drowning	The arrest is asphyxial : give 5 rescue breaths/2 minutes of CPR before calling for help if alone; rescue breathing may be started in shallow water; do not try to drain water from the lungs; expect vomiting.
Electrocution / lightning	Switch off the current first ; do not touch the casualty until the supply is isolated. Cardiac arrest is common — CPR and early AED; treat burns at entry and exit points.
Trauma	Control catastrophic bleeding first, protect the cervical spine with manual in-line support, use jaw thrust to open the airway.
Pregnancy (after 20 weeks)	Compress slightly higher on the sternum ; have someone displace the uterus to the left manually; transport urgently — saving the mother saves the baby.
Hypothermia / cold water drowning	Handle gently; continue CPR for much longer and warm the casualty — 'not dead until warm and dead'.
Poisoning / overdose	Avoid mouth-to-mouth contact if a corrosive or organophosphate poison is on the lips — use a mask or compression-only CPR .
Choking-related arrest	Start with compressions; check the mouth for the dislodged object after each cycle.
Suspected spinal injury	Open the airway with a jaw thrust and keep the head in line; log-roll with helpers if the casualty must be turned.
COVID-19 / infectious risk	Place a cloth over the mouth and nose, use compression-only CPR with a mask on the rescuer, and use the AED normally.

4.13 Signs of Effective CPR, Complications and After-Care

Signs that CPR is effective	Complications of CPR
Chest visibly rises with each rescue breath.	Fracture of ribs or sternum (commonest, especially in the elderly)
A carotid/femoral pulse can be felt with each compression.	Vomiting and aspiration of stomach contents
Skin colour improves; cyanosis lessens.	Injury to the liver, spleen or stomach from thrusts placed too low (over the xiphoid)
Pupils begin to constrict and react to light.	Pneumothorax , lung contusion, fat embolism

Signs that CPR is effective	Complications of CPR
Spontaneous breathing, coughing, movement or a return of consciousness = ROSC (return of spontaneous circulation) .	Gastric distension from over-ventilation; bruising of the chest wall

- **Complications must never stop you from giving CPR** — ‘a broken rib is better than a dead casualty’.
- After ROSC: place in the **recovery position**, keep warm, give oxygen if trained, monitor breathing and pulse continuously, keep nil by mouth and arrange **urgent hospital transfer** — arrest may recur.
- Record the **time of arrest, the time CPR was begun, the number of shocks and the time of ROSC** for the doctor.
- Rescuers should be offered emotional support afterwards — resuscitation is stressful even when successful.

4.14 Chapter Recap — One-Line Revision

- CPR sequence is **C-A-B**; start on any unresponsive casualty who is **not breathing normally**.
- Adult: **rate 100–120/min, depth 5–6 cm, ratio 30 : 2**, full recoil, pauses **< 10 s**, change rescuer every **2 minutes**.
- Child: depth **1/3 of chest (≈ 5 cm)**; Infant: **two fingers, 4 cm, 5 breaths first**, brachial pulse; two professional rescuers use **15 : 2**.
- Hand position = **lower half of the sternum**; infant = **just below the nipple line**.
- **AED pads**: below the right collar bone and on the left mid-axillary line; resume compressions immediately after the shock.
- **Choking**: conscious — **5 back blows + 5 abdominal thrusts**; infant — **back blows + chest thrusts, never abdominal**; unconscious — **CPR**.
- **Recovery position** for the unconscious casualty who is breathing; change sides every 30 minutes.
- Brain damage in **3–4 minutes**; survival falls **7–10 % per minute**; defibrillation is the definitive treatment of **ventricular fibrillation**.
- Commonest complication of CPR = **fractured ribs**; commonest fatal error = **not starting CPR at all**.

PART III • INJURIES, BLEEDING AND SHOCK

CHAPTER 5

Wounds

Classification and types of wounds, signs, general and special management, wound healing and its complications, tetanus, minor wounds, and burns and scalds.

SYLLABUS COVERED IN THIS CHAPTER

- Wounds — definition, classification, types (incised, lacerated, contused, abrasion, punctured, gunshot, penetrating, avulsion, amputation, crush), signs and symptoms and first-aid management of each.
- Wound healing, infection, tetanus and gas gangrene; special wounds of the scalp, eye, chest and abdomen; minor wounds; burns and scalds.

5.1 Definition and Classification

DEFINITION

- A **wound** is a **break in the continuity of the skin, mucous membrane or the surface of any tissue of the body**, through which blood escapes and germs may enter.
- Every wound therefore carries **three dangers: (1) bleeding, (2) infection and (3) shock** — together with possible damage to deeper structures (nerves, tendons, vessels, bones, organs).

Basis of classification	Types
Skin intact or broken	Open wound (skin broken — cut, abrasion, puncture) and Closed wound (skin unbroken — bruise, contusion, internal injury)
Cause	Mechanical (cut, blow, crush), thermal (burn, scald, frostbite), chemical (acid, alkali), electrical, radiational, bite
Depth / structures involved	Superficial (skin only), deep (muscle, vessel, nerve), penetrating (enters a cavity), perforating (enters and comes out)
Contamination	Clean (surgical), contaminated (dirty, soil, rust), infected (pus present)
Bleeding	External (visible) and internal (concealed)
Healing	Healing by first intention (clean edges, stitched) and by second intention (gaping/infected, heals with a scar)

5.2 Types of Wounds — Complete Table

Type of wound	How caused	Features	Special danger
Incised (clean cut)	Sharp edge — knife, razor, glass, blade	Clean, straight edges ; bleeds freely and profusely because vessels are cut straight across	Severe bleeding ; deep cut may divide tendons and nerves
Lacerated (tear)	Blunt force, machinery, barbed wire, animal claws, fall on rough ground	Ragged, irregular, torn edges ; bleeds less than an incised wound	Greater risk of infection ; more tissue destruction
Abrasion (graze)	Rubbing or sliding — fall on a road, friction burn	Only the superficial skin is scraped off ; raw, painful, oozing surface with embedded dirt	Infection and tattooing by grit; very painful
Contusion (bruise) — a closed wound	Blunt blow, fall, crush	Skin unbroken ; bleeding into the tissues — swelling, discoloration (red — purple — blue — green — yellow), pain, tenderness	May conceal a fracture or internal injury
Punctured (stab)	Nail, needle, spike, knife, bayonet, fork, thorn, injection	Small external opening but deep track ; little external bleeding	Deep infection, especially TETANUS and gas gangrene ; may injure internal organs — internal bleeding
Gunshot (bullet/firearm)	Bullet, pellet, blast fragment	Small entry wound, larger ragged exit wound (entry may show burning/blackening); may be blind (no exit)	Massive internal damage, foreign bodies, infection ; medico-legal case — inform police
Penetrating / perforating	Sharp or high-velocity object	Penetrating — object enters a body cavity (chest, abdomen, skull); perforating — enters and leaves	Injury to the heart, lung, gut, liver — internal haemorrhage and shock
Avulsion / degloving	Machinery, road accident, animal bite	A flap of skin or tissue is torn away or peeled off, partly or completely	Heavy bleeding; tissue may die — preserve the flap

Type of wound	How caused	Features	Special danger
Amputation (traumatic)	Machine, railway, road accident, blast	A part (finger, hand, limb, ear) is completely severed	Severe bleeding and shock; the part may be re-implanted if properly preserved
Crush injury	Heavy weight, building collapse, vehicle wheel	Extensive damage of muscle with swelling; skin may look almost normal	Crush syndrome — release of muscle toxins (myoglobin, potassium) — kidney failure and cardiac arrest
Bite wound (animal/human)	Dog, cat, monkey, snake, human teeth	Punctured + lacerated wound with saliva contamination	Rabies , tetanus, severe infection; human bites are the most heavily infected of all
Burn and scald wound	Dry heat/flame (burn), moist heat/steam (scald), chemicals, electricity, radiation	Redness, blisters, charring; loss of skin	Fluid loss — shock , infection, airway burn

MOST-ASKED DISCRIMINATORS

- Wound with **clean, straight edges that bleeds most profusely = incised**.
- Wound with **ragged torn edges and greater risk of infection = lacerated**.
- Wound with a **small opening but deep track — highest risk of tetanus = punctured**.
- **Closed** wound = **contusion (bruise)**.
- **Superficial** wound with only skin scraped off = **abrasion**.
- Wound in which **entry is small and exit large = gunshot**.

5.3 Signs and Symptoms of a Wound

Symptoms (told by the casualty)	Signs (seen by the first aider)
Pain at the site, often throbbing	Bleeding — arterial, venous or capillary
Faintness, giddiness, thirst	Visible break in the skin , gaping of the edges
Feeling of cold and weakness	Swelling, bruising, discoloration
Nausea and restlessness	Loss of function of the part; deformity if a bone is broken
Anxiety and fear	Signs of shock — pale, cold clammy skin, rapid weak pulse, rapid shallow breathing
—	Foreign bodies, dirt, glass in the wound; escape of tissue fluid or organs

5.4 General First-Aid Management of Wounds

1. **Reassure** the casualty and lay him down (this alone prevents fainting).
2. **Wash your hands** and wear gloves; do not touch the wound.
3. **Expose the wound** — remove or cut clothing as necessary, without dragging.
4. **Control the bleeding** — direct pressure over a dressing, **elevate** the part, and rest. (See Chapter 6.)
5. **Do not remove** a blood clot or a foreign body that is deeply embedded; do not probe the wound.
6. **Clean a minor wound** with clean running water or normal saline, washing **from the centre outwards**; dry with a sterile swab and apply an antiseptic (povidone-iodine) around — **not into** — the wound.
7. **Cover with a sterile dressing** larger than the wound and bandage firmly but not tightly.
8. **Immobilise and support** the injured part (a sling for the arm, splint for the leg); keep the part **raised** if possible.
9. **Treat for shock** — keep the casualty warm, flat, legs raised, and give **nothing by mouth** (moisten the lips if thirsty).
10. **Seek medical help** and ensure **anti-tetanus prophylaxis**.
11. Record the time, the appearance of the wound and the treatment given.

WOUNDS THAT MUST ALWAYS BE SEEN BY A DOCTOR

- Bleeding that cannot be controlled, or **spurting** bleeding.
- A wound that **gapes** and needs stitching, or is longer than 2–3 cm.
- **Deep, punctured, crushed or bite** wounds; any dirty wound contaminated with soil, dung or rust.
- Wounds with an **embedded foreign body**, glass or grit that cannot be flushed out.
- Wounds over a **joint, on the face, palm, sole or genitals**, or with loss of feeling/movement (nerve or tendon).

injury).

- Any wound in a **diabetic, an elderly person, or anyone on blood-thinning medicine**, and any wound showing **signs of infection**.
- Wounds where **tetanus immunisation is not up to date**.

5.5 Wound Healing and Its Complications

Phases of healing

Phase	Time	What happens
1. Haemostasis	Seconds to minutes	Vessels constrict; platelets plug; fibrin clot forms
2. Inflammatory	0–3 days	Redness, heat, swelling, pain; white cells clear bacteria and debris
3. Proliferative	3 days–3 weeks	New capillaries, granulation tissue, collagen; edges contract; epithelium grows over
4. Maturation (remodelling)	3 weeks–2 years	Scar forms, shrinks and pales; regains up to 80 % of original strength

Factors that promote healing	Factors that delay healing
Clean wound with apposed edges; good blood supply	Infection , foreign body, dead tissue (slough)
Rest and immobilisation of the part	Movement, repeated trauma, poor blood supply
Good nutrition — protein, vitamin C, vitamin A, zinc, iron	Malnutrition, anaemia, deficiency of vitamin C (scurvy)
Youth and general good health	Diabetes mellitus , obesity, old age, cancer, kidney/liver disease
Proper dressing and moist wound environment	Smoking , steroids, chemotherapy, radiotherapy; oedema

Complications of wounds

Complication	Points to remember
Haemorrhage	Primary (at the time of injury), reactionary (within 24 hours, as BP rises) and secondary (after 7–10 days, due to infection)
Shock	From blood and fluid loss or from pain
Infection	Signs: increasing pain, redness, heat, swelling, pus, foul smell, red streaks up the limb (lymphangitis), tender swollen glands, fever and malaise
Tetanus (lock-jaw)	See below — the most feared complication of a dirty punctured wound
Gas gangrene	<i>Clostridium perfringens</i> in deep crushed, dirty wounds — foul-smelling gas in tissues, crackling under the skin, rapidly fatal
Septicaemia / toxæmia	Spread of germs or toxins into the blood — high fever, rigors, collapse
Gangrene	Death of tissue from loss of blood supply (too-tight bandage, tourniquet left on, crush, diabetes)
Scar, keloid and contracture	Excessive scar tissue; contracture across a joint limits movement
Damage to deeper structures	Nerve (numbness, paralysis), tendon (loss of movement), vessel, bone (compound fracture), organ

Tetanus — must-know facts

Point	Detail
Organism	<i>Clostridium tetani</i> — an anaerobic, spore-forming bacillus found in soil, dust, dung and rusty metal
Entry	Deep punctured wounds, dirty lacerations, burns, animal bites, the umbilical stump of a newborn (tetanus neonatorum)
Incubation period	7–10 days (range 3 days – 3 weeks); the shorter the period, the worse the prognosis
Toxin	Tetanospasmin — acts on the nervous system
Features	Lock-jaw (trismus) , stiff neck, risus sardonicus (fixed grin), painful generalised muscle spasms, arching of the back (opisthotonos), spasm of respiratory muscles → death; spasms provoked by light, noise or touch
First-aid prevention	Thorough cleaning of every wound, removal of dirt and dead tissue, leaving deep wounds open (not sealed) , and prompt medical referral
Immunisation	Tetanus toxoid (TT) — active immunity, part of DPT in infancy with boosters; a booster is given if the last dose was more than 5 years ago (10 years for a clean minor wound). ATS / tetanus immunoglobulin gives immediate passive immunity in an unimmunised casualty

5.6 Special Wounds and Their Management

Wound with an embedded foreign body

- **Do not remove** a large or deeply embedded object — it may be plugging a bleeding vessel.
- Control bleeding by pressing **on either side** of the object, not over it.
- Build up a **ring pad** (or two rolled bandages) around the object, higher than the object, and bandage over the pads so that no pressure falls on the object itself.
- If the object is very long, support it; do not let it move. Arrange urgent hospital transfer and note the time.
- Small surface splinters and grit may be picked out with sterilised tweezers in the direction of entry.

Scalp wounds

- Bleed **profusely** because the scalp is very vascular and its vessels cannot retract; the wound edges gape.
- Apply a **sterile pad with firm direct pressure** and bandage; do not press hard if you suspect an underlying **depressed skull fracture** — use a ring pad instead.
- Look for signs of **head injury** (unconsciousness, vomiting, bleeding/CSF from ear or nose, unequal pupils) — all such casualties go to hospital.

Eye wounds

- Lay the casualty on his **back**, keep the head still, and tell him not to move the eyes.
- Cover with a **sterile eye pad or clean dressing**; do not apply pressure and do not try to remove an embedded object.
- **Never** wash out a penetrating eye injury, never remove a protruding object, never allow rubbing, and never apply ointment.
- Cover **both** eyes if possible (movement of one eye moves the other), and transfer lying down to an eye hospital.
- For a loose particle: flush with clean water from the inner to the outer corner, or lift it off with a moist swab; for chemical splash, **irrigate with running water for at least 20 minutes**.

Chest wounds (penetrating / 'sucking' wound)

- A penetrating chest wound may **suck air into the chest cavity** with each breath (open pneumothorax), collapsing the lung.
- Signs: **bubbling/hissing/sucking sound at the wound, frothy blood, extreme breathlessness, blue lips, coughing of bright red frothy blood**, and later a rapid weak pulse with distended neck veins.
- Treatment: **cover the wound immediately with the palm of your gloved hand**, then apply a **sterile pad covered with plastic/cling film, taped on three sides only** so that air can escape but not enter ('flutter valve').
- Place the casualty in a **half-sitting position, leaning towards the injured side**, support the arm in a sling, treat for shock and call an ambulance urgently. Do **not** give anything by mouth.
- If breathing worsens after sealing (tension pneumothorax), **release the dressing** at once.

Abdominal wounds with protruding intestine (evisceration)

- Lay the casualty **on his back with knees bent and supported** to relax the abdominal wall.
- **Do NOT try to push the organs back and do not touch them.**
- Cover the protruding organs with a **large sterile dressing moistened with clean water/saline, then a plastic sheet or cling film**, and fix loosely — the object is to keep them **moist and clean**.
- **Nothing by mouth** — not even water (surgery will be needed); moisten the lips only.
- If the casualty coughs or vomits, support the abdomen with your hands over the dressing.
- Treat for shock and arrange **immediate** hospital transport.

Amputation and avulsion — care of the severed part

1. Control bleeding of the stump with **direct pressure and elevation**; cover with a bulky sterile dressing and bandage firmly. A tourniquet is used **only** if bleeding is life-threatening and uncontrollable.
2. **Preserve the amputated part** — do **not** wash it with antiseptic or water and do **not** put it directly on ice.
3. Wrap the part in a **clean, dry (or lightly moistened saline) gauze**, place it in a **clean plastic bag**, seal it, and put that bag in **another container of ice and water (cold, about 4 °C)**.
4. Label with the casualty's name and the time of injury and send it **with the casualty** to hospital.
5. Re-implantation is possible for up to about **6 hours (12–18 hours if well cooled)**; for the avulsed flap, replace it in position and cover.

Crush injury

- Release the casualty as quickly as possible if trapped for **less than 15 minutes**; if trapped longer, **do not release without medical help** if it can be avoided — sudden release floods the blood with toxins (**crush syndrome**) and can cause cardiac arrest and kidney failure.
- Support and immobilise the part, treat bleeding and shock, keep the casualty warm and note the **exact time of the crush** and of release.
- Watch for swelling and loss of pulse distally (**compartment syndrome**) — remove tight clothing, jewellery and bandages.

Gunshot and blast wounds

- Treat as a **penetrating wound**: control bleeding, cover entry **and exit** wounds, immobilise, treat shock, urgent transport.
- Do not probe for the bullet; do not remove clothing unnecessarily — **preserve evidence** and **inform the police** (medico-legal case).
- Blast injuries may cause hidden **lung, ear and abdominal** damage even without external wounds.

5.7 Minor Wounds — Practical First Aid

Condition	Management
Small cut / abrasion	Wash hands, clean with running water or saline, dry, apply antiseptic and an adhesive dressing; check tetanus status
Blister (friction)	Do not prick an intact blister; cover with a soft padded dressing. If burst, clean, do not remove the loose skin and cover
Splinter	Clean the area, sterilise tweezers, grasp the splinter close to the skin and pull out along the line of entry ; if it is deep, broken off or under a nail — leave it and refer
Fish hook	If the barb is exposed, cut it off and back the hook out; if not, leave it in , pad round it and refer
Torn or crushed nail	Cover with a dressing, support the finger and refer; a blood blister under a nail (subungual haematoma) needs medical release
Bruise / contusion	RICE — Rest, I ce (cold compress for 10–20 minutes wrapped in cloth), C ompression with a crepe bandage, E levation. Never massage a fresh bruise
Graze with embedded grit	Flush with plenty of clean water; do not scrub; cover with a non-adherent dressing

5.8 Burns and Scalds

A **burn** is an injury caused by **dry heat** (flame, hot metal, sun, friction, electricity, radiation); a **scald** is caused by **moist heat** (boiling water, steam, hot oil, tea). The first-aid treatment of both is the same.

Classification by depth

Degree	Old name	Layers involved	Appearance and features
First degree	Superficial	Epidermis only	Red, dry, painful , no blister; heals in 3–7 days without a scar (e.g. sunburn)
Second degree	Partial thickness	Epidermis + part of dermis	Blisters, moist, red, very painful ; heals in 2–3 weeks, may scar
Third degree	Full thickness	Whole skin (± fat)	White, waxy, brown or charred, dry and leathery ; PAINLESS (nerve endings destroyed); needs grafting
Fourth degree	—	Muscle, tendon, bone	Charred, blackened; often after electrical burns — amputation may be needed

Extent — the Rule of Nines (adult)

Body part	% of body surface
Head and neck	9 %
Each arm (front + back)	9 % each (total 18 %)
Front of trunk (chest + abdomen)	18 %
Back of trunk	18 %
Each leg	18 % each (total 36 %)
Perineum / genitals	1 %
Palm of the casualty's own hand	1 % — useful for small, patchy burns

Body part	% of body surface
Child modification	Head 18 % , each leg 14 % (a child's head is relatively larger)

WHEN IS A BURN 'SERIOUS'? (SEND TO HOSPITAL)

- Any burn over **more than 10 % of the body in a child or 15–20 % in an adult** (some texts: >9 % child, >18 % adult).
- **All third-degree (full-thickness) burns**, however small.
- Burns of the **face, neck, mouth or throat** (airway risk), **hands, feet, genitals** or across a **joint**.
- **Circumferential** burns (right round a limb or the chest).
- **Electrical, chemical, inhalation and blast** burns.
- Burns in **infants, the elderly, pregnant women** and in casualties with diabetes, heart or lung disease.
- Any burn with **shock**, or with suspected non-accidental injury.

First aid for burns and scalds — step by step

1. **Stop the burning process** — remove the casualty from danger or the danger from the casualty. Smother flames with a blanket; make the casualty **STOP, DROP, WRAP and ROLL** on the ground; switch off the current in an electrical burn.
2. **Cool the burn with cool (not ice-cold) running water for at least 10–20 minutes** — this is the single most useful measure and is worthwhile up to 3 hours after the injury. Use clean water; never use ice, butter, oil, toothpaste, mud, ink, turmeric, cow dung or leaves.
3. **Remove rings, watches, bangles, belts and tight clothing** from the area before it swells, but **never remove clothing stuck to the burn**.
4. **Cover the burn with a sterile non-adherent dressing, clean plastic (cling film) or a clean cotton sheet** — lengthwise, not wound round the limb. For a burnt hand, a clean plastic bag may be used.
5. **Do not break blisters**, do not apply any ointment, powder or lotion, and do not use cotton wool or fluffy dressings.
6. **Treat for shock** — lay the casualty down, raise the legs, keep him warm (burns lose heat), reassure, and **give frequent small sips of cool water or ORS** if he is fully conscious and the burn is not going to need immediate surgery.
7. **Monitor the airway** — hoarse voice, cough, soot round the mouth or nose, singed nasal hair and breathing difficulty mean **inhalation injury**: give oxygen if trained and transport at once, sitting up.
8. Arrange **urgent hospital transfer** for all serious burns; keep the burnt part **elevated** where possible and record the time, cause, extent and depth.

Special burn	Additional first aid
Chemical burn	Wear gloves; brush off dry powder first , then flood with copious running water for at least 20 minutes ; remove contaminated clothing while flooding; do not attempt to neutralise acid with alkali or vice-versa; identify the chemical for the hospital
Chemical in the eye	Irrigate the eye with gently running water for 20 minutes, holding the lids open, with the affected eye lower so that the chemical does not run into the other eye; cover and refer
Electrical burn	Isolate the current first . Look for entry and exit wounds (both must be dressed), expect deep tissue damage, fractures and cardiac arrest ; all electrical burns need hospital assessment
Inhalation / airway burn	Move to fresh air, sit the casualty up, loosen clothing, give oxygen if trained, watch for swelling of the throat; transport urgently — the airway can close within minutes
Sunburn	Move into the shade, cool the skin with water or cool compresses, give sips of water, apply calamine or after-sun lotion; do not break blisters
Friction burn	Treat as an abrasion plus a burn — cool and cover
Radiation burn	Remove contaminated clothing, wash the skin, cover and refer; protect yourself

BURNS — THE CLASSICAL 'NEVER' LIST

- **Never** apply ice or ice-cold water directly (causes further tissue damage).
- **Never** apply oil, ghee, butter, toothpaste, turmeric, ink, mud, cow dung, egg white, leaves or any home remedy.
- **Never** prick or break blisters, and never peel off dead skin.
- **Never** use cotton wool, fluffy or adhesive dressings on a burn.
- **Never** remove clothing that is stuck to the burn.
- **Never** give alcohol, and never give anything by mouth to an unconscious or severely burnt casualty awaiting surgery.

- **Never** cough or breathe over a burn — burnt skin has lost its barrier to infection.

5.9 Chapter Recap — One-Line Revision

- Wound = break in the continuity of skin/mucous membrane; three dangers = **bleeding, infection, shock**.
- **Incised** → bleeds most; **lacerated** → most infected edges; **punctured** → tetanus risk; **contusion** → closed wound; **abrasion** → superficial only.
- Treatment order: **stop bleeding** → **prevent infection** → **immobilise** → **treat shock** → **refer**.
- Tetanus: *Clostridium tetani*, anaerobic spore-former, incubation **7–10 days**, toxin **tetanospasmin**, signs **trismus**, **risus sardonicus**, **opisthotonos**.
- **Sucking chest wound** → seal with plastic **taped on three sides**, half-sitting, leaning to the injured side.
- **Protruding intestine** → knees bent, cover with moist sterile dressing, **never push back**, nothing by mouth.
- **Amputated part** → dry sterile wrap → sealed plastic bag → bag in ice water; **never directly on ice**.
- Burns: cool with running water for **10–20 minutes**, cover with cling film, **never** apply oil/ice/toothpaste, never prick blisters.
- **Rule of nines**: head 9, each arm 9, each leg 18, front trunk 18, back 18, perineum 1; palm = 1 %; in a child, head 18 and each leg 14.
- Third-degree burn is **painless, white or charred and dry** — always needs hospital care.

CHAPTER 6

Haemorrhage (Bleeding)

Types of bleeding, effects of blood loss, natural arrest of haemorrhage, control of external bleeding, pressure points, tourniquets, internal bleeding and bleeding from special sites.

SYLLABUS COVERED IN THIS CHAPTER

- Haemorrhage — definition, classification (arterial, venous, capillary; external, internal, concealed; primary, reactionary, secondary), signs and symptoms, effects of blood loss.
- Control of bleeding — direct and indirect pressure, pressure points, pressure bandage, tourniquet; internal haemorrhage; bleeding from the nose, ear, mouth, lung, stomach, varicose veins and tooth socket.

6.1 Definition and Classification

DEFINITION

- **Haemorrhage** means the **escape of blood from a ruptured or cut blood vessel** (*haima* = blood, *rhegnumi* = to burst forth). **Severe bleeding is second only to airway obstruction as a cause of preventable death** and must be controlled immediately.
- The average adult has **5–6 litres** of blood (about 8 % of body weight; 70–80 ml/kg). A **child** has about 2–2.5 litres and an **infant** only about **250–350 ml** — so even a small loss is dangerous in a baby.

Basis	Types
Source (type of vessel)	Arterial, venous and capillary bleeding
Site	External (visible, from a wound) and internal (concealed — into a body cavity or tissues). Internal bleeding may become revealed when blood appears at a natural opening (nose, mouth, ear, urine, stool, vagina)
Time of occurrence	Primary — at the moment of injury; Reactionary (intermediate) — within 24 hours , when blood pressure rises, a clot is dislodged or a ligature slips; Secondary — after 7–14 days , due to infection eroding a vessel
Amount	Mild, moderate and severe (life-threatening/catastrophic)

Arterial, venous and capillary bleeding — the master comparison

Feature	Arterial	Venous	Capillary
Source	Cut artery — carries blood from the heart	Cut vein — carries blood towards the heart	Ruptured capillaries
Colour of blood	Bright red (scarlet) — oxygenated	Dark red (bluish/purple) — deoxygenated	Bright to dark red, mixed
Flow	Spurts in jets, synchronised with each heart beat ; pulsating	Steady, copious, welling-up flow ; may gush but does not spurt	Slow oozing from the whole raw surface
Pressure	High — difficult to control	Low — easier to control	Very low
Danger	Greatest — rapid, massive loss; death in minutes from a large artery	Serious if a large vein; air may be sucked into a neck vein (air embolism)	Usually slight, but large grazes can lose much fluid; risk of infection
Example	Wound of the femoral, brachial or carotid artery	Bleeding varicose vein, cut on the back of the hand	Graze, abrasion, shallow cut
Control	Direct pressure , elevation, pressure point; tourniquet only as a last resort	Direct pressure and elevation (very effective)	Direct pressure and a simple dressing

6.2 Effects of Blood Loss

Blood lost	Approximate volume (adult)	Effects
Up to 500 ml (10 %)	Half a litre — the amount given by a blood donor	Usually no symptoms; quickly replaced by the body
500–1,000 ml (10–20 %)	Up to 1 litre	Slight rise in pulse rate, pallor, mild thirst and giddiness; compensated — blood pressure still normal
1,000–1,500 ml (20–30 %)	1–1.5 litres	Definite shock — pulse rapid and weak, cold clammy skin, thirst, restlessness, rapid breathing, falling blood pressure, reduced urine

Blood lost	Approximate volume (adult)	Effects
1,500–2,000 ml (30–40 %)	1.5–2 litres	Severe shock — marked pallor, air hunger, confusion, systolic BP below 90, feeble pulse, cold extremities
More than 2,000 ml (> 40 %)	Over 2 litres	Profound, often fatal shock — unconsciousness, absent peripheral pulses, cardiac arrest

- A loss of about **one-third of the blood volume may be fatal** if it is rapid and untreated.
- **Rapid** loss is far more dangerous than the same loss occurring slowly, because there is no time for compensation.
- Children, the elderly and thin persons tolerate blood loss poorly.
- The body compensates by **constricting skin vessels (pallor)**, **increasing the heart rate**, **increasing breathing and shifting fluid from the tissues** — these are exactly the signs of shock.

Signs and symptoms of severe bleeding

Symptoms (complained of)	Signs (observed)
Faintness and giddiness	Visible bleeding, blood-soaked clothing
Thirst (a very important symptom)	Pale, cold, clammy (sweating) skin ; blue lips
Nausea, sometimes vomiting	Rapid, weak, thready pulse (later imperceptible)
Restlessness, anxiety, 'feeling of impending doom'	Rapid, shallow, sighing breathing (air hunger) , yawning
Dimness of vision, noises in the ears	Dilated pupils; sunken eyes with dark rings
Feeling cold, weakness	Falling blood pressure, reduced urine output, restlessness → confusion → unconsciousness

6.3 How the Body Arrests Bleeding Naturally

1. **Retraction and contraction** of the cut vessel ends (an artery retracts into its sheath and its muscular wall constricts).
2. **Fall in blood pressure** as blood is lost, which slows the flow.
3. **Platelets adhere** to the damaged wall and form a plug.
4. **Clotting (coagulation)** — prothrombin → thrombin, then **fibrinogen** → **fibrin**, forming a mesh that traps blood cells; normal clotting time **3–8 minutes**.
5. Clot retraction and, finally, organisation of the clot by fibrous tissue.

Clotting needs vitamin K, calcium, platelets and clotting factors. It is delayed in haemophilia, in liver disease, in patients on **aspirin, heparin or warfarin**, and when a wound is repeatedly disturbed — which is why a dressing once applied should **not** be removed.

6.4 Control of External Bleeding — the Practical Sequence

MNEMONIC: 'S-E-R' / 'R-E-D' — THE THREE PILLARS

- **Rest** — lay the casualty **down** with the head low; rest reduces the pulse rate and the force of bleeding.
- **Elevation** — **raise the bleeding part above the level of the heart** (unless a fracture is suspected) to reduce the blood flow into it.
- **Direct pressure** — press **firmly on the wound** over a sterile dressing, with the fingers or palm, for **at least 10 minutes without releasing**.

1. **Lay the casualty down** and reassure him; call for help (112/108).
2. **Expose the wound** and look for embedded foreign bodies.
3. **Apply direct pressure** with a sterile dressing (or your gloved hand, or the casualty's own hand) for **10 minutes continuously**.
4. **Elevate** the injured part and support it.
5. Apply a **pressure (firm) dressing**: pad + bandage, tight enough to stop bleeding but not to cut off circulation; **check the pulse beyond** it.
6. If blood soaks through, **do not remove the dressing** — add another pad on top and bandage more firmly. If it soaks through a second time, remove them and re-apply accurate direct pressure over the bleeding point.
7. **Indirect pressure** at a **pressure point** may be used for a short time (**not more than 10–15 minutes**) when direct pressure fails on a limb.
8. **Immobilise** the part (sling or splint) — movement restarts bleeding.

9. **Treat for shock:** keep flat, legs raised, warm (cover with a blanket), loosen tight clothing, **nothing by mouth**, and reassure.
10. **Monitor** breathing, pulse and level of response every 10 minutes and record; be ready to give CPR.
11. For **catastrophic limb bleeding that cannot be controlled**, apply **wound packing with a haemostatic dressing** or a **tourniquet** (see below), and get the casualty to hospital urgently.

Pressure points (indirect digital pressure)

A pressure point is a place where a **main artery runs close to the skin over a bone**, so that it can be pressed against the bone to stop the flow beyond. Pressure is applied with the **thumb or fingers** and cannot be maintained for long.

Artery	Where to press	Controls bleeding from
Temporal	In front of the ear, against the skull	Scalp and temple
Facial	On the lower border of the jaw, about 3 cm in front of its angle	Face, cheek and lip
Carotid	Side of the neck, beside the windpipe, pressing backwards — never press both sides and never press longer than necessary	Severe bleeding from the neck (last resort)
Subclavian	Behind the middle of the collar bone, pressing downwards and backwards against the first rib	Shoulder and upper arm
Brachial	On the inner side of the upper arm , in the groove between the biceps and triceps, pressing the artery against the humerus	Forearm and hand (the classical arm pressure point)
Radial / Ulnar	At the wrist, thumb side and little-finger side	Hand and palm
Femoral	In the groin , at the mid-point of the fold, pressing firmly backwards against the pelvis with the thumbs or heel of the hand	Thigh and the whole lower limb (the most important pressure point)
Popliteal	Behind the knee	Leg below the knee
Posterior tibial / Dorsalis pedis	Behind the inner ankle / front of the ankle	Foot
Abdominal aorta	Firm pressure through the abdominal wall (medical use only)	Massive pelvic/lower-limb bleeding

The tourniquet — rules and dangers

Point	Detail
What it is	A firm band applied round a limb to compress all vessels and completely stop the blood flow beyond it
When justified	Only for life-threatening limb bleeding that cannot be stopped by direct pressure and packing; traumatic amputation ; a trapped or multiple-casualty situation where the rescuer cannot maintain pressure
Where applied	5 cm above the wound (never over a joint), on a single bone segment (upper arm or thigh); never on the forearm or lower leg if it can be avoided; never over a fracture
How applied	Wide (at least 5 cm) flat band or a commercial windlass tourniquet, tightened just enough to stop the bleeding and abolish the distal pulse; padded underneath if improvised
Documentation	Write the time of application on the casualty's forehead or on a label and tell the ambulance staff; the tourniquet must be clearly visible — never covered by a bandage or clothing
Time limit	It should not be left on for more than 1–2 hours ; do not loosen and re-tighten it periodically in first aid — releasing it restarts bleeding and can cause shock. Only a doctor should release it
Dangers	Gangrene and loss of the limb, nerve and muscle damage, crush-type toxic release on removal, severe pain, and renewed bleeding if it slips; it may also increase bleeding if too loose (it then blocks only the veins)
Improvised	A narrow-fold triangular bandage, belt, tie or cloth with a stick twisted into it (windlass) — never use wire, rope, string or a thin cord

MODERN PRACTICE POINT

- For ordinary bleeding the tourniquet is **obsolete** — direct pressure, elevation and a pressure dressing control almost all bleeding.
- But in **catastrophic haemorrhage** (blast, amputation, severe machinery injury) a correctly applied tourniquet is **life-saving**; the modern rule is '**life before limb**'.
- **Wound packing** (pushing a haemostatic or plain gauze firmly into a deep wound and holding pressure for 3 minutes) is used for junctional wounds (groin, armpit, neck) where a tourniquet cannot be applied.

6.5 Internal (Concealed) Haemorrhage

Bleeding into a body cavity or into the tissues, where it cannot be seen. It is **dangerous precisely because it is hidden** — suspect it in every serious injury and in any casualty with **unexplained shock**.

Causes	Sites
Blunt injury to the chest or abdomen (road accident, fall, blow, crush)	Chest — haemothorax; abdomen — rupture of the liver, spleen, kidney , mesenteric vessels
Fracture of large bones (femur may bleed 1–2 litres; pelvis 2–3 litres)	Thigh, pelvis, retroperitoneum
Rupture of an aneurysm, ectopic pregnancy, peptic ulcer	Abdomen, pelvis, gastro-intestinal tract
Bleeding disorders, anticoagulant drugs	Brain (intracranial haemorrhage), joints, muscles
Penetrating wounds, stab and gunshot injuries	Any cavity — skull, chest, abdomen, pelvis

Signs of internal bleeding

- **All the signs of shock without visible blood loss** — pale, cold, clammy skin; rapid weak pulse; rapid shallow breathing; thirst; restlessness; giddiness; falling level of consciousness.
- **Pain, tenderness, rigidity or guarding** of the abdomen; **distension**; ‘board-like’ abdomen.
- Pattern **bruising** (seat-belt mark, tyre mark), swelling of a limb, deformity.
- **Revealed** internal bleeding — blood at a natural opening: **haemoptysis** (coughed-up bright red frothy blood — lung), **haematemesis** (vomited blood, dark like coffee grounds — stomach), **malaena** (black tarry stool — upper gut), **haematuria** (blood in urine — kidney/bladder), bleeding from the ear or nose (**skull fracture**), vaginal bleeding (pregnancy/uterine).
- **Air hunger** — restless gasping for air — is a grave sign.

First aid for internal bleeding

1. **Lay the casualty down**, head low and turned to one side; **raise the legs** (shock position) unless there is a head or chest injury or a leg fracture.
2. **Loosen** tight clothing at the neck, chest and waist.
3. **Keep warm** with a blanket, but do **not** use hot-water bottles (they dilate skin vessels and worsen shock).
4. **Give nothing by mouth** — no water, no food, no medicine, no alcohol. Moisten the lips with a wet cloth.
5. **Do not apply any external pressure** on the abdomen or chest; do not massage.
6. **Immobilise fractures** — this greatly reduces internal blood loss.
7. **Record and monitor** the pulse, breathing and level of response every **10 minutes** (every 5 minutes if serious); note the time.
8. Arrange the **most urgent possible transport** to hospital — internal bleeding needs surgery. Preserve any vomited or passed blood for the doctor.
9. Be prepared to give **CPR**.

6.6 Bleeding from Special Sites

Site / type	Recognition	First aid
Nose (epistaxis)	Bleeding from one or both nostrils; may follow a blow, picking, sneezing, high BP or heat	Sit up, head bent slightly forward; pinch the soft part of the nose for 10 minutes continuously, breathe through the mouth; cold compress over the nose; do not blow the nose for 4 hours. Refer if bleeding lasts more than 20–30 minutes , is very heavy, or follows a head injury (may be CSF/skull fracture)
Ear	Blood or straw-coloured fluid from the ear canal	Never plug the ear. Cover lightly with a sterile pad, lay the casualty in the recovery position with the bleeding ear downwards so that it drains; refer at once (suspect fractured base of skull or a ruptured eardrum)
Mouth / tooth socket	Bleeding after extraction, cut lip or tongue	Sit up with the head forward; place a thick gauze pad on the socket and ask the casualty to bite firmly for 10–20 minutes ; do not rinse the mouth, do not suck or probe the socket, avoid hot drinks; refer if it continues
Lung (haemoptysis)	Bright red, frothy blood coughed up , breathlessness	Half-sitting position, leaning to the affected side; loosen clothing; reassure; nothing by mouth ; urgent hospital transfer; do not let the casualty talk
Stomach (haematemesis)	Vomited blood — bright red or dark ‘coffee-ground’ , often with abdominal pain and black tarry stools	Lay flat with head turned to one side, knees bent; nothing by mouth ; keep the vomit for the doctor; treat shock; urgent transfer

Site / type	Recognition	First aid
Varicose vein (leg)	Profuse, dark, welling bleeding from the leg of an elderly person	Lay the casualty down and raise the leg high (well above the heart), apply firm direct pressure with a pad and bandage; remove garters/tight clothing; refer
Palm of the hand	Deep cut with heavy bleeding; often a divided tendon	Place a pad in the palm, ask the casualty to clench the fist over the pad , bandage the fist tightly, and support in an elevation sling
Scalp	Profuse bleeding from a vascular gaping wound	Firm direct pressure with a sterile pad and bandage; use a ring pad if a depressed fracture is suspected; refer
Vaginal / obstetric	Bleeding in pregnancy, after childbirth or from injury	Lay down, knees raised, place a sanitary pad (do not pack the vagina), keep warm, nothing by mouth, urgent transfer; save all pads and clots
Amputation / catastrophic limb bleeding	Spurting blood, part severed	Direct pressure/packing; if uncontrolled apply a tourniquet , note the time, preserve the severed part, treat shock, urgent transfer
Bleeding from a stoma or under a nail	Local ooze	Gentle direct pressure with a sterile pad; refer if persistent

6.7 Blood Replacement — What the First Aider Should Know

- Lost blood is finally replaced in hospital by **intravenous fluids (normal saline, Ringer lactate) and blood transfusion** — never by anything given by mouth at the scene.
- **Blood groups:** A, B, AB, O with Rh factor. **O negative = universal donor; AB positive = universal recipient.** Cross-matching is essential; a mismatched transfusion causes a fatal reaction.
- A healthy donor of 18–65 years weighing over **45–50 kg** can give **350–450 ml** every **3 months** (men) / 4 months (women). **World Blood Donor Day — 14 June**; National Voluntary Blood Donation Day — 1 October.
- Autologous transfusion = the patient's own blood; **massive transfusion** = replacement of the whole blood volume in 24 hours.

6.8 Chapter Recap — One-Line Revision

- **Arterial** = bright red, spurting in jets; **venous** = dark red, steady flow; **capillary** = slow ooze.
- Time types: **primary** (at once), **reactionary** (within 24 h), **secondary** (7–14 days, due to infection).
- Adult blood volume **5–6 litres**; loss of **1–1.5 litres** causes definite shock; loss of **one-third** may be fatal.
- Control = **Rest + Elevation + Direct pressure** for **10 minutes**; then a pressure dressing.
- **Never remove a blood-soaked dressing** — add another pad over it.
- Chief pressure points: **brachial** (arm, inner upper arm against the humerus) and **femoral** (groin, against the pelvis).
- Tourniquet = **last resort only, 5 cm above the wound**, note the **time**, keep it **visible**, never loosen in first aid.
- Internal bleeding = **shock without visible blood**; treat with the shock position, warmth, **nothing by mouth** and urgent transfer.
- **Epistaxis** — sit up, head forward, pinch the soft part for 10 minutes. **Bleeding ear** — never plug; lie with that ear **downwards**.
- **Haemoptysis** = coughed bright frothy blood (lung); **haematemesis** = vomited coffee-ground blood (stomach); **malaena** = black tarry stool.

CHAPTER 7

Shock

Meaning and mechanism, all types of shock, stages, signs and symptoms, general and specific management, anaphylaxis and the difference between shock and fainting.

SYLLABUS COVERED IN THIS CHAPTER

- Shock — definition, types (hypovolaemic, cardiogenic, neurogenic, anaphylactic, septic, psychogenic), causes, stages, signs and symptoms and complete first-aid management.
- Anaphylactic shock and the use of adrenaline; difference between shock and fainting; position of the casualty and the 'nothing by mouth' rule.

7.1 Meaning and Mechanism

DEFINITION

- **Shock** is a **condition of acute circulatory failure in which the flow of oxygenated blood through the tissues becomes inadequate to maintain normal cell function** — in simple words, '*insufficient blood is reaching the vital organs*'.
- Shock is **not** the same as fright or emotional 'shock'; it is a **progressive, life-threatening physical state** that will end in death if not reversed.
- It is produced whenever there is (a) **loss of blood or fluid volume**, (b) **failure of the heart to pump**, or (c) **excessive widening of the blood vessels**. Blood pressure = cardiac output × peripheral resistance; shock follows a fall in either factor.

As blood flow falls, the body compensates through **adrenaline**: the heart beats faster, the skin vessels constrict (**pale, cold, clammy skin**), breathing quickens and fluid is drawn from the tissues (**thirst**). If the cause continues, compensation fails, blood pressure falls, the brain and kidneys suffer, and the process becomes **irreversible**.

7.2 Classification of Shock

Type	Mechanism	Common causes
Hypovolaemic (oligaemic) shock — the commonest type	Loss of blood, plasma or body fluid — reduced circulating volume	Haemorrhage (external or internal), burns and scalds (plasma loss), severe vomiting and diarrhoea (cholera, gastro-enteritis), excessive sweating, crush injury, intestinal obstruction, peritonitis, diabetic coma (dehydration)
Cardiogenic shock	Failure of the heart to pump effectively	Heart attack (myocardial infarction) , severe arrhythmias, heart failure, cardiac tamponade, myocarditis, poisoning of the heart, electric shock
Neurogenic (vasogenic) shock	Loss of nervous control of the vessel walls — sudden widespread vasodilatation	Spinal cord injury (above T6), severe head injury , severe pain, spinal anaesthesia
Anaphylactic shock	Severe, generalised allergic reaction with massive vasodilatation and leaking capillaries	Bee/wasp stings, penicillin and other drugs, peanuts, egg, fish, shellfish, latex, vaccines, radiological dyes
Septic (toxic) shock	Bacterial toxins in the blood damage vessels and cause dilatation and leakage	Severe infection — septicæmia, peritonitis, infected wounds, burns, septic abortion, pneumonia
Obstructive shock	Mechanical obstruction to blood flow in or around the heart	Tension pneumothorax , cardiac tamponade, massive pulmonary embolism
Psychogenic shock (fainting/syncope)	Temporary reflex vagal slowing of the heart and dilatation of vessels — transient reduction of blood to the brain	Fear, pain, sight of blood, bad news, prolonged standing, hot crowded rooms, hunger

Primary and secondary shock

Primary (neurogenic/psychogenic) shock	Secondary (true) shock
Occurs immediately at the time of injury or fright.	Develops gradually, minutes to hours after the injury.
Due to a nervous reflex — pain, fear, sight of blood.	Due to actual loss of blood or fluid , or to toxins.
Usually mild and temporary — fainting; recovers when the casualty lies down.	Progressive and dangerous — needs urgent treatment and often surgery/transfusion.

Primary (neurogenic/psychogenic) shock	Secondary (true) shock
Skin is pale and moist, pulse slow then normal.	Skin cold and clammy, pulse rapid and thready , BP falling.

CAUSES OF SHOCK — QUICK LIST FOR REVISION

- **Loss of body fluid** — haemorrhage, burns, vomiting, diarrhoea, excessive sweating, crush injury.
- **Heart conditions** — heart attack, heart failure, arrhythmia.
- **Severe pain and fear**; extensive injury, multiple fractures.
- **Infection and poisoning; allergy (anaphylaxis).**
- **Spinal or severe head injury; electric shock**; heat stroke; hypoglycaemia; abdominal emergencies.

7.3 Stages of Shock

Stage	What happens	Clinical picture
1. Compensated (initial / non-progressive)	Adrenaline maintains blood pressure by a faster heart rate and constricted skin vessels	Casualty is alert or anxious , skin pale and cool, pulse fast, blood pressure still normal , thirst; urine output falling
2. Decompensated (progressive)	Compensation fails; blood pressure falls; vital organs become starved of oxygen	Restless, confused, drowsy ; skin cold, clammy, grey-blue; pulse very rapid and weak ; BP low ; breathing rapid and shallow (air hunger); very little urine
3. Irreversible (refractory)	Cell and organ death; heart and brain damage cannot be reversed even if the cause is treated	Unconscious , pulse and BP unrecordable, breathing irregular → cardiac arrest and death

7.4 Signs and Symptoms of Shock

System	Signs and symptoms
Skin	Pale (pallor) , cold, clammy skin with sweating; blue-grey (cyanosed) lips, ears and nail beds; capillary refill more than 2 seconds
Pulse	Rapid, weak and thready ; later slow, irregular and finally imperceptible at the wrist
Breathing	Rapid, shallow, sighing or gasping ('air hunger'); yawning
Blood pressure	Falling — systolic below 90 mm Hg is a danger sign
Brain / behaviour	Giddiness and faintness, restlessness and anxiety , 'feeling of impending doom', confusion, dimness of vision, noises in the ears, drowsiness → unconsciousness
Digestive	Intense thirst , nausea and vomiting
Eyes	Sunken, dull eyes with dark rings; dilated pupils
Kidneys	Reduced or absent urine output (< 30 ml/hour)
Temperature	Subnormal; the casualty feels cold and may shiver

MNEMONIC FOR THE SIGNS OF SHOCK — 'COLD AND CLAMMY'

- **C** – Cold, clammy, pale skin
- **O** – Oliguria (little urine)
- **L** – Low blood pressure
- **D** – Dizziness, drowsiness, dimness of vision
- **A** – Anxiety and restlessness
- **N** – Nausea and vomiting
- **D** – Dry mouth and intense **thirst**
- **Plus**: rapid weak pulse and rapid shallow breathing — the two most reliable early signs.

7.5 First-Aid Management of Shock

The golden rule is: **treat the cause, and treat the casualty at the same time**. Shock cannot be reversed at the scene unless the cause (usually bleeding) is controlled.

1. **Treat the cause** — stop bleeding, cool burns, relieve pain, give the adrenaline auto-injector in anaphylaxis, treat the heart attack, immobilise fractures.
2. **Lay the casualty down** on a blanket, with the **head low and turned to one side** (to prevent inhalation of vomit).

- Raise the legs 20–30 cm (8–12 inches)** — the 'shock position' — so that blood drains towards the heart and brain. **Do not raise the legs** if there is a fracture of the leg, a head, chest or abdominal injury, or a suspected spinal injury.
- Loosen tight clothing** at the neck, chest and waist.
- Keep the casualty warm** — cover with a blanket or coat, over and under him. **Do NOT use hot-water bottles, heaters, massage or a fire** — direct heat brings blood to the skin and away from the vital organs.
- Reassure constantly** — a calm confident manner reduces fear, pain and the adrenaline response.
- Give NOTHING by mouth** — no water, tea, food, medicine or alcohol (the casualty may need an anaesthetic, may vomit and inhale, or may have an abdominal injury). If very thirsty, **moisten the lips** with a wet cloth. *Exception:* in **burns** or where transport will be very long, small sips of **water/ORS** may be allowed to a fully conscious casualty.
- Do not allow the casualty to move, sit up, smoke or walk.**
- Monitor and record** breathing, pulse and level of response every **10 minutes** (5 minutes if severe).
- Arrange urgent transport** to hospital, in the lying position, without jolting, and stay with the casualty. Be ready to start **CPR**.

SHOCK — DO'S AND DON'TS AT A GLANCE

- **DO** lay flat, raise the legs, keep warm, reassure, give oxygen if trained, and treat the cause.
- **DON'T** give anything by mouth; **DON'T** apply direct heat; **DON'T** let the casualty sit up or move; **DON'T** leave the casualty alone.
- **DON'T** give alcohol (it dilates vessels and worsens shock) or any sedative/painkiller on your own.
- **DON'T** raise the legs in a head, chest, abdominal, spinal or leg injury — keep the casualty flat instead.

7.6 Anaphylactic Shock (Anaphylaxis)

A **severe, sudden and potentially fatal allergic reaction** affecting the whole body, usually within **seconds to 30 minutes** of exposure. It is a **medical emergency** — death may occur in minutes from airway swelling or circulatory collapse.

Common triggers	Signs and symptoms
Insect stings — bee, wasp, hornet, ant	Airway: swelling of the lips, tongue, throat and face; hoarseness; difficulty in breathing and swallowing ; wheeze and stridor
Drugs — penicillin and other antibiotics, aspirin, NSAIDs, anaesthetics, vaccines, X-ray contrast dyes	Skin: widespread red blotchy rash, hives (urticaria) , itching, flushing, swelling around the eyes
Foods — peanuts and tree nuts, egg, milk, fish, shellfish, soy, wheat, sesame	Circulation: rapid weak pulse, falling blood pressure , pallor, collapse, unconsciousness
Others — latex, pollen, exercise, cold, idiopathic	Gut: nausea, vomiting, abdominal cramps, diarrhoea; also anxiety and a feeling of impending doom

- Remove the trigger** if possible (scrape out a bee sting, stop the drug/food).
- Call an ambulance immediately** — say the word 'anaphylaxis'.
- Give adrenaline (epinephrine) without delay** using the casualty's **auto-injector (EpiPen)** into the **outer middle third of the thigh**, through clothing if necessary, holding it in place for about 3 seconds (new devices) to 10 seconds (older devices). Adult dose **0.3–0.5 mg (1:1000)**; child **0.15 mg**. A **second dose may be given after 5–15 minutes** if there is no improvement.
- Position:** if breathing is the main problem, let the casualty **sit up**; if he feels faint or is shocked, **lay him flat with the legs raised**; if unconscious and breathing, use the **recovery position**. **Never let an anaphylactic casualty stand or walk suddenly.**
- Loosen tight clothing**; help with the casualty's own inhaler if he is asthmatic; give **nothing by mouth**.
- Monitor** continuously and be ready to begin **CPR** — cardiac arrest can follow within minutes.
- Every casualty must go to hospital** even if he recovers, because the reaction can **return after a few hours (biphasic reaction)**.

WHY ADRENALINE?

- It **constricts blood vessels** (raises blood pressure), **relaxes the air passages** (relieves wheeze) and **reduces swelling** — reversing all the dangerous effects at once.
- It is the **drug of choice** and must be given **early**; antihistamines and steroids are secondary and act too slowly.
- Route for anaphylaxis = **intramuscular, in the outer thigh** (not intravenous by a first aider).

7.7 Points Special to Other Types of Shock

Type	Additional first-aid points
Haemorrhagic / hypovolaemic	Control bleeding first — nothing else works until the bleeding stops; immobilise fractures; keep flat with legs raised
Burn shock	Cool the burn, cover it, and give small sips of water or ORS if fully conscious; expect shock in any burn over 15 %
Cardiogenic (heart attack)	Half-sitting position (not flat, as breathing is difficult), absolute rest, loosen clothing, reassure; give one dispersible aspirin 300 mg to chew if the casualty is conscious and not allergic; help with his own nitroglycerine spray/tablet; be ready for CPR
Neurogenic / spinal	Do not move the casualty; support the head and neck in line with the body; keep flat; keep warm
Septic	Suspect when shock follows infection, a wound, an abortion or burns with fever or a very low temperature ; urgent hospital transfer — antibiotics are needed
Psychogenic (fainting)	Lay the casualty down and raise the legs ; fresh air; loosen clothing; recovery is usually within a few minutes; sips of water only after full recovery

7.8 Shock versus Fainting — the Classical Difference

Point	Fainting (syncope)	Shock (true/secondary)
Cause	Temporary reduction of blood flow to the brain — fear, pain, standing long, heat, hunger, sight of blood	Actual failure of circulation — bleeding, fluid loss, heart failure, allergy, infection
Onset	Sudden , often with warning (giddiness, nausea, blurring)	Gradual and progressive , often after injury
Duration	Brief — a few seconds to 2 minutes	Prolonged; worsens unless treated
Pulse	Slow and weak at first, then normal	Rapid, weak and thready , getting worse
Skin	Pale, cool, moist — but recovers colour quickly	Pale, cold, clammy and stays so
Blood pressure	Falls briefly, then returns to normal	Progressively falls
Consciousness	Brief complete loss, then rapid full recovery	Restless → confused → drowsy → unconscious
Response to lying down with legs raised	Recovers quickly and completely	Little or no improvement — needs hospital treatment
Danger	Usually harmless (danger is from the fall)	Life-threatening

7.9 Chapter Recap — One-Line Revision

- Shock = **acute failure of circulation** with inadequate oxygen supply to the tissues; the commonest type is **hypovolaemic**.
- Types: **hypovolaemic, cardiogenic, neurogenic, anaphylactic, septic, obstructive, psychogenic**.
- Cardinal signs: **pale, cold, clammy skin + rapid weak thready pulse + rapid shallow breathing + thirst + restlessness + falling BP**.
- Position of a shocked casualty = **flat, head low and turned to one side, legs raised 20–30 cm**.
- Keep warm with a blanket — never with direct heat; nothing by mouth; never give alcohol**.
- Monitor pulse, breathing and response every **10 minutes**; treat the cause first.
- Anaphylaxis** → **adrenaline 0.5 mg IM in the outer thigh** at once, call the ambulance, sit up if breathless / lie flat if faint, hospital even after recovery.
- Heart attack (cardiogenic shock)** → half-sitting, rest, **300 mg aspirin chewed**, ready for CPR.
- Fainting** recovers within minutes on lying down; **true shock does not** — that is the examination difference.

CHAPTER 8

Electric Shock

Mechanism and severity of electrical injury, low- and high-voltage accidents, lightning strike, safe rescue, complete first aid, electrical burns and prevention.

SYLLABUS COVERED IN THIS CHAPTER

- Electric shock — meaning, causes, factors deciding severity, effects on the body, signs and symptoms.
- Safe release of the casualty from the current, first aid including resuscitation and burn care, lightning injury, and prevention / electrical safety.

8.1 Meaning and Mechanism

DEFINITION

- **Electric shock** is the sudden and violent effect produced on the body when an **electric current passes through it**. The current enters at the point of contact, travels through the tissues along the path of least resistance and leaves at the point of **earthing**, causing **burns at the entry and exit points** and disturbing the nerves, muscles and heart on the way.
- The body is a **conductor** because it is largely water and salt; **wet skin has about 25–100 times less resistance than dry skin**, so shocks are far more dangerous when the skin, floor or clothing is wet.
- Current follows the shortest path: **hand-to-hand or hand-to-opposite foot paths cross the heart and are the most dangerous**.

Causes of electrical injury

- **Domestic (low voltage — up to 1,000 V, usually 220–240 V AC)**: faulty wiring, bare or frayed wires, broken plugs and sockets, wet hands on switches, using electrical appliances in the bathroom, children inserting objects into sockets, defective geysers, iron, heaters and immersion rods.
- **Industrial and high voltage (over 1,000 V)**: overhead power lines, transformers, substations, welding sets, cranes touching lines, railway overhead equipment.
- **Lightning**: a natural discharge of millions of volts lasting a fraction of a second.
- **Others**: electric fences, stun devices, medical equipment and faulty earthing.

Factors that decide the severity of an electric shock

Factor	Effect on severity
Type of current	AC (alternating current) is about 3–5 times more dangerous than DC at the same voltage, because it causes sustained muscular contraction ('locking on') and readily provokes ventricular fibrillation — DC tends to throw the casualty clear
Amount of current (amperage)	The real killer . About 1 mA is just felt; 10–15 mA produces 'let-go' failure (muscles lock on); 50–100 mA can cause ventricular fibrillation (fatal); several amperes cause deep burns and cardiac standstill
Voltage	Higher voltage drives more current through the body; high-voltage contact causes deep charring, blast injury and may throw the casualty some distance
Resistance	Wet skin, sweat, water, bare feet and metal contact reduce resistance and increase the current; dry skin, rubber shoes and dry clothing protect
Duration of contact	The longer the contact, the greater the damage — hence the first action is always to break the contact
Path taken through the body	Hand to hand, or hand to opposite foot (across the chest) — most dangerous because the current crosses the heart; current through the head affects the brain and breathing centre
Area of contact and general health	A small contact area concentrates the current; children, the elderly and those with heart disease are more vulnerable

8.2 Effects, Signs and Symptoms

System / part	Effects
Heart	Ventricular fibrillation (the commonest cause of death), other arrhythmias, cardiac arrest, chest pain; arrhythmia may appear hours later
Breathing	Respiratory arrest from spasm/paralysis of the respiratory muscles or damage to the respiratory centre; breathing may stop while the heart still beats

System / part	Effects
Muscles	Violent tetanic (sustained) contraction — the casualty may be unable to let go of the wire; dislocations and fractures of the spine or limbs from the spasm or from a fall
Skin	Entry and exit burns — small, round, painless, charred, depressed craters , often on the hand (entry) and foot (exit); flash and flame burns from arcing and burning clothing
Deep tissues	Extensive hidden destruction of muscle and nerve along the current's path — the external wound looks small but the inside is cooked ('iceberg injury')
Nervous system	Unconsciousness, confusion, fits, temporary paralysis, loss of memory, deafness, blindness, later neuritis
Kidneys	Acute kidney failure from destroyed muscle (myoglobin) — as in crush injury
General	Shock , internal injuries, secondary injuries from a fall (head injury, fractures), and psychological shock

Recognition at the scene: an **unconscious or dazed casualty near an electrical appliance, wire or pole**, with **burn marks at two points**, muscle rigidity, and often no breathing or pulse. Look also for a **fallen wire, smell of burning, blown fuse or sparking switch**.

8.3 First Aid — Breaking the Contact Safely

THE FIRST AND MOST IMPORTANT RULE

- **DO NOT TOUCH THE CASUALTY UNTIL YOU ARE CERTAIN THAT THE CONTACT WITH THE CURRENT IS BROKEN.** A rescuer who touches a live casualty becomes the next casualty.
- First **switch off** the current at the main switch, plug, fuse or circuit breaker — this is always the best method.
- If the switch cannot be reached, **insulate yourself and push, pull or lever the casualty away** from the source using a **dry non-conductor**: dry wooden stick, broom handle, bamboo, plastic chair, rubber mat, thick dry newspaper, folded dry cloth, dry rope or the casualty's dry clothing.
- **Stand on dry insulating material** (rubber mat, thick dry newspaper, dry wooden board, rubber-soled shoes) and use **one hand only**, keeping the other hand in your pocket.
- **Never use anything wet or metallic**; never touch the casualty's skin or wet clothing with bare hands; never use bare hands even with a dry cloth wrapped round them if the voltage is high.

Low-voltage accidents (household — up to 1,000 V)

1. **Switch off** at the mains/socket, or pull out the plug — do not pull on the cable.
2. If that is impossible, stand on dry insulating material and **use a dry wooden/plastic object to push the casualty's limb away** from the source, or to move the wire away.
3. You may also **loop a dry rope or dry cloth round the casualty's feet or arms and drag him away**.
4. Only then begin the **DRABC** assessment.

High-voltage accidents (over 1,000 V — power lines, substations, railway lines)

1. **Do not approach.** High voltage can **arc (jump) several metres** and the ground itself becomes live ('ground current/step potential').
2. Keep yourself and all bystanders **at least 18 metres (about 20 yards) away**, or further if instructed, until the supply authority confirms the power is **switched off and earthed**.
3. **Telephone the electricity supply company/police/fire brigade immediately**; only they can isolate the supply.
4. **Nothing** — not dry wood, not rubber — gives protection at these voltages.
5. If the casualty is in a vehicle touching a line, tell the occupants **to stay inside** until the power is cut.
6. Begin treatment only when the officials declare the line dead.

If the casualty is in water or the area is flooded

- **Switch off the supply before entering the water** — water conducts electricity over a wide area.
- Do not enter water containing a live cable; use a dry non-conducting pole from dry ground.

8.4 First Aid After the Contact Is Broken

1. **Check response and breathing (DRABC).** Electrocuting very commonly causes **cardiac and respiratory arrest** — if the casualty is not breathing normally, **start CPR at once** and **use an AED as soon as it arrives** (the rhythm is often a shockable ventricular fibrillation).
2. **Call 112/108** and mention electrocution, so that the ambulance brings a defibrillator.
3. If the casualty is unconscious but breathing, place him in the **recovery position**, supporting the head and neck (assume a **spinal injury** if he fell or was thrown).

4. **Treat the burns** — look for **both entry and exit wounds**; cool with clean water for 10–20 minutes if the burn is fresh and small, then cover each with a **sterile non-adherent dressing or clean cling film**. Do not apply ointment or break blisters.
5. **Treat for shock** — lay flat, keep warm, reassure, nothing by mouth.
6. **Immobilise** any suspected fracture or dislocation caused by the muscle spasm or fall.
7. **Monitor** breathing, pulse and consciousness continuously — arrhythmias may occur up to 24–48 hours later.
8. **Send every electrical casualty to hospital**, even if he seems perfectly well — deep tissue damage and delayed heart rhythm problems are invisible at the scene.

ELECTRICAL BURNS — WHY THEY ARE SPECIAL

- They are usually **deep (third or fourth degree)** but look **small** on the surface; there is always an **entry and an exit** wound.
- They are often **painless** at first because the nerve endings are destroyed.
- The real damage is to the **muscles, nerves and vessels beneath** — hence swelling, kidney failure and amputation may follow.
- **Never underestimate an electrical burn**; always refer to hospital.

8.5 Lightning Strike

- Lightning carries **millions of volts** but lasts only a few milliseconds; it may strike directly, by side-flash from a tree, by ground current, or by contact with a struck object.
- Effects: instantaneous **cardiac and respiratory arrest**, unconsciousness, **fern-like (Lichtenberg) skin markings**, superficial burns, ruptured eardrums, temporary blindness or deafness, paralysis of the legs (keraunoparalysis), and injuries from being thrown.
- **A casualty struck by lightning carries no electric charge — he is safe to touch immediately.**
- **Reverse triage** applies: in a multiple lightning casualty situation, attend **first to those who appear dead** (not breathing), because they can often be revived by prompt CPR, while those who are moving and moaning will usually survive.
- Move everyone to safe shelter first — **lightning can strike the same place twice.**
- Give prolonged CPR; recovery after lightning arrest is common if CPR is started at once.

LIGHTNING SAFETY RULES (30/30 RULE)

- If the time between the flash and the thunder is **less than 30 seconds**, the storm is within 10 km — take shelter at once; wait **30 minutes** after the last thunder before going out.
- **Safe places:** a substantial building with wiring and plumbing, or a closed metal-topped vehicle (windows up).
- **Unsafe:** open ground, hill tops, isolated trees, verandas, sheds, bus shelters, water bodies, swimming pools, metal fences, tractors and open vehicles; also avoid using corded telephones and touching plumbing.
- If caught in the open with no shelter: crouch on the balls of the feet, heels together, head down, hands over the ears, **do not lie flat**, and keep away from tall objects; groups should spread out.
- Put down umbrellas, fishing rods, golf clubs and metal tools.

8.6 Prevention — Electrical Safety

- Have all wiring installed and repaired by a **qualified electrician**; use **ISI-marked** appliances, plugs and cables.
- Ensure proper **earthing** of every appliance, and fit **fuses, MCBs (miniature circuit breakers) and an ELCB/RCCB (earth-leakage circuit breaker)**.
- **Switch off and unplug** before repairing or cleaning any appliance; use the **'lock-out, tag-out'** system at work.
- **Never touch switches, plugs or appliances with wet hands** or while standing in water or barefoot on a wet floor; keep electrical devices out of bathrooms.
- Replace frayed cords; do not overload sockets or use joined/taped wires; do not run wires under carpets.
- Fit **safety covers on sockets** where there are small children; keep them away from appliances and cords.
- Keep **at least 3 metres** away from fallen overhead wires and report them; never fly kites or erect poles near power lines; do not build or climb near transformers.
- Display **'DANGER — ELECTRIC SHOCK'** notices, keep a **dry wooden stick, rubber mat and an insulated hook** near switchboards, and put up the **first-aid chart for electric shock** at work sites.
- Use **dry chemical or CO₂ extinguishers** for electrical fires — **never water**; switch off the supply first.

8.7 Chapter Recap — One-Line Revision

- First action in electric shock = **switch off / break the contact safely**; never touch a live casualty.
- **AC is more dangerous than DC**; the amount of **current (mA)** kills, not the voltage alone; **50–100 mA causes ventricular fibrillation**.
- **Wet skin reduces resistance** and multiplies the danger; the hand-to-hand/hand-to-foot path across the heart is the most dangerous.
- High voltage — keep **at least 18 metres away** and wait for the supply authority; no insulator protects you.
- Commonest cause of death = **ventricular fibrillation** → immediate **CPR + AED**.
- Electrical burns have **entry and exit wounds**, are **deep but look small**, and are often **painless**.
- Lightning casualties **carry no charge** — safe to touch; use **reverse triage** and treat the apparently dead first.
- Every electrical casualty goes to **hospital** — rhythm disturbances can appear many hours later.
- Use **CO₂/dry powder** extinguishers on electrical fires, never water.

PART IV • BREATHING EMERGENCIES AND BONE INJURIES

CHAPTER 9

Different Methods of Artificial Respiration

Expired-air (mouth-to-mouth, mouth-to-nose, mouth-to-stoma) methods, the manual methods of Schafer, Holger Nielsen, Sylvester and Eve, mechanical methods, their technique, rates, advantages and drawbacks.

SYLLABUS COVERED IN THIS CHAPTER

- Different methods of artificial respiration — classification, indications, complete technique of each method with rates, advantages, disadvantages and choice of method.
- Signs of effectiveness, complications, when to start and when to stop, and artificial respiration in special situations.

9.1 Meaning, Aim and Indications

DEFINITION

- **Artificial respiration (artificial ventilation)** is the **procedure by which air is made to enter and leave the lungs of a casualty whose natural breathing has stopped or is failing**, in order to supply oxygen to the blood and remove carbon dioxide, until natural breathing returns.
- It is also called **rescue breathing, artificial ventilation** or (when combined with chest compressions) part of **CPR**.
- **Aim:** to keep the vital organs — especially the **brain** — supplied with oxygen. **Time is everything:** breathing must be restored within **3–4 minutes** to avoid brain damage.

Indications — when is artificial respiration needed?

- **Asphyxia from any cause** — drowning, choking, hanging, strangulation, suffocation, smothering, crush of the chest.
- **Electric shock and lightning.**
- **Gas and fume poisoning** — carbon monoxide, LPG, sewer gas, smoke inhalation.
- **Poisoning** by drugs that depress the respiratory centre — opium/morphine, barbiturates, alcohol, organophosphates, snake venom (cobra).
- **Head and spinal injury**, stroke, cardiac arrest.
- **Diseases and conditions** — severe asthma, status epilepticus, polio, tetanus, anaphylaxis.
- **Newborn babies who fail to breathe** after birth (asphyxia neonatorum).

Artificial respiration is needed whenever breathing has stopped (**apnoea**). If the **heart has also stopped**, artificial respiration alone is useless — it must be combined with **chest compressions (CPR)**.

9.2 Classification of the Methods

Group	Methods	Remarks
A. Expired-air (mouth-to-...) methods — direct methods	1. Mouth-to-mouth 2. Mouth-to-nose 3. Mouth-to-mouth-and-nose (infants) 4. Mouth-to-stoma 5. Mouth-to-mask / face shield	The methods of choice — most effective, can be started anywhere, deliver 16 % oxygen and a large volume, and allow the rescuer to watch the chest
B. Manual (indirect / pressure) methods	1. Schafer's prone-pressure method 2. Holger Nielsen's back-pressure arm-lift method 3. Sylvester's chest-pressure arm-lift method 4. Eve's rocking method 5. Nielsen–Callaghan and other historical variants	Obsolete as first choice ; used only when mouth-to-mouth is impossible (severe facial injury, corrosive poison on the lips, gas contamination, rescuer unable to make a seal)
C. Mechanical methods	Bag-valve-mask (Ambu bag), pocket mask with oxygen, mechanical ventilator, anaesthetic machine, historical Pulmotor and iron lung (tank respirator)	Used by trained personnel and in hospital; deliver controlled volumes and up to 100 % oxygen

9.3 Expired-Air Methods — Technique

Preparation common to all methods

1. Remove the casualty from danger (and the danger from the casualty).
2. Lay him **flat on the back on a firm surface**; loosen clothing at the neck, chest and waist.
3. **Clear the mouth and throat** — turn the head to one side, hook out vomit, weed, sand, blood, broken/loose dentures and food with two fingers wrapped in cloth. Well-fitting dentures are left in place.
4. **Open the airway** — one hand on the forehead to tilt the head back, two fingers under the chin to lift it (**head tilt-chin lift**). Use a **jaw thrust** without head tilt if a neck injury is suspected.
5. **Check breathing** for not more than **10 seconds** — look, listen and feel.

1. Mouth-to-mouth respiration (the 'kiss of life')

1. Kneel beside the casualty's head. Keep the airway open with head tilt-chin lift.
2. **Pinch the soft part of the nose** closed with the thumb and index finger and let the mouth fall open.
3. Take a normal breath, place your **lips widely around the casualty's open mouth making an air-tight seal**.
4. **Blow steadily for about 1 second**, watching the chest rise ($\approx 500\text{--}600\text{ ml}$ — enough for a **visible chest rise**, no more).
5. Lift your mouth away, turn your head to watch the **chest fall** and breathe in again.
6. Give **2 breaths**, then 30 chest compressions if there is no pulse; if the heart is beating and only breathing has stopped, continue at **10–12 breaths per minute (1 breath every 5–6 seconds) in an adult and 12–20 per minute (1 every 3 seconds) in a child or infant**.
7. Re-check for normal breathing and circulation after every **1–2 minutes** (about 10 breaths).
8. If the chest does not rise: **re-adjust the head tilt, check the seal, look for an obstruction** — do not attempt more than 2 breaths before resuming compressions.

2. Mouth-to-nose respiration

- **Indications:** injury of the mouth or lips, **jaw clenched or wired (trismus)**, mouth cannot be sealed (a small child's or a very large adult's), rescue from water, corrosive poison on the mouth, and when the rescuer's mouth is smaller than the casualty's.
- **Technique:** close the mouth by pressing the chin upwards, seal your lips round the **nose** and blow. Open the mouth during expiration to let air escape (the soft palate may act as a valve).
- Slightly more difficult and slower, but equally effective.

3. Mouth-to-mouth-and-nose (for infants and small babies)

- Keep the head in a **neutral** position — do **not** over-extend the neck.
- Cover the **whole of the baby's mouth and nose** with your mouth (or seal the nose and use the mouth alone in a bigger child).
- Blow **only the air held in your cheeks (small puffs)** for about 1 second, just enough to make the chest rise; give **5 initial breaths** and then **12–20 breaths per minute**.
- **Never blow hard into a baby's lungs** — they can be torn (barotrauma).

4. Mouth-to-stoma and 5. mouth-to-mask

- **Mouth-to-stoma:** for a casualty who has had a **laryngectomy or tracheostomy** (a hole in the front of the neck). Seal the casualty's **mouth and nose** with your hand and blow directly into the stoma after wiping it clean.
- **Mouth-to-mask / face shield:** a transparent mask with a **one-way valve** — avoids direct contact with the casualty's mouth, protects from infection and vomit, and allows **supplementary oxygen**. This is the recommended method for trained first aiders.

ADVANTAGES OF EXPIRED-AIR METHODS OVER MANUAL METHODS

- Deliver a **much larger volume of air (500–800 ml)** compared with 100–300 ml by manual methods.
- Can be given in **any position** — in water, in a confined space, on a stretcher, while the casualty is being extricated.
- The rescuer **sees, hears and feels** the effectiveness (chest rise) immediately.
- **No equipment** is needed and one rescuer can also give chest compressions.
- Can be **combined with chest compressions** — manual methods cannot.
- Expired air still contains **16 % oxygen** and 4 % CO_2 , enough to oxygenate the casualty's blood.

9.4 Manual Methods of Artificial Respiration

These older methods work by alternately **compressing and expanding the chest from the outside**. They are much less efficient and are now used only when mouth-to-mouth is impossible — but they are **regularly asked in examinations**, so learn the positions, the movements and the rates.

1. Schafer's method (prone-pressure method)

Point	Detail
Position of casualty	Prone (face down) , head turned to one side, one arm extended above the head, the other bent with the hand under the cheek; a folded blanket under the chest
Position of rescuer	Astride or kneeling beside the casualty's thighs , facing the head
Movement	Place the palms on the lower ribs on either side of the spine (over the loins), thumbs almost touching, arms straight . Swing forward with the weight of the body, pressing steadily downwards and inwards for about 2–3 seconds (expiration) , then swing back releasing the pressure completely for 2–3 seconds (inspiration)
Rate	12–15 cycles per minute
Advantages	Simple, needs no assistance, useful in drowning because the prone position lets water, vomit and weed drain out of the mouth
Disadvantages	Only expiration is active (inspiration is passive) so the air exchanged is small (100–300 ml); cannot be used with a chest, spine or abdominal injury or in pregnancy; the airway is not under direct control; no chest compressions possible

2. Holger Nielsen's method (back-pressure arm-lift method)

Point	Detail
Position of casualty	Prone , arms bent with the hands one over the other under the forehead , head turned slightly to one side
Position of rescuer	Kneeling on one knee at the casualty's head , the other foot beside his elbow
Movement (4 phases)	(a) Place your palms on the casualty's shoulder blades , thumbs touching; (b) rock forward with straight arms and press steadily downwards — expiration ; (c) rock back, slide the hands to just above the elbows; (d) raise the arms upwards and towards you until resistance is felt — this expands the chest, giving inspiration — then lower the arms
Rate	12 complete cycles per minute (about 5 seconds per cycle)
Advantages	Both inspiration and expiration are active — therefore more efficient than Schafer's ; drainage of fluid from the mouth is still possible; less tiring for the rescuer
Disadvantages	Cannot be used if the arms or shoulders are injured , or with chest/spinal injuries; the rescuer must be at the casualty's head; still far less effective than mouth-to-mouth

3. Sylvester's method (chest-pressure arm-lift method)

Point	Detail
Position of casualty	Supine (on the back) , with a folded blanket or cushion under the shoulders so that the head is slightly lower and the neck extended; the tongue must be drawn forward and the mouth cleared
Position of rescuer	Kneeling at the casualty's head , facing his feet
Movement	Grasp the casualty's wrists and cross them over the lower chest, pressing firmly downwards — expiration (2 seconds); then sweep the arms outwards, upwards and backwards above the head as far as possible — inspiration (2–3 seconds)
Rate	12 cycles per minute
Advantages	Useful when the casualty cannot be turned prone — e.g. injuries of the back or when found lying on the back; both phases active; the face is visible
Disadvantages	Danger of the tongue falling back and of inhaling vomit or water , because the casualty is on his back — hence unsuitable for drowning unless an assistant holds the tongue/head; cannot be used with chest, rib or arm injuries

4. Eve's rocking method

Point	Detail
Principle	The casualty is strapped to a stretcher or see-saw frame pivoted at its centre and rocked head-down and foot-down; the weight of the abdominal organs moves the diaphragm up and down, causing expiration and inspiration
Movement / angle	Tilted about 45° head-down (expiration) and 45° foot-down (inspiration)
Rate	10–12 rocks per minute

Point	Detail
Advantages	Very little effort for the rescuer — can be continued for hours; drains fluid from the air passages , so it was widely used in drowning and in poliomyelitis; the casualty can be transported while being ventilated
Disadvantages	Needs special apparatus (stretcher/rocking board) ; unsuitable for chest, spine or abdominal injuries and in pregnancy; slow to set up

5. Mechanical methods

- **Bag-valve-mask (Ambu bag)** — self-inflating bag with a non-rebreathing valve and mask; delivers 400–600 ml per squeeze and up to **100 % oxygen** with a reservoir; best used by **two rescuers** (one holds the mask with both hands, the other squeezes).
- **Pocket mask with oxygen inlet** — simple, effective, protects the rescuer.
- **Mechanical ventilator** — hospital machine giving controlled volume, rate and oxygen concentration through an endotracheal tube.
- **Historical:** the **Pulmotor** (an early automatic resuscitator) and the **iron lung/tank respirator** (negative-pressure chamber used in polio epidemics).
- **Oxygen cylinders with masks** are *not* artificial respiration — they only enrich the air of a casualty who is still breathing.

9.5 Comparison of All Methods — Master Table

Method	Position of casualty	Rate/min	Air moved	Chief use
Mouth-to-mouth	Supine	10–12 (adult), 12–20 (child)	500–800 ml	Method of choice in all cases
Mouth-to-nose	Supine	10–12	500–800 ml	Mouth injured, jaw clenched, rescue in water
Mouth-to-mouth-and-nose	Supine, neutral head	12–20	Cheek puffs only	Infants under 1 year
Mouth-to-stoma	Supine	10–12	500–800 ml	Tracheostomy/laryngectomy
Schafer's (prone pressure)	Prone	12–15	100–300 ml	Drowning, when mouth-to-mouth impossible
Holger Nielsen's (back pressure–arm lift)	Prone	12	300–500 ml	Best of the manual methods
Sylvester's (chest pressure–arm lift)	Supine	12	300–500 ml	When the casualty cannot be turned prone
Eve's rocking	Strapped to a rocking stretcher	10–12	Variable	Prolonged ventilation, drainage of fluid
Bag-valve-mask	Supine	10–12 (1 squeeze every 5–6 s)	400–600 ml + O ₂	Trained rescuers, ambulance

9.6 Effectiveness, Complications and Stopping

Signs that it is working	Complications and errors
The chest rises and falls with each breath.	Gastric distension from blowing too hard or too fast → vomiting and aspiration
Colour improves — blueness of lips and face disappears.	Over-inflation → rupture of alveoli (barotrauma), especially in children
The pulse becomes stronger and more regular.	Failure of the chest to rise — blocked airway, poor seal, insufficient head tilt
Spontaneous breathing, coughing, swallowing or movement returns.	Cross-infection — use a face shield/pocket mask
Pupils constrict and consciousness begins to return.	Rescuer hyperventilation and giddiness from blowing too hard/fast

- **When breathing returns:** stop ventilating, place the casualty in the **recovery position**, keep him warm, watch him continuously (breathing may stop again) and send him to hospital.
- **Continue artificial respiration until** natural breathing returns, qualified help takes over, the casualty's heart stops (then start full CPR), or you are exhausted — **never abandon a casualty because 'enough time has passed'**. In drowning, electrocution, lightning and hypothermia, prolonged efforts have succeeded after an hour or more.

- **Never give artificial respiration to a casualty who is breathing normally;** and never delay it to look for equipment.

SPECIAL SITUATIONS

- **Drowning** — begin rescue breaths as soon as the airway is clear, even while still in shallow water; do **not** waste time trying to expel water from the lungs; expect vomiting.
- **Corrosive poison or contact poison (organophosphate) on the mouth** — do not do direct mouth-to-mouth; use a **mask, bag-valve device or a manual method**.
- **Gas-filled space** — remove the casualty into fresh air first; never enter without protection.
- **Suspected spinal injury** — open the airway with a **jaw thrust**, keeping the head in line.
- **Facial injury/severe bleeding from the mouth** — use mouth-to-nose or a manual method, keeping the airway clear.
- **Newborn baby** — gentle puffs at **30–60 per minute**; resuscitation ratio for a newborn is **3 compressions : 1 breath**.

9.7 Chapter Recap — One-Line Revision

- Aim of artificial respiration = to **supply oxygen to the brain** within **3–4 minutes**.
- **Mouth-to-mouth is the method of choice** — largest volume, no equipment, can be combined with chest compressions; rate **10–12/min** in adults (1 breath every 5–6 seconds).
- **Schafer's** — prone, pressure over the loins, **12–15/min**, only expiration active.
- **Holger Nielsen's** — prone, **back pressure + arm lift, 12/min**, the **most efficient manual method**.
- **Sylvester's** — supine, **chest pressure + arm lift, 12/min**, risk of the tongue falling back and of aspiration.
- **Eve's rocking** — rocking stretcher at 45°, **10–12/min**, least tiring and drains fluid.
- **Infants** — mouth-to-mouth-and-nose with **cheek puffs**, neutral head, 12–20/min; newborn ratio **3 : 1**.
- Expired air contains **16 % oxygen** — that is why expired-air methods work.
- Commonest complication = **stomach inflation and vomiting** from too forceful breaths.

CHAPTER 10

Asphyxia

Meaning and causes of asphyxia, its stages and signs, general management, and the first aid of drowning, hanging, strangulation, suffocation, gas poisoning, smoke inhalation, traumatic asphyxia, asthma and hyperventilation.

SYLLABUS COVERED IN THIS CHAPTER

- Asphyxia — definition, causes and classification, stages, signs and symptoms, general principles of treatment.
- First aid in drowning, hanging, strangulation, throttling, suffocation, smothering, gas and smoke inhalation, traumatic asphyxia, choking, asthma, croup and hyperventilation.

10.1 Meaning and Definition

DEFINITION

- **Asphyxia** (Greek *a* = without, *sphysis* = pulse) is the condition in which the **supply of oxygen to the blood and tissues is cut off or greatly reduced, with an accumulation of carbon dioxide in the body**, because of **interference with breathing**.
- Other names: **suffocation**; the state of oxygen lack in the tissues is **hypoxia/anoxia**, and the state of excess carbon dioxide is **hypercapnia**.
- Asphyxia is the **commonest cause of preventable death** in first aid — and the first aider's first duty (the 'A' and 'B' of DRABC) is to correct it.

Death from asphyxia occurs in **3–5 minutes**; the **brain** is the first organ to suffer because it has no store of oxygen.

10.2 Causes and Classification of Asphyxia

Group	Mechanism	Examples
1. Obstruction of the air passages (mechanical)	Air cannot reach the lungs	Choking on food or a foreign body; the tongue falling back in an unconscious casualty (commonest of all); swelling of the throat in burns, scalds of the mouth, anaphylaxis , croup, diphtheria; blood, vomit or water in the airway; drowning; hanging, strangulation, throttling
2. Closure of the external openings (smothering)	Mouth and nose blocked from outside	Pillow, plastic bag, cloth, hand over the face; a baby sleeping face down or overlaid by an adult; sand, grain, earth or a landslide burying the face
3. Compression of the chest (traumatic asphyxia)	The chest cannot expand	Crushing by a wall, vehicle, machinery or earth; being crushed in a stampede or crowd ; being pinned under a heavy weight
4. Lack of oxygen in the surrounding air (atmospheric/environmental)	The air itself is deficient in oxygen or contains a poisonous gas	Confined spaces — wells, tanks, silos, sewers, mines, ship holds; carbon monoxide from charcoal fires, geysers, car exhaust; LPG, CNG, sewer gas (H₂S), ammonia, chlorine ; smoke in a fire; high altitude; deep-sea diving accidents
5. Paralysis of the respiratory centre or muscles	The brain or nerves cannot drive breathing	Electric shock and lightning ; poisoning by opium/morphine, barbiturates, alcohol, organophosphates, cobra venom ; severe head or spinal injury ; stroke; tetanus; poliomyelitis; status epilepticus
6. Disease of the lungs and circulation	Gas exchange in the lungs fails	Severe asthma , pneumonia, pulmonary oedema, collapsed lung (pneumothorax), pleural effusion, emphysema, pulmonary embolism, heart failure, severe anaemia

10.3 Stages, Signs and Symptoms

Stage	What happens	Signs
1. Stage of dyspnoea (breathlessness)	Rising CO ₂ stimulates the respiratory centre	Rapid, deep, noisy and difficult breathing ; distress and anxiety; rapid pulse; casualty fights for air, uses the neck and shoulder muscles, and cannot speak in full sentences
2. Stage of convulsions	Oxygen lack begins to affect the brain	Blueness (cyanosis) of the lips, ear lobes, tongue and nail beds; congested, swollen face with prominent veins ; bloodshot, bulging eyes ; frothing at the mouth; convulsions and involuntary passing of urine/stool ; loss of consciousness
3. Stage of exhaustion / apnoea	The respiratory centre becomes paralysed	Breathing becomes slow, shallow, gasping (agonal) and then stops ; pulse becomes feeble and irregular; pupils dilate and become fixed ; muscles relax; cardiac arrest and death

- **Cardinal signs of asphyxia** — **difficult noisy breathing, cyanosis (blueness), congested swollen face, frothing at the mouth, restlessness and fighting for breath, falling level of consciousness, and finally stoppage of breathing.**
- In **carbon monoxide poisoning** the skin may be **cherry-red instead of blue** — a classical exception.
- In **anaemia or severe blood loss**, cyanosis may not be evident even with severe hypoxia.

10.4 General Principles of Treatment

1. **Remove the cause of asphyxia, or the casualty from the cause** — taking care of your own safety (gas, fire, electricity, water, confined space).
2. **Open and clear the airway** — head tilt–chin lift; remove vomit, blood, weed, dentures, foreign bodies.
3. **Give fresh air / oxygen** — loosen tight clothing at the neck, chest and waist; keep bystanders away so that air circulates.
4. **Start artificial respiration at once** if breathing has stopped, and **full CPR** if there is no pulse.
5. Once breathing returns, place the casualty in the **recovery position** and keep him warm.
6. **Treat the associated injuries** — burns, fractures, wounds — and **treat for shock**.
7. **Monitor** breathing, pulse and consciousness continuously; breathing may stop again.
8. **Send every casualty to hospital**, even if he recovers — **secondary (delayed) lung oedema** may develop hours later, especially after drowning or smoke/gas inhalation.

10.5 Drowning

Drowning is asphyxia caused by the **immersion of the mouth and nose in a liquid**. Even a few centimetres of water in a bucket or tub can drown an infant.

Term / type	Meaning
Wet drowning	Water is actually inhaled into the lungs (about 85–90 % of cases) — washes away surfactant, causes lung oedema
Dry drowning	Sudden spasm of the larynx (laryngospasm) on contact with water stops breathing; little or no water enters the lungs (10–15 %)
Secondary (delayed) drowning	Deterioration with breathlessness and lung oedema 1–72 hours after apparently successful rescue — the reason every near-drowning casualty must go to hospital
Immersion syndrome	Sudden cardiac arrest from vagal inhibition on plunging into very cold water
Near-drowning	Survival after a drowning episode
Fresh water vs sea water	Fresh water is hypotonic — absorbed into the blood, may haemolyse RBCs; sea water is hypertonic — draws fluid into the alveoli. First aid is the same for both

Rescue — protect yourself first

- **‘REACH – THROW – ROW – GO’** is the order of rescue: **reach** out with a stick, rope or clothing; **throw** a rope, float, tube, empty bottle or plank; **row** a boat; and only as a last resort **go** (swim) — and only if you are a trained swimmer.
- Never jump in alone if you cannot swim well; take a float with you; approach the casualty from behind.
- **Suspect a spinal injury** in a diving accident — support the head and neck in line and float the casualty out on his back.
- Keep the casualty **horizontal** while lifting him from the water if possible (vertical lifting can cause collapse).

First aid for a drowning casualty

1. Get the casualty out of the water as quickly and safely as possible; **if he is not breathing, give 5 rescue breaths even in shallow water.**
2. Lay him on his **back on a firm surface, head slightly lower** if possible, and **clear the mouth** of weed, sand, vomit and dentures.
3. **Open the airway and check breathing.** If absent, begin **CPR with 5 initial rescue breaths, then 30 : 2**. Call 112/108 (if alone, give **1 minute of CPR first**, then call).
4. **Do NOT waste time trying to drain water from the lungs** — do not hold the casualty upside down, do not press on the abdomen, do not use the Heimlich manoeuvre for water.
5. **Expect vomiting** — turn the head to one side, clear the mouth and continue.
6. As soon as breathing returns, place in the **recovery position** and **treat for hypothermia** — remove wet clothing, dry the casualty, wrap in blankets, protect from wind; handle gently.
7. **Give nothing by mouth** while unconscious; when fully conscious, warm sweet drinks may be given.

8. Send every casualty to hospital, even one who appears fully recovered — because of **secondary drowning**.
9. In cold-water drowning, **continue resuscitation for a long time** — survival has followed 30–60 minutes of submersion in cold water, especially in children.

10.6 Hanging, Strangulation and Throttling

Term	Meaning
Hanging	Constriction of the neck by a ligature, the weight of the body providing the force
Strangulation	Constriction of the neck by a ligature or other force other than the body weight (rope, scarf, wire, cord round a baby's neck)
Throttling (manual strangulation)	Constriction of the neck by hands
Mugging	Constriction by a forearm across the throat from behind

- Signs: **a mark, groove or bruising round the neck**; congested, swollen, blue face; **bloodshot eyes with tiny haemorrhages (petechiae)**; protruding tongue; froth at the mouth; unconsciousness; absent or gasping breathing; possible **fracture of the cervical spine or larynx**.
1. **Take the weight of the body at once** and **cut or remove the constriction immediately** — cut **above the knot** and **preserve the knot for the police** (it is a medico-legal case).
 2. **Support the head and neck** — assume a **spinal injury** and do not bend the neck.
 3. Lay the casualty down, **open the airway (jaw thrust)** and check breathing; start **CPR** if needed.
 4. If breathing, place in the **recovery position** and watch continuously — **swelling of the throat can obstruct the airway even hours later**.
 5. **Do not disturb the scene** more than necessary; note the position of the body and the time; **inform the police**.
 6. **Every casualty must go to hospital**, however well he seems.

10.7 Suffocation and Smothering

- Causes: a **plastic bag or sheet over the face**, a pillow or cloth, a hand, a baby's face pressed into soft bedding or overlaid by a sleeping adult, being buried under **sand, grain, earth or a landslide**, and being trapped in a **collapsed building or a refrigerator/box**.
- Treatment: **remove the obstruction from the face at once**; clear the mouth and nose of sand, mud or vomit; open the airway; start artificial respiration/CPR if needed; treat injuries; recovery position; hospital.
- In burial under earth or grain, **clear the chest as well as the face** so that the chest can expand, and expect crush injury.
- **Prevention**: keep plastic bags away from children, never let a baby sleep on a soft pillow or with a plastic sheet, remove doors from disused refrigerators, and fence wells and sand pits.

10.8 Asphyxia from Gases, Fumes and Smoke

Gas / fume	Source and features	First aid
Carbon monoxide (CO)	Charcoal <i>angeethi/sigri</i> and gas geysers in closed rooms, car exhaust, coal stove, fire smoke. Colourless and odourless ; binds haemoglobin 200–250 times more strongly than oxygen. Features: headache, giddiness, nausea, confusion, cherry-red skin , collapse; whole families are affected together	Ventilate the room from outside; switch off the source ; drag the casualty into fresh air (do not enter without protection); give 100 % oxygen if available; CPR if needed; hospital — may need hyperbaric oxygen
LPG / CNG / cooking gas	Leaking cylinder or pipe; heavier than air (LPG) so it collects near the floor; explosion risk	Do not switch on or off any light or fan and do not strike a match ; open doors and windows, close the cylinder valve, remove the casualty into fresh air, call the fire brigade/gas agency
Sewer gas (hydrogen sulphide)	Manholes, septic tanks, drains; smell of rotten eggs which quickly deadens the sense of smell; causes instant collapse	Never enter a manhole/tank to rescue without breathing apparatus and a lifeline — many would-be rescuers die; call the fire brigade; ventilate; haul the casualty out with a rope and harness; give oxygen and CPR
Smoke (fire)	Hot smoke, soot, CO and cyanide; burns of the airway. Features: soot round the mouth and nose, singed nasal hairs, hoarse voice, barking cough, black sputum, breathlessness	Move to fresh air (crawl low under smoke); sit the casualty up; oxygen ; watch for airway swelling — transfer urgently; treat burns
Chlorine, ammonia, nitrous fumes, pesticide fumes	Industrial leaks, swimming-pool chemicals, cold storages, spraying in closed spaces; intense irritation of eyes and throat, coughing, choking, pulmonary oedema	Move upwind and uphill; wash eyes and skin with plenty of water; sit the casualty up; oxygen ; strict rest and hospital (oedema may be delayed 6–24 hours)

Gas / fume	Source and features	First aid
Confined spaces — wells, silos, tanks, holds, mines	Air deficient in oxygen or full of CO ₂ /methane; a candle will not burn in it	Never enter alone or without breathing apparatus and a lifeline. Test the atmosphere, ventilate, wear a harness, have two persons outside; remove the casualty and resuscitate in the open air

RESCUER SAFETY IN GAS ATMOSPHERES

- **More rescuers die in confined-space accidents than victims.** Never enter a well, tank, manhole or gas-filled room without **breathing apparatus, a safety harness and lifeline, and at least two attendants outside.**
- Take a **deep breath and hold it** only for a very brief snatch rescue close to the entrance — never rely on it.
- **Do not use a naked flame** to test or light a gas-filled space; switch nothing on or off.
- Bring the casualty into **fresh air** before beginning any treatment.

10.9 Traumatic Asphyxia and Other Causes

Traumatic asphyxia (crush asphyxia) — Prolonged crushing of the chest or abdomen — a wall, vehicle, earth fall or **crowd stampede**. Signs: **deep blue-purple congestion of the face, neck and upper chest with bloodshot eyes**, while the skin below the crush is normal. Treatment: relieve the pressure, note the time, give oxygen, treat for shock and crush injury, urgent transfer.

Choking — The commonest cause of sudden asphyxia in a conscious person — see Chapter 4 (back blows, abdominal thrusts, CPR if unconscious).

Anaphylaxis / burns of the airway — Swelling of the tongue and throat closes the airway — **adrenaline** and urgent transfer; sit the casualty up.

Asthma attack — Spasm and swelling of the bronchi — **difficulty in breathing out, wheeze, use of accessory muscles, inability to speak in sentences**. First aid: **sit the casualty up leaning slightly forward**, keep calm, help him use his own **reliever (blue salbutamol) inhaler — 2 puffs, repeat every 2 minutes up to 10 puffs**, loosen clothing, fresh air; call the ambulance if there is no relief in 5–10 minutes, if speech is impossible, or if the casualty becomes exhausted or blue. **Never make an asthmatic lie flat.**

Croup / diphtheria (children) — Barking cough, hoarse voice, noisy breathing (stridor) — sit the child up, calm him, provide humid air (steam from a hot shower in the bathroom), **never examine the throat with a spoon**, urgent medical help.

Hyperventilation — Over-breathing from anxiety or panic — dizziness, tingling of the hands and lips, cramps of the hands and feet (tetany), a feeling of suffocation. First aid: **be firm and reassuring, lead the casualty to a quiet place, ask him to breathe slowly (e.g. in for 4, out for 6)**; breathing into cupped hands may help. **Do not use a paper bag** (obsolete and dangerous) and always exclude a genuine cause such as asthma, diabetes or a heart problem.

High altitude / mountain sickness — Thin air — headache, breathlessness, nausea, sleeplessness; **descend at once**, rest, oxygen if available.

10.10 Chapter Recap — One-Line Revision

- Asphyxia = **interference with breathing causing oxygen lack and CO₂ accumulation**; death in **3–5 minutes**.
- Six groups of causes: **airway obstruction, smothering, chest compression, bad atmosphere, paralysis of the respiratory centre, and lung disease**.
- Three stages: **dyspnoea → convulsions → exhaustion/apnoea**; cardinal signs are **noisy difficult breathing, cyanosis, congested face and froth at the mouth**.
- Treatment principle: **remove the cause → clear and open the airway → fresh air/oxygen → artificial respiration/CPR → recovery position → hospital**.
- In **carbon monoxide** poisoning the skin is **cherry-red**, not blue; CO binds haemoglobin **200–250 times** more strongly than oxygen.
- Drowning: rescue order **reach-throw-row-go**; give **5 rescue breaths first**; **never try to drain water from the lungs**; always hospitalise (**secondary drowning**).
- Hanging/strangulation: **take the weight, cut above the knot, preserve the knot, suspect a neck injury, inform the police**.
- **Never enter a well, tank or manhole** without breathing apparatus and a lifeline.
- Asthma: **sit up, own blue inhaler 2 puffs**, never lie flat; hyperventilation: reassure and slow the breathing, no paper bag.

CHAPTER 11

Fractures and Dislocations

Types and signs of fractures, general and special management, splints and immobilisation, fractures of individual bones, dislocations, sprains and strains.

SYLLABUS COVERED IN THIS CHAPTER

- Fractures — definition, causes, classification and types, signs and symptoms, complications and complete first-aid management; splints and immobilisation.
- Fractures of the skull, jaw, spine, ribs, collar bone, arm, forearm, hand, pelvis, thigh, knee, leg and foot.
- Dislocation — definition, common sites, signs, management; sprain and strain and the RICE treatment; difference between fracture, dislocation, sprain and strain.

11.1 Definition and Causes of Fracture

DEFINITION

- A **fracture** is a **break or crack in the continuity of a bone** (or of a cartilage).
- **Causes — three mechanisms:** (1) **Direct violence** — the bone breaks at the exact spot where the force is applied (a blow, a bullet, a wheel passing over a limb). (2) **Indirect violence** — the bone breaks at a distance from the point of impact (falling on the outstretched hand breaks the **collar bone or wrist**; falling from a height on the feet fractures the **spine**). (3) **Muscular action** — a violent muscle pull snaps the bone (fracture of the **patella** or of a vertebral process while throwing, or during a fit).
- Contributory factors: **old age and osteoporosis, rickets, bone tumour/cyst, long-standing disease (pathological fracture), repeated stress (stress fracture in athletes and soldiers), and the softness of a child's bone (greenstick).**

11.2 Types of Fracture

Type	Description	Significance
Simple (closed) fracture	The bone is broken but the skin over it is intact — no wound communicating with the fracture	Less risk of infection; still bleeds internally
Compound (open) fracture	There is a wound leading down to the fracture , or the bone end protrudes through the skin; may also communicate with a body cavity	More dangerous — severe bleeding and a high risk of infection, including tetanus and gas gangrene
Complicated fracture	The fracture is associated with injury to an important structure — a large blood vessel, nerve, joint, or an organ such as the brain, lung, liver, spleen, bladder or spinal cord	Needs the most urgent hospital care; may cause paralysis or internal bleeding
Comminuted fracture	The bone is broken into more than two pieces (crushed/splintered)	Common in crush and road injuries; difficult to set
Greenstick fracture	An incomplete break — the bone bends and cracks on one side only, like a green twig	Occurs only in children , whose bones are soft and elastic
Impacted fracture	One broken end is driven firmly into the other	Common at the neck of the femur and the wrist in the elderly; may allow limited use, so it is easily missed
Depressed fracture	A piece of bone is driven inwards	Typically of the skull — presses on the brain
Transverse / oblique / spiral fracture	The line of break runs straight across / obliquely / in a spiral round the bone	Spiral fractures follow a twisting force
Avulsion fracture	A fragment is pulled off by a tendon or ligament	Seen at the ankle, finger and pelvis
Pathological fracture	A diseased bone (tumour, cyst, tuberculosis, osteoporosis) breaks with little or no force	Suspect in the elderly or in cancer patients
Stress (fatigue) fracture	A hairline crack from repeated overuse	Long marches, running, sports — the metatarsals and tibia
Fissured / hairline fracture	A mere crack, with no displacement	Easily missed on examination
Fracture-dislocation	A fracture close to a joint with dislocation of that joint	Shoulder, ankle, hip — needs hospital reduction

MOST-ASKED DISTINCTIONS

- Skin **intact** = **simple/closed**; skin **broken over the fracture** = **compound/open** — the most dangerous because of infection and bleeding.
- Bone in **more than two pieces** = **comminuted**.
- Fracture **only in children, bone bent and cracked on one side** = **greenstick**.
- Fracture with **injury to a nerve, vessel or organ** = **complicated**.
- One end **driven into the other** = **impacted**; piece **driven inwards** = **depressed** (skull).

11.3 Signs and Symptoms of a Fracture

Learn the table below; of all the signs, **deformity, crepitus, unnatural movement and loss of function are the most significant**, while **localised tenderness is the most constant**.

Sign / symptom	Explanation
Pain and tenderness	Sharp pain at the site, made worse by movement; localised tenderness on gentle pressure is the most reliable sign
Loss of power / function	The casualty cannot use the part, cannot bear weight or lift the limb
Swelling and bruising	From bleeding into the tissues; appears within minutes to hours and may mask deformity
Deformity	The limb looks bent, twisted, shortened or out of the normal line ; compare with the sound limb
Crepitus (bony grating)	A grating sound or feeling when the broken ends rub — never test for it deliberately , it causes pain and further damage
Irregularity	A step, gap or bony point felt along the bone; the bone end may be visible in an open fracture
Unnatural movement	Movement where there should be none
Signs of shock	Pale, cold, clammy skin with a rapid weak pulse — marked in fractures of the femur and pelvis
History / the casualty's own account	A snap may have been heard or felt; the mechanism of injury suggests the fracture

REMEMBER

- **Always suspect a fracture** if the mechanism of injury or the casualty's account suggests one — **treat as a fracture until an X-ray proves otherwise**.
- A fracture is **not** necessarily painful at first (shock, alcohol, an impacted fracture), and the casualty may even walk on a fractured bone.
- **Never move a casualty to look for a fracture** and never test the movement of a suspected fracture.

11.4 Complications of Fractures

Immediate / early	Late
Severe bleeding — a fractured femur can bleed 1–2 litres, the pelvis 2–3 litres	Delayed union, non-union or mal-union of the bone
Shock — from blood loss and pain	Stiffness of joints, muscle wasting, shortening of the limb
Injury to nerves — numbness, tingling, paralysis (e.g. radial nerve with a humerus fracture)	Avascular necrosis (death of bone — head of the femur, scaphoid)
Injury to blood vessels — loss of pulse, cold pale limb — gangrene	Osteomyelitis (bone infection) after an open fracture
Injury to organs — brain (skull), lung (ribs), bladder and urethra (pelvis), spinal cord (spine)	Myositis ossificans, arthritis of the neighbouring joint
Infection, tetanus and gas gangrene in open fractures	Volkmann's ischaemic contracture (after elbow fractures)
Fat embolism — fat from the marrow enters the blood, causing breathlessness and confusion 24–72 hours later	Compartment syndrome — increasing pain, tense swelling, numbness, loss of pulse; an emergency
Crush syndrome and kidney failure after prolonged compression	Deep-vein thrombosis and pulmonary embolism from immobility

11.5 General First-Aid Management of Fractures

GOLDEN RULES OF FRACTURE FIRST AID

- **'Treat the casualty where he lies'** — do not move him unless there is danger, and never until the fracture is immobilised.
- **Steady and support** the injured part with your hands at once, above and below the fracture, until it is immobilised.
- **Immobilise the fracture before moving the casualty**; immobilise **the joints above and below** the fracture.
- **Never attempt to reduce (set) a fracture** or push a protruding bone back.
- **Never test for crepitus**, never make the casualty walk or move the part to 'see if it is broken'.
- **Treat bleeding and cover open wounds first**, then immobilise.
- **Nothing by mouth** — surgery and anaesthesia are likely.

1. **Reassure** the casualty and tell him to keep still.
2. **Steady and support** the injured limb with your hands; a helper maintains gentle support throughout.
3. **Check and treat the airway, breathing and severe bleeding first** (a fracture is never treated before a life-threat).
4. **Open (compound) fracture**: cover the wound with a **sterile dressing**, control bleeding by pressure **around** the wound, place a **ring pad** round any protruding bone and bandage over the pads — **never press on the bone end** and never push it in.
5. **Immobilise** the fracture in the most comfortable position: by **bandaging the injured limb to a sound part of the body** (arm to chest, injured leg to the sound leg) or by applying a **splint**.
6. Apply **broad-fold bandages above and below the fracture** — **never over it**; tie the knots on the **uninjured side** (or over the splint).
7. **Pad hollows and bony points** (ankle, knee, armpit, between the legs) with cotton wool, folded cloth or a small cushion.
8. **Raise and support** the injured part after immobilising, where possible, to reduce swelling; apply a **cold compress** over a closed fracture through cloth.
9. **Check the circulation beyond the bandages** immediately and every 10 minutes — feel the pulse, look for pallor, blueness, coldness, tingling and numbness; **loosen at once** if any of these appear.
10. **Treat for shock** — lie the casualty down, keep him warm, reassure; do **not** raise the legs if a leg is fractured.
11. **Transport** carefully, on a stretcher for lower-limb, pelvic, spinal or multiple fractures; record the time and findings, and hand over to the doctor.

Splints — the rules of immobilisation

- A **splint** is a rigid or semi-rigid support applied to hold a fractured bone still. **Types**: wooden (Thomas splint, Cramer wire, Bohler), inflatable (air splint), moulded plastic/vacuum splints, cardboard, and the **body's own 'anatomical splint'** (the sound limb or the trunk).
- **Improvised splints**: umbrella, walking stick, broom, cricket bat, bamboo, rolled newspaper or magazine, thick cardboard, wooden plank, pillow, folded blanket, rifle, hockey stick.
- A **splint must be**: (a) **rigid**, (b) **well padded**, (c) **long enough to immobilise the joints above and below the fracture**, and (d) **applied over clothing** and secured firmly with broad bandages — but not so tightly as to stop circulation.
- **Never apply a bandage directly over the fracture site**; use at least **two** ties, one above and one below the fracture.
- Check the **colour, warmth, sensation and pulse** of the fingers/toes before and after splinting.
- In the field, **body bandaging** (tying the injured limb to the trunk or to the sound limb) is usually quicker and sufficient.

11.6 Fractures of Individual Bones

Fracture	Recognition	First aid
Skull (vault and base)	Wound or bruise of the scalp, soft/boggy swelling or a depression; bleeding or clear straw-coloured fluid (CSF) from the ear or nose; bruising round the eyes ('panda eyes') or behind the ear (Battle's sign) ; blood-shot eye; unconsciousness, unequal pupils, vomiting, fits; deteriorating level of response	Do not press on the injury; use a ring pad for scalp bleeding; if unconscious and breathing, recovery position with the bleeding ear downwards ; never plug the ear or nose ; keep the head and neck in line; nothing by mouth; monitor level of response every 10 minutes; urgent hospital transfer
Lower jaw (mandible)	Pain on speaking and swallowing, dribbling of blood-stained saliva, irregular teeth or a step in the jaw line, difficulty in opening or closing the mouth	Do not bandage the jaw if the casualty is drowsy or bleeding into the mouth (risk of choking); let him support the jaw with a soft pad in his own hand; if fully conscious apply a narrow-fold bandage under the chin and over the top of the head ; sit him up with the head forward to let blood and saliva drain; nothing by mouth

Fracture	Recognition	First aid
Spine (vertebral column) — the most dangerous fracture	Fall from a height, diving accident, heavy blow on the back, road accident. Pain in the neck or back, tenderness over the spine; loss of feeling, tingling or 'pins and needles'; weakness or paralysis of the limbs; inability to move the legs; loss of control of bladder and bowel; difficulty in breathing (high cervical injury)	Do NOT move the casualty at all unless life is in danger. Steady and support the head in the neutral position with both hands (manual in-line stabilisation), place rolled blankets/sandbags along the body, maintain the airway with a jaw thrust , and wait for the ambulance. If the casualty must be turned or lifted, use the log-roll with 4–5 helpers and a scoop/spine board , keeping the head, neck and trunk in one straight line
Ribs (and flail chest)	Sharp pain at the site, worse on breathing, coughing or moving; shallow breathing; tenderness; in multiple fractures, paradoxical breathing (the injured part moves in on inspiration and out on expiration); coughing of frothy blood means lung injury	Simple fracture: support the arm of the injured side in an arm sling and let the casualty sit in the position he finds easiest ; do not strap the chest tightly. Multiple/flail/penetrating injury: half-sitting, leaning towards the injured side , cover any open wound (three-sided dressing), oxygen if trained, urgent transfer; nothing by mouth
Collar bone (clavicle) — the commonest fracture	Fall on the shoulder or outstretched hand; pain, the casualty supports the elbow of the affected side and leans the head towards it ; swelling or a step felt over the collar bone; shoulder appears dropped	Place the arm of the injured side in an elevation sling (or a St John's arm sling if elevation is too painful), and secure the arm to the chest with a broad-fold bandage tied on the sound side; pad the armpit
Upper arm (humerus)	Pain, swelling, deformity of the upper arm; the arm cannot be raised; may injure the radial nerve (wrist drop)	Pad the armpit, support the forearm in a collar-and-cuff sling so that the weight of the arm acts as traction, and secure the upper arm to the chest with a broad-fold bandage above and below the fracture; a padded splint may be added on the outer side
Forearm (radius and ulna) and wrist	Pain, swelling, deformity; a 'dinner-fork' deformity at the wrist (Colles' fracture) in the elderly after a fall on the outstretched hand	Support the forearm with a well-padded splint from elbow to fingers (rolled newspaper/magazine is ideal), bandage above and below, and place the arm in an arm sling with the hand slightly raised ; remove rings and bangles
Hand and fingers	Swelling, bruising, deformity, inability to move the fingers; often crush injury	Remove rings at once; place a soft pad in the palm, wrap the hand in a soft dressing and support in an elevation sling ; a finger may be strapped to its neighbour ('buddy strapping')
Pelvis	Crush injury or fall; pain in the groin, hip or back, worse on movement; inability to stand or walk; blood-stained urine or inability to pass urine (bladder/urethral injury); severe shock from heavy internal bleeding	Do not move unnecessarily and do not test the movement of the legs . Lay the casualty flat on his back , place padding between the knees and tie the knees and ankles together with broad and narrow bandages, or use a pelvic binder/folded sheet round the pelvis; treat for shock ; nothing by mouth; stretcher transport only
Thigh (femur)	Severe pain, marked deformity with the leg turned outwards and shortened , inability to move the leg, severe shock from internal bleeding of 1–2 litres	Steady the limb with gentle traction if trained; place the sound leg beside it with padding between the legs; tie figure-of-eight at the ankles and feet, then broad bandages above and below the fracture, at the knees and thighs ; ideally apply a long splint from the armpit to beyond the foot (Thomas splint in ambulance use); treat shock; stretcher transport
Knee and patella	Pain, swelling, inability to straighten or bear weight; the kneecap may be felt in two pieces with a gap	Do not force the knee straight — immobilise in the position found, with a padded splint behind the leg from buttock to heel; raise and support the limb; cold compress; stretcher transport
Leg (tibia and fibula)	Pain, swelling, deformity; often an open fracture because the tibia lies just under the skin	Dress any wound first; splint from above the knee to beyond the foot, or bandage to the sound leg with padding between; raise the limb; stretcher transport
Ankle and foot	Pain, swelling, bruising, inability to bear weight; difficult to distinguish from a severe sprain	Remove the shoe only if it can be done easily; support the foot and ankle with a pillow or blanket splint bandaged in figure-of-eight turns ; raise the limb; cold compress; do not let the casualty walk

11.7 Dislocation

DEFINITION

- A **dislocation (luxation)** is the **displacement of one or more bones from their normal position at a joint**, with tearing or stretching of the capsule and ligaments.
- A **subluxation** is a **partial or incomplete** dislocation.
- Caused by **indirect violence, a wrench or a sudden muscular action**; it may **recur** easily once the ligaments have been stretched (recurrent dislocation of the shoulder).

- **Commonest sites:** the **shoulder** (the most frequently dislocated joint, because it is the most mobile), then the **fingers and thumb, jaw (mandible), elbow, patella, hip** (needs great force) and the **vertebrae** (very dangerous — may damage the cord).

- A **pulled elbow (radial head subluxation)** is common in small children who are jerked up by one arm.

Signs and symptoms of dislocation

- **Severe sickening pain** at the joint, and a feeling that the joint is 'out'.
- **Marked deformity** — the joint looks abnormal, with an unusual prominence or hollow; compare with the other side.
- **Complete loss of movement (fixed joint)** — movement is impossible and any attempt causes intense pain.
- **Swelling, bruising and tenderness** around the joint; the limb may look **longer or shorter**.
- **No crepitus** and **no unnatural mobility** (unlike a fracture); possible numbness or loss of pulse if a nerve or vessel is compressed.

First aid for a dislocation

1. **Never attempt to reduce (put back) a dislocation** — nerves and vessels may be trapped, and there may also be a fracture.
2. **Support and immobilise the joint in the most comfortable position** — an arm sling or collar-and-cuff for the shoulder or elbow; a padded splint for a knee; soft padding and a bandage for a finger.
3. Use a **cold compress/ice pack wrapped in cloth** for 10–20 minutes to reduce pain and swelling.
4. **Do not move the joint** and do not let the casualty try to use it.
5. **Check the circulation and sensation** beyond the joint; loosen any bandage that tightens with swelling; remove rings.
6. **Nothing by mouth** (an anaesthetic is likely), treat for shock, and arrange hospital transfer for X-ray and reduction.

Fracture versus dislocation — comparison

Point	Fracture	Dislocation
What is injured	The bone is broken	The joint — bone ends are displaced
Site	Anywhere along a bone	Only at a joint
Deformity	Present, along the bone; limb may be shortened	Marked, at the joint , with abnormal contour
Movement	Painful; unnatural movement possible	Movement impossible — joint fixed
Crepitus	Present (do not test for it)	Absent
Pain	Severe, at the fracture site	Severe, sickening, at the joint
Length of limb	Usually shortened	May be lengthened or shortened
Recurrence	Does not recur after healing	May recur (habitual dislocation)
Treatment	Immobilise; do not reduce	Immobilise in a comfortable position; do not reduce

11.8 Sprain and Strain

	Sprain	Strain
Definition	Injury (stretching or tearing) of the ligaments and tissues round a joint , caused by a sudden wrench or twist	Over-stretching or tearing of a muscle or its tendon from a sudden violent or unaccustomed effort
Common sites	Ankle (commonest), wrist, knee, thumb, shoulder	Back, thigh (hamstring), calf, shoulder
Signs	Pain and tenderness around the joint , swelling, bruising, difficulty and pain on movement, but some movement is possible	Sudden sharp pain in the muscle , tenderness, stiffness, cramp, swelling, bruising, loss of power
First aid	RICE (below) and referral if severe — it may really be a fracture	RICE , rest of the muscle, gentle support, gradual return to activity

THE RICE (OR PRICE / RICER) TREATMENT — FOR ALL SPRAINS, STRAINS AND BRUISES

- **P – Protect** the injured part from further injury.
- **R – Rest** the part; stop the activity at once; do not walk on a sprained ankle.
- **I – Ice**: apply a cold compress or an ice pack **wrapped in a cloth (never directly on the skin)** for **10–20 minutes, repeated every 2–3 hours** for the first 24–48 hours.
- **C – Compression**: apply a **crepe/elastic bandage** in figure-of-eight turns, firm but not tight, extending well above and below the joint.
- **E – Elevation**: raise and support the limb **above the level of the heart** to reduce swelling.
- **R – Referral** for X-ray if pain is severe, weight-bearing is impossible, there is marked deformity or swelling, or

there is no improvement.

- **Avoid H-A-R-M for the first 48–72 hours:** Heat, Alcohol, Running/exercise and Massage — all increase bleeding and swelling.

Quick comparison of the four bone-and-joint injuries

Injury	Structure damaged	Key feature
Fracture	Bone	Crepitus, deformity, unnatural movement, point tenderness
Dislocation	Joint (bone ends displaced)	Joint fixed and deformed, no crepitus
Sprain	Ligament round a joint	Pain and swelling round the joint, some movement possible
Strain	Muscle or tendon	Sudden pain in the muscle during effort, cramp, weakness

11.9 Chapter Recap — One-Line Revision

- Fracture = break in the continuity of a bone; caused by **direct**, **indirect** violence or **muscular action**.
- **Simple** = skin intact; **compound/open** = wound over the fracture (most dangerous); **comminuted** = more than two pieces; **greenstick** = children only; **complicated** = nerve/vessel/organ injured.
- Signs: **pain, tenderness, swelling, deformity, loss of function, crepitus, irregularity, unnatural movement, shock** — never test for crepitus.
- Golden rules: **treat where he lies, steady and support, immobilise joints above and below, never reduce, bandage above and below — never over the fracture, check circulation, nothing by mouth.**
- **Collar bone** → elevation sling; **humerus** → collar-and-cuff sling; **forearm/wrist** → padded splint + arm sling; **femur** → long splint/tie to the sound leg + treat shock; **pelvis** → tie knees and ankles, flat, stretcher.
- **Spinal injury** → do **not** move; support the head in line; log-roll with helpers; jaw thrust for the airway.
- **Skull fracture** → CSF/blood from ear or nose, panda eyes, Battle's sign; **never plug the ear**; recovery position with the bleeding ear down.
- Dislocation = displacement at a **joint**; commonest = **shoulder**; **movement impossible, no crepitus**; never reduce it.
- **Sprain** = **ligament**; **strain** = **muscle/tendon**; both treated by **RICE**, and avoid **HARM** for 48–72 hours.

PART V • MEDICAL EMERGENCIES AND POISONING

CHAPTER 12

Unconsciousness and Fainting

Levels of consciousness, the AVPU scale and Glasgow Coma Scale, all causes of unconsciousness, general care of the unconscious casualty, fainting, head injury, stroke, diabetic emergencies and heart attack.

SYLLABUS COVERED IN THIS CHAPTER

- Unconsciousness — definition, degrees/levels, causes, signs and symptoms, general first-aid management, monitoring and transport.
- Fainting — causes, warning signs, treatment and prevention; recovery position; special causes of unconsciousness — head injury (concussion and compression), stroke, diabetic coma and hypoglycaemia, heart attack and infantile convulsions.

12.1 Consciousness and Unconsciousness

DEFINITIONS

- **Consciousness** is the state of being **awake and aware of oneself and one's surroundings**, and of responding normally to stimuli. It depends on an intact **brain (cerebrum + brain stem)** receiving a constant supply of **oxygen and glucose**.
- **Unconsciousness** is a condition in which the casualty is **insensible — unaware of himself and of his surroundings — and does not respond to stimuli such as speech, touch or pain**, because of interruption of normal brain activity.
- **Coma** is deep, prolonged unconsciousness from which the casualty cannot be roused.
- **Why it is dangerous:** the unconscious casualty **loses the cough and swallowing reflexes and muscle tone**, so the **tongue falls back** and blocks the airway, and vomit or blood may be inhaled. **Unconsciousness itself can kill through the airway** — which is why airway care always comes first.

12.2 Levels (Degrees) of Consciousness

Level	Description
Fully conscious (alert)	Awake, aware, oriented in time, place and person; answers questions correctly
Confused / disoriented	Awake but muddled, cannot answer correctly, restless
Drowsy (somnolent)	Sleepy but can be roused by speech, and then answers slowly
Stupor	Roused only by painful stimuli; responds by groaning or withdrawing
Coma (unconscious)	Cannot be roused at all; no response to voice or pain; reflexes are lost

The AVPU scale — the first aider's quick check

Letter	Meaning	How tested
A	Alert	Eyes open spontaneously; talks normally
V	Responds to Voice	Opens the eyes or moves when spoken to loudly
P	Responds to Pain	Moves or groans only when pinched on the ear lobe/shoulder or when the nail bed is pressed
U	Unresponsive	No response of any kind — a true coma

The Glasgow Coma Scale (GCS) — for reference

Eye opening (E) — 4	Verbal response (V) — 5	Motor response (M) — 6
4 Spontaneous	5 Oriented	6 Obeys commands
3 To speech	4 Confused conversation	5 Localises pain
2 To pain	3 Inappropriate words	4 Withdraws from pain
1 No eye opening	2 Incomprehensible sounds	3 Abnormal flexion
	1 No verbal response	2 Abnormal extension
		1 No motor response

- **Total = 3 (worst) to 15 (normal).** GCS **13–15** = mild, **9–12** = moderate, **8 or less** = **severe head injury/coma** (such a casualty cannot protect his own airway).
- A **falling level of response** is the most important single sign of a worsening head injury — record it with the **time** every 10 minutes.

12.3 Causes of Unconsciousness

MNEMONIC 1 — 'FISH SHAPED' (ST JOHN AMBULANCE)

- **F** – Fainting
- **I** – Infantile convulsions
- **S** – Shock
- **H** – Head injury
- **S** – Stroke (cerebro-vascular accident)
- **H** – Heart attack
- **A** – Asphyxia
- **P** – Poisoning (including alcohol and drugs)
- **E** – Epilepsy
- **D** – Diabetes (hypoglycaemia or diabetic coma)

MNEMONIC 2 — 'AEIOU-TIPS' (MEDICAL PRACTICE)

- **A** – Alcohol and drug intoxication
- **E** – Epilepsy, Electrolyte disorder, Encephalopathy
- **I** – Insulin (hypoglycaemia) and diabetic coma
- **O** – Opium/Overdose of drugs, Oxygen lack
- **U** – Uraemia (kidney failure) and other metabolic causes
- **T** – Trauma (head injury), Tumour, Temperature (heat stroke, hypothermia)
- **I** – Infection (meningitis, malaria, encephalitis, septicaemia)
- **P** – Poisoning, Psychiatric (hysteria)
- **S** – Stroke, Shock, Syncope, Seizure

12.4 General Management of an Unconscious Casualty

Whatever the cause, the **treatment of unconsciousness is the same**. The priority is the **airway**.

1. **Check for danger**, then check the **response** (shake and shout, AVPU).
2. **Shout for help; ask a bystander to dial 112/108.**
3. **Open the airway** — head tilt and chin lift (jaw thrust if a spinal injury is suspected); **clear the mouth** of vomit, blood, loose dentures or food.
4. **Check breathing** for up to **10 seconds**. If absent or abnormal — **start CPR**. If present — go on.
5. **Control any severe bleeding** and treat obvious serious injuries.
6. **Place the casualty in the recovery position** (unless a spinal injury is suspected, when the head is held in line and the casualty is log-rolled only with helpers).
7. **Loosen tight clothing** at the neck, chest and waist; remove spectacles and dentures.
8. **Keep the casualty warm** with a blanket over and under him, and protect him from the sun and rain.
9. **Give NOTHING by mouth** — no water, no food, no tablets; the casualty cannot swallow and will choke.
10. **Do not leave the casualty alone**; do not attempt to rouse him by slapping, shaking, shouting, or throwing water on the face; do not give smelling salts to an unconscious casualty.
11. **Look for clues to the cause** — medical bracelet or card (diabetes, epilepsy, allergy), medicines or an inhaler in the pocket, needle marks, smell of the breath (alcohol, acetone, kerosene, insecticide), injuries, pupil size, and the story of bystanders. Keep any container, tablet strip or vomit for the doctor.
12. **Monitor and record every 10 minutes** — level of response (AVPU), breathing rate, pulse rate, pupils, colour and temperature; note the **time** of every change.
13. **Arrange urgent transport** to hospital in the recovery position, on a stretcher, and hand over a written report.

THE FOUR 'NEVERS' OF UNCONSCIOUSNESS

- **Never** give anything by mouth.
- **Never** leave the casualty lying on his back, unattended.

- **Never** try to wake him up forcibly, and never put a pillow under the head (it bends the neck and blocks the airway).
- **Never** assume alcohol is the cause — a 'drunk' may in fact have a head injury, hypoglycaemia or a stroke.

12.5 Fainting (Syncope)

DEFINITION

- **Fainting** is a **brief and temporary loss of consciousness caused by a momentary reduction of the blood supply to the brain**, from which the casualty recovers quickly and completely when he lies down.
- Mechanism: a **reflex (vagal) slowing of the heart with dilatation of the blood vessels**, so that blood pools in the legs and abdomen and the brain is briefly starved of oxygen — also called **psychogenic, primary or neurogenic shock**.

Causes of fainting

- **Emotional** — fear, fright, anxiety, bad news, the **sight of blood or of an injection**, severe pain.
- **Postural** — **standing still for a long time** (parades, queues, guard duty), or **standing up suddenly** after lying or sitting long (postural hypotension), especially in the elderly.
- **Environmental** — a **hot, crowded, ill-ventilated room**, prolonged sun exposure.
- **Physical** — **hunger, fatigue, exhaustion, dehydration, sleeplessness, low blood pressure, anaemia**, after an illness or a hot bath, straining at stool, severe coughing, pregnancy.
- **Medical** — heart disease and irregular rhythms, blood loss, hypoglycaemia, some medicines (for blood pressure), and after taking alcohol.

Signs and symptoms

Warning (premonitory) symptoms	Signs during the faint
Giddiness and light-headedness; 'everything going black'	Pale, cold, moist (clammy) skin ; sweating
Blurring or dimness of vision, spots before the eyes	Slow, weak pulse (this is characteristic — compare with the rapid pulse of true shock)
Ringling in the ears, nausea, yawning	Shallow breathing; brief loss of consciousness (seconds to about 2 minutes)
Weakness of the legs, a feeling of falling	The casualty falls to the ground — the fall itself may cause injury; there may be a few twitches
Sweating, feeling hot then cold	Rapid, complete recovery once flat , with no confusion afterwards

First aid for fainting

1. **If the casualty feels faint** (still conscious): make him **sit or lie down at once** and **raise the legs above the level of the heart** (20–30 cm), or sit him with the **head bent forward between the knees**.
2. **If the casualty has fainted**: lay him flat on his back, **raise and support the legs**.
3. **Ensure plenty of fresh air** — open windows and doors, ask onlookers to move back, fan the casualty.
4. **Loosen tight clothing** at the neck, chest and waist.
5. **Check breathing**; if it is absent or abnormal, treat as cardiac arrest and start CPR. If the casualty does not regain consciousness within **1–2 minutes**, treat him as an **unconscious casualty** — recovery position and call an ambulance.
6. **On recovery**, reassure him, help him to sit up **gradually**, and give **sips of cool water** (sweet drinks if he is hungry or diabetic). Do not let him stand up suddenly or walk away at once.
7. Look for and treat any **injury caused by the fall**, and find out the cause.
8. **Refer to a doctor** if the faint was prolonged, if there were convulsions, if it happened while lying down or during exercise, if there was chest pain or palpitation, if the casualty is elderly or pregnant, or if fainting is recurrent — these suggest a heart or brain cause.

PREVENTION OF FAINTING — AND WHAT NOT TO DO

- **Do not** crowd round the casualty, do not raise the head, do not make him sit or stand, and **do not throw water on the face** or slap him.
- **Do not** give anything by mouth while he is unconscious.
- Prevention: avoid standing still for long (flex the calf muscles and shift weight), get up slowly, eat and drink

regularly, avoid hot crowded places, sit down at the first warning sign.

12.6 Head Injury — Concussion and Compression

Point	Concussion (cerebral concussion)	Compression (cerebral compression)
Nature	'Shaking up' of the brain — a temporary and reversible disturbance of brain function	Pressure on the brain from bleeding, a depressed fracture, swelling or a tumour — serious and progressive
Onset	Immediately after the blow	May develop minutes, hours or even days after the injury (after a lucid interval)
Consciousness	Brief loss (seconds to a few minutes) with complete recovery	Deteriorating level of response — drowsy → confused → unconscious
Pupils	Equal and reacting	Unequal — one pupil dilated and not reacting to light
Pulse	Rapid and weak at first, becoming normal	Slow and full/bounding
Breathing	Shallow	Noisy, snoring, slow and irregular
Face and skin	Pale, cold, clammy	Flushed, hot, dry ; temperature rises
Other features	Giddiness, nausea, vomiting, headache, loss of memory of the event (amnesia) , mild confusion — the casualty may say 'what happened?' repeatedly	Severe headache, vomiting, fits, weakness or paralysis on one side, rising blood pressure
Outcome	Usually recovers fully, but must be watched for 24 hours and taken to hospital if any warning sign appears	A surgical emergency — needs immediate hospital treatment

First aid in head injury

1. Assume that **every casualty with a head injury may also have a neck (cervical spine) injury** — support the head in line with the body and do not move him unnecessarily.
2. Maintain the **airway and breathing**; place an unconscious but breathing casualty in the **recovery position** with support to the head and neck; be ready for CPR.
3. **Control scalp bleeding** with direct pressure over a dressing (use a **ring pad** if a depressed fracture is suspected).
4. **Never plug the ear or nose** if blood or clear fluid escapes; cover lightly and let it drain, with the affected ear **downwards**.
5. **Give nothing by mouth**; do not give aspirin (it increases bleeding).
6. **Record the level of response (AVPU/GCS), pupils, pulse and breathing every 10 minutes** with the time — this record is invaluable to the doctor.
7. Arrange **urgent hospital transfer** for anyone who has been unconscious, has vomited more than once, has a severe headache, fits, weakness, unequal pupils, bleeding from the ear/nose, a wound of the scalp, or who is on blood-thinning medicine, elderly or intoxicated.
8. Advise a casualty with even a minor head injury to be **observed for 24–48 hours** and to return at once if headache, vomiting, drowsiness, double vision or confusion develops.

12.7 Stroke (Cerebro-Vascular Accident)

- **Stroke** = sudden interruption of the blood supply to a part of the brain, either by a **clot (ischaemic — about 80 %)** or by **bleeding (haemorrhagic — about 20 %)**. A **TIA (transient ischaemic attack, 'mini-stroke')** produces the same signs, which clear within 24 hours — but it is a serious warning.
- Risk factors: **high blood pressure (the most important)**, diabetes, smoking, high cholesterol, heart disease, obesity, alcohol, age, family history.

RECOGNITION — THE 'F-A-S-T' TEST

- **F – Face**: has the face **fallen on one side**? Can the casualty smile? Is the mouth or eye drooping?
- **A – Arms**: can he **raise both arms** and keep them up?
- **S – Speech**: is his speech **slurred or muddled**? Can he understand you?
- **T – Time**: **Time to call 112/108 at once** — note the **time when the symptoms began** (clot-dissolving treatment works only within about 4½ hours).
- Other signs: sudden **weakness or numbness of one side**, loss of vision, severe sudden headache ('the worst headache of my life'), giddiness, loss of balance, confusion, dribbling, loss of bladder control, unequal pupils, unconsciousness.

1. **Call an ambulance immediately** and say that you suspect a stroke.
2. If conscious: lay the casualty down with the **head and shoulders slightly raised and supported**, head turned to the side; loosen clothing; reassure him — he may understand everything even if he cannot speak.
3. If unconscious but breathing: **recovery position, paralysed side uppermost**.
4. **Nothing by mouth** — not even water or medicine (swallowing is unsafe).
5. Wipe away dribbling; keep the airway clear; monitor every 10 minutes; be ready for CPR.
6. **Do not** give aspirin (unlike a heart attack) — the stroke may be haemorrhagic.

12.8 Diabetic Emergencies

Point	Hypoglycaemia (low blood sugar) — the true emergency	Hyperglycaemia / diabetic ketoacidosis (high blood sugar)
Cause	Too much insulin , missed meal, unaccustomed exercise, alcohol, vomiting	Too little insulin , infection, over-eating, undiagnosed diabetes
Onset	Sudden — minutes	Gradual — hours to days
Skin	Pale, cold, profusely sweating	Warm, dry, flushed ; dry tongue
Breathing	Normal or shallow, rapid	Deep, rapid, sighing (Kussmaul) breathing with a sweet acetone/nail-polish smell of the breath
Pulse	Rapid, strong ('bounding')	Rapid and weak
Behaviour	Confused, aggressive, trembling, behaving as if drunk , hungry, headache, blurred vision, fits, then unconsciousness	Extreme thirst , frequent passing of urine, nausea and vomiting, abdominal pain, weakness, drowsiness → coma
First aid (conscious)	Give sugar at once — 3–4 teaspoons of sugar or glucose in water, a sweet drink, glucose gel, honey, 3–4 sweets or biscuits; repeat in 10–15 minutes if no improvement; then give a starchy snack. Improvement is usually dramatic	Do not give sugar . Help the casualty take his own insulin if he is able, give sips of water, and arrange hospital care urgently
First aid (unconscious)	Nothing by mouth — recovery position, call an ambulance immediately (hospital gives IV glucose/glucagon)	Recovery position, call an ambulance immediately

THE GOLDEN RULE OF DIABETIC EMERGENCIES

- If you are **not sure** whether the blood sugar is high or low in a **conscious** casualty, **give sugar**: it will save a hypoglycaemic casualty within minutes and will do little harm to a hyperglycaemic one in the short term.
- **Hypoglycaemia is the greater and more immediate danger** — it can cause permanent brain damage within an hour.
- **Never** give anything by mouth to an unconscious diabetic, and never inject insulin yourself.

12.9 Heart Attack, Angina and Cardiac Emergencies

	Angina pectoris	Heart attack (myocardial infarction)
Cause	Temporary lack of blood to the heart muscle on exertion	Sudden blockage of a coronary artery — part of the heart muscle dies
Pain	Central chest pain/tightness brought on by exertion or emotion , relieved by rest in a few minutes and by the casualty's own tablet/spray	Severe crushing, vice-like central chest pain, often at rest , lasting more than 15–20 minutes and not relieved by rest
Radiation	May spread to the left arm, jaw or neck	Spreads to the left arm (or both arms), neck, jaw, back or upper abdomen
Other signs	Breathlessness, anxiety	Ashen grey, cold clammy skin, profuse sweating, nausea and vomiting, breathlessness, giddiness, irregular pulse, a feeling of impending death ; may collapse suddenly with cardiac arrest
First aid	Stop the activity and sit the casualty down ; help him take his own nitroglycerine tablet/spray; if the pain persists beyond 10–15 minutes treat it as a heart attack	Call 112/108 at once ; sit the casualty in a half-sitting (W-position) posture with the knees bent and supported; loosen clothing; give one 300 mg dispersible aspirin to chew slowly (if conscious, not allergic, and over 16); help with his own nitroglycerine; keep him calm and still; be ready to start CPR and use an AED

- **Never** allow a casualty with chest pain to walk, drive or exert himself; **never** leave him alone.

- 'Silent' heart attacks with little or no pain occur in **diabetics, the elderly and women** — breathlessness, fainting or sudden sweating may be the only sign.
- **Cardiac arrest** (no response, no normal breathing) → immediate **CPR and AED** (Chapter 4).

12.10 Infantile (Febrile) Convulsions

- Occur in children of **6 months to 5–6 years** with a **rapidly rising fever** (from any infection).
- Signs: **violent twitching of the face and limbs, stiffness with arching of the back, clenched fists, holding of the breath with a red or blue face, rolled-up eyes, frothing at the mouth, hot flushed skin and sweating**; the child may pass urine.
- **First aid**: protect the child from injury (put soft padding around, never restrain); **do not put anything in the mouth**; remove clothing and bedding to **cool the child**; sponge with tepid (not cold) water; when the fit stops, place him in the **recovery position**; give **paracetamol** when he is fully awake and able to swallow; and **always seek medical advice**.
- **Call an ambulance** if the fit lasts more than 5 minutes, if fits recur, if the child does not regain consciousness, if it is his first fit, or if a rash or neck stiffness suggests **meningitis**.

12.11 Chapter Recap — One-Line Revision

- Unconsciousness = **insensibility with no response to stimuli**; the danger is the **airway** — the tongue falls back.
- Levels: alert → confused → drowsy → stupor → coma; quick scale **AVPU**; detailed scale **GCS (3–15; ≤ 8 = severe)**.
- Causes mnemonic = **FISH SHAPED** or **AEIOU-TIPS**.
- Management of every unconscious casualty: **airway → breathing → bleeding → recovery position → warmth → nothing by mouth → monitor every 10 minutes → hospital**.
- **Fainting** = brief loss of consciousness from reduced blood to the brain, with a **slow weak pulse**; lay flat and **raise the legs**; recovery in 1–2 minutes.
- **Concussion** = temporary, recovers, **equal pupils, rapid pulse**; **compression** = progressive, **unequal pupils, slow full pulse, noisy breathing** — a surgical emergency.
- **Stroke** → **FAST** test, note the time, nothing by mouth, **no aspirin**, recovery position with the paralysed side uppermost.
- **Hypoglycaemia** — sudden, pale, sweating, aggressive → **give sugar**; **hyperglycaemia** — gradual, dry, flushed, acetone breath, deep breathing → **hospital**.
- **Heart attack** → half-sitting, **300 mg aspirin chewed**, call 112/108, ready for CPR; **angina** is relieved by rest and the casualty's own spray.
- **Febrile convulsion** → protect from injury, cool the child, nothing in the mouth, recovery position afterwards.

CHAPTER 13

Epilepsy and Hysteria

Epilepsy — causes, types of seizure, the phases of a major fit, first aid, status epilepticus; hysteria — features, management and the classical differences from an epileptic fit.

SYLLABUS COVERED IN THIS CHAPTER

- Epilepsy — definition, causes, types (major and minor), aura, phases of a fit, signs and symptoms, complete first aid, status epilepticus and when to call an ambulance.
- Hysteria — definition, causes, signs and symptoms, first-aid management, and the difference between an epileptic and a hysterical fit.

13.1 Epilepsy — Meaning and Causes

DEFINITIONS

- **Epilepsy** is a **chronic disorder of the brain characterised by recurrent (two or more) unprovoked seizures**, caused by sudden, excessive and disorderly electrical discharges of the brain cells.
- A **seizure (fit, convulsion)** is the **event** — a sudden burst of abnormal electrical activity producing involuntary movements, abnormal sensations, or loss of consciousness. A person may have a **single provoked seizure** (fever, head injury, low sugar, poison) **without having epilepsy**.
- **Aura** — a brief warning sensation felt by some casualties just before a fit: a strange smell or taste, flashing lights, a rising feeling in the stomach, giddiness or a sense of fear. It gives the casualty time to lie down.
- **Ictal** = during the fit; **post-ictal** = the period of recovery after it.

Group of causes	Examples
Idiopathic / primary (commonest)	No demonstrable cause; often begins in childhood or adolescence; hereditary tendency
Structural brain disease	Head injury and birth injury, brain tumour, stroke, scar from an old injury or operation, cerebral palsy, congenital malformation
Infections	Meningitis, encephalitis, cerebral malaria, neurocysticercosis (pork tapeworm — a very common cause in India) , brain abscess, tuberculoma
Metabolic / toxic	Low blood sugar (hypoglycaemia) , low calcium or sodium, kidney and liver failure, alcohol and drug withdrawal , poisoning (lead, insecticides, camphor), high fever in children, eclampsia of pregnancy, oxygen lack (asphyxia, cardiac arrest)
Precipitating (trigger) factors	Missed anti-epileptic medicine (the commonest trigger), lack of sleep, fatigue, stress, flashing or flickering lights , television/video games, alcohol, menstruation, fever, hunger

13.2 Types of Seizure

Type	Features
Generalised tonic-clonic (major fit, 'grand mal')	The classical fit — sudden loss of consciousness with stiffening and then jerking of the whole body; the type described in every first-aid syllabus
Absence seizure (minor fit, 'petit mal')	Chiefly in children — a sudden, brief (5–30 second) 'blank stare' with stopping of activity, fluttering eyelids and slight twitching of the lips or fingers; no fall, no convulsion , and the child resumes what he was doing as if nothing happened. Often mistaken for day-dreaming or inattention
Myoclonic seizure	Sudden brief jerks of the arms or legs, usually without loss of consciousness
Atonic ('drop attack')	Sudden loss of muscle tone — the casualty collapses limply; injuries are common
Tonic seizure	Generalised stiffening only
Focal (partial) seizure — simple	Twitching or abnormal sensation in one part (hand, face) with consciousness preserved
Focal seizure — complex (psychomotor)	Altered awareness with automatic behaviour — lip smacking, chewing, plucking at clothes, aimless wandering; the casualty appears dazed and does not remember it
Status epilepticus	A fit lasting more than 5 minutes, or repeated fits without recovery of consciousness in between — a life-threatening emergency
Febrile convulsion	Fit provoked by high fever in a child of 6 months to 6 years — not epilepsy

13.3 Phases of a Major (Tonic-Clonic) Fit

Phase	Duration	What is seen
1. Aura (warning)	Seconds	Strange smell, taste, sound, light, giddiness, epigastric sensation or fear — present in only some casualties
2. Tonic phase	10–30 seconds	Sudden loss of consciousness and a fall (the casualty may utter a sharp 'epileptic cry' as air is forced out); the whole body becomes rigid , the back arches, the jaws clench, the breathing stops and the face and lips become blue (cyanosed) ; the pupils dilate
3. Clonic (convulsive) phase	1–2 minutes (rarely up to 5)	Violent jerking movements of the limbs and body; noisy difficult breathing; frothing or foaming at the mouth , often blood-stained from a bitten tongue ; the eyes roll up; involuntary passing of urine (sometimes stool) ; profuse sweating
4. Post-ictal (recovery) phase	Minutes to hours	The muscles relax and breathing becomes normal; the casualty is deeply unconscious, then dazed, confused, exhausted, with headache and muscle aches ; he does not remember the fit ; he may fall into a deep sleep for several hours; occasionally there is automatic behaviour or temporary weakness of a limb

13.4 First Aid During and After an Epileptic Fit

What to DO

1. **Stay calm, note the time** the fit began and stay with the casualty throughout.
2. **Protect him from injury** — clear away furniture, hot objects, sharp objects, glass and spectacles; **place something soft under the head** (a folded coat, cushion or your hands); loosen tight clothing at the neck.
3. **Let the fit take its course** — do not interfere with the movements.
4. **Ask onlookers to move back** and protect the casualty's **privacy and dignity** (cover him if he has passed urine).
5. **When the jerking stops**, open the airway, check breathing and place him in the **recovery position**; wipe away froth and saliva.
6. **Note the details for the doctor** — what happened before the fit, which part started moving first, how long it lasted, whether he passed urine or bit his tongue, and how long the recovery took.
7. **Stay until he is fully recovered and oriented** — reassure him gently, tell him where he is, and let him rest or sleep; arrange for someone to take him home.
8. Treat any **injury** sustained in the fall.

WHAT NEVER TO DO DURING A FIT — GUARANTEED EXAM QUESTION

- **Do NOT put anything in the mouth** — no spoon, no cloth, no finger, no key, no wooden stick. It breaks teeth, injures the mouth and may block the airway. (The old teaching about preventing tongue biting is **wrong and dangerous**.)
- **Do NOT restrain the casualty** or try to hold down the limbs — it causes fractures and dislocations.
- **Do NOT move** the casualty unless he is in danger (fire, water, traffic, machinery).
- **Do NOT throw water on the face**, slap, shake or shout at him.
- **Do NOT give anything by mouth** — no water, no tablets — until he is **fully conscious** and able to swallow.
- **Do NOT try to bring him round** with smelling salts or by making him smell an onion or a shoe.
- **Do NOT leave him alone** while he is drowsy or confused.

When to call an ambulance (112 / 108)

- The fit lasts **more than 5 minutes** (status epilepticus) or fits **recur one after another** without full recovery.
- It is the casualty's **first known fit**, or the cause is unknown.
- The casualty is **injured** in the fit, or has **breathing difficulty** after it.
- He **does not regain consciousness** within 10–15 minutes, or remains confused for a long time.
- The casualty is **pregnant, diabetic, elderly, a child**, or the fit happened **in water**.
- There is a **head injury**, high fever with a rash or stiff neck (**meningitis**), or a known poisoning/overdose.

Minor fits (absence seizures)

- Remove any source of danger; **do not interrupt or shake** the casualty.
- Talk to him quietly, stay with him until he is fully alert, and reassure him — he may not know he has had an attack.

- Guide him gently away from danger if he wanders; advise medical assessment, especially in a child whose 'day-dreaming' is frequent.

Living with epilepsy — advice the first aider can give

- **Take the medicines regularly and never stop them suddenly;** keep a record of fits.
- **Get enough sleep,** avoid alcohol and avoid known triggers such as flickering lights and computer/TV strain.
- **Avoid dangerous situations** — swimming alone, climbing heights, cooking on an open flame, operating machinery, and driving (until certified fit).
- **Take showers rather than baths,** keep bathroom doors unlocked, use guards on fires and heaters.
- **Carry a medical card or bracelet** and tell friends, teachers and colleagues what to do.

13.5 Hysteria (Conversion / Dissociative Disorder)

DEFINITION

- **Hysteria is a psychological (functional) disorder in which emotional conflict is unconsciously converted into physical symptoms** — fits, paralysis, blindness, aphonia (loss of voice), over-breathing or dramatic collapse — **without any organic disease of the body.**
- It is not pretending (that is **malinger**ing, which is conscious). The casualty genuinely experiences the symptoms, but they **serve a purpose** — **attention, escape or sympathy** and are almost always produced **in the presence of other people.**
- Commoner in **young women and adolescents**, in emotional, suggestible personalities and after a quarrel, bereavement, fright, examination stress or family conflict.

Signs and symptoms of a hysterical attack

- **Dramatic, theatrical behaviour** — shouting, screaming, crying, laughing, abusing, tearing clothes or hair, rolling on the ground.
- **Apparent fit:** wild, purposeless, **thrashing and writhing movements** that are **not the regular jerking of epilepsy**; arching of the back; the attack **worsens when watched** and stops when the audience leaves.
- **Consciousness is not truly lost** — the eyelids resist opening and flutter, the pupils react normally, and the casualty avoids being hurt (the arm held above the face does not fall on it).
- **No injury, no tongue bite, no passing of urine, no cyanosis, and no true post-ictal confusion or sleep.**
- Other presentations: **over-breathing (hyperventilation) with tingling and cramp of the hands, a lump in the throat (globus), sudden inability to speak, walk, see or move a limb**, with normal reflexes.
- The attack occurs **only before an audience** and often has an obvious **emotional trigger.**

First aid for hysteria

1. **Be firm, calm, kind and confident** — neither scold nor sympathise excessively.
2. **Remove the audience** — ask onlookers to leave quietly; this alone often ends the attack.
3. **Protect the casualty from injury** but do **not** restrain him forcibly.
4. **Speak quietly and reassuringly**, addressing the casualty by name; ask simple questions and give simple instructions ('breathe slowly with me').
5. Do **not** slap, shake, shout at, throw water over or otherwise humiliate the casualty; these worsen the attack and are unprofessional.
6. Do **not** discuss the emotional cause in front of others; do not give medicines.
7. **Never assume hysteria until all physical causes are excluded** — always check the **airway, breathing, pulse, pupils and blood sugar.** Epilepsy, hypoglycaemia, head injury, poisoning, tetany and heart disease can all look 'hysterical'.
8. After the attack, allow the casualty to rest, and **advise medical / psychiatric assessment and counselling** — the underlying stress needs treatment; attacks are otherwise likely to recur.

13.6 Epileptic Fit versus Hysterical Fit — the Classical Table

Point	Epileptic fit	Hysterical fit
Nature	Organic — abnormal electrical discharge in the brain	Functional/psychological — emotional in origin
Onset	Sudden , any time, may be preceded by an aura	Gradual, after an emotional upset , and always with an audience
Time and place	Anywhere, any time, even during sleep or when alone	Only in the presence of others , never during sleep
Cry	A sharp ' epileptic cry ' at the onset	Continuous shouting, screaming, crying or talking

Point	Epileptic fit	Hysterical fit
Fall	Falls suddenly and heavily , and may be injured	Sinks down carefully in a safe place — rarely injured
Consciousness	Truly lost	Not truly lost — aware of the surroundings
Movements	Regular, rhythmic tonic then clonic convulsions	Irregular, purposeless, dramatic thrashing , resisting help
Eyes and pupils	Eyes roll up; pupils dilated and not reacting ; eyelids can be opened easily	Eyelids tightly closed, flutter and resist opening ; pupils normal and reacting
Face	Blue (cyanosed) , congested	Normal colour or flushed
Tongue bite	Common — blood-stained froth	Never
Incontinence	Passing of urine (and sometimes stool) is common	Absent
Duration	1-2 minutes , then deep sleep	Prolonged — may last many minutes to hours, and stops when the audience leaves
After the fit	Confused, exhausted, headache, muscle pain, deep sleep; no memory of the fit	Recovers abruptly and completely ; often remembers the event; no true sleep
Response to a firm word or removing the audience	None	Attack subsides
Self-injury / dangerous acts	May burn or injure himself	Avoids injury — will not let a raised arm fall on his own face

13.7 Chapter Recap — One-Line Revision

- Epilepsy = **recurrent unprovoked seizures** from abnormal electrical discharges of the brain; a **single** provoked fit is not epilepsy.
- Major fit phases: **aura** — **tonic (10-30 s, rigid, blue, breathing stops)** — **clonic (1-2 min, jerking, froth, tongue bite, incontinence)** — **post-ictal (confusion, sleep, no memory)**.
- Commonest trigger = **missed medicine**; commonest identifiable cause in India = **neurocysticercosis**.
- **Never put anything in the mouth, never restrain, never throw water, never give anything by mouth till fully conscious.**
- **Status epilepticus = fit > 5 minutes or repeated fits** — call an ambulance at once.
- After the fit → **recovery position**, note the timing and details, stay until fully oriented.
- **Absence (petit mal)** = brief blank stare in children, no fall, no convulsion.
- **Hysteria** = emotional conflict converted into physical symptoms; occurs **only before an audience**; **no tongue bite, no incontinence, no cyanosis, no injury, pupils normal**.
- Management of hysteria = **firm, calm reassurance + remove the audience**; never slap or throw water; always exclude physical causes first.

CHAPTER 14

Poisons including Food Poisoning

Classification of poisons, routes of entry, general principles of management, individual poisons and their antidotes, alcohol and drug poisoning, food poisoning and its prevention.

SYLLABUS COVERED IN THIS CHAPTER

- Poisons — definition, classification, routes of entry, general signs and symptoms, general first-aid management, when to induce vomiting and when not.
- Individual poisons — corrosives, kerosene and petroleum, insecticides, dhatura, opium, sedatives, alcohol, paracetamol, aspirin, carbon monoxide, rat poison and aluminium phosphide, plant poisons, and their antidotes.
- Food poisoning — types, causes, incubation period, signs, first aid, ORS and prevention; botulism and mushroom poisoning.

14.1 Definition and Classification of Poisons

DEFINITIONS

- A **poison** is any substance which, when taken into or applied to the body in sufficient quantity, damages health or destroys life by its chemical action.
- **Toxicology** is the study of poisons; an **antidote** is a substance that counteracts a poison; the **lethal dose** is the smallest amount that can kill.
- Poisoning may be **accidental** (commonest in children), **suicidal** (commonest in adults in India — insecticides and medicines), **homicidal**, or **occupational/environmental**.
- **Every case of poisoning is a medico-legal case** — preserve the container, the vomit and any remaining substance, and inform the police where suicide or homicide is suspected.

Basis	Classes and examples
Action on the body	1. Corrosives — destroy tissue on contact: strong acids (sulphuric, nitric, hydrochloric, carbolic acid/phenol) and strong alkalis (caustic soda/potash, ammonia, lime, drain cleaner). 2. Irritants — inflame the gut: arsenic, copper sulphate, zinc, croton oil, cantharides, food poisons. 3. Systemic / neurotics — act on the nervous system: opium, dhatura, alcohol, barbiturates, organophosphates, strychnine, snake venom. 4. Cardiac poisons — oleander, aconite, digitalis, nicotine. 5. Asphyxiants — carbon monoxide, carbon dioxide, cyanide, hydrogen sulphide.
Route of entry	Ingested (swallowed — commonest), inhaled (gases, fumes, vapours), absorbed through the skin, eyes or mucous membrane (insecticides, phenol), injected (drugs, stings, bites)
Source	Household (bleach, detergent, acid, kerosene, camphor, naphthalene, phenyl), agricultural (pesticides, weed killers, rat poison), medicinal (overdose of tablets), industrial (lead, mercury, arsenic, solvents), plant and animal (dhatura, oleander, mushroom, snake, scorpion), food (bacterial and chemical)
Speed of action	Acute (single large dose — rapid effects) and chronic (small repeated doses — lead, arsenic, mercury)

14.2 Recognition — General Signs and Symptoms

Route	Typical signs and symptoms
Swallowed	Burns, stains or blisters round the mouth and lips ; peculiar smell of the breath (kerosene, alcohol, insecticide, bitter almonds); nausea, vomiting, abdominal pain, diarrhoea; burning pain from mouth to stomach; drowsiness, delirium, convulsions, unconsciousness; shock
Inhaled	Breathlessness, cough, tightness of the chest, cyanosis (or cherry-red skin in CO poisoning) , headache, giddiness, confusion, collapse; often several people affected together
Absorbed (skin/eye)	Redness, burning, blistering, swelling, itching or numbness of the contaminated area; sweating; later systemic effects
Injected	Puncture marks, local pain and swelling; rapid systemic effects; syringes or needles nearby
Clues at the scene	Open bottles, blister packs, tablets, a suicide note, a farm sprayer, a smell in the room, vomit, spilt liquid, a charcoal fire or gas geyser in a closed room

PUPIL SIGNS — A GIFT TO THE EXAMINER

- **Pin-point (constricted) pupils** — opium/morphine/heroin, organophosphate insecticides, mushroom (muscarine), some sedatives.
- **Widely dilated pupils** — dhatura/atropine/belladonna, cocaine, amphetamine, alcohol, antihistamines,

cyanide, deep unconsciousness and cardiac arrest.

- **Cherry-red skin** → **carbon monoxide**; **blue (cyanosed)** → asphyxiant poisons; **yellow (jaundiced)** → phosphorus, paracetamol, mushroom (late).
- **Smell of the breath**: bitter almonds → **cyanide**; garlic → **phosphorus/arsenic/aluminium phosphide**; kerosene → petroleum; acetone/sweet → diabetic coma; alcohol → ethanol (but beware — a drunk man may also be injured or hypoglycaemic).

14.3 General First-Aid Management of Poisoning

The four aims are: (1) **remove the poison or the casualty from it**, (2) **prevent further absorption**, (3) **maintain the vital functions (airway, breathing, circulation)**, and (4) **get the casualty and the poison to hospital quickly**.

1. **Protect yourself first** — wear gloves; do not touch a corrosive or insecticide with bare hands; ventilate a gas-filled room from outside; never enter a confined space without breathing apparatus.
2. **Remove the casualty from the poison**, or the poison from the casualty — fresh air for gases, remove contaminated clothing, wash the skin with plenty of water and soap for **at least 15–20 minutes**, irrigate eyes with running water for 20 minutes.
3. **Assess DRABC**. If unconscious but breathing → **recovery position** (this also helps vomit to drain). If not breathing → **CPR**, using a **face mask or shield** if there is poison on the lips (never mouth-to-mouth with corrosives, cyanide or organophosphates).
4. **Ask what, when, how much and why** — and **keep the evidence**: container, label, tablets, plant, vomit. This is the single most useful thing you can do for the doctor.
5. **Do NOT induce vomiting** (see below) and do **not** give salt water, mustard water or any 'universal antidote'.
6. **If the poison is corrosive** and the casualty is fully conscious, wash the mouth out and give **small sips of cold water or milk** to dilute and soothe (about 250 ml in an adult; some authorities allow milk or water only if hospital is more than 30 minutes away).
7. **Nothing by mouth** in an unconscious, convulsing or drowsy casualty.
8. **Keep the casualty warm and quiet**, treat for shock, and **monitor breathing, pulse and level of response every 10 minutes**.
9. **Telephone 112/108 and the poison information centre (1800-11-6117, AIIMS)** for advice; transport to hospital at once, in the recovery position, with a written note of the time and amount taken.
10. For **suspected suicide**, stay with the casualty, remove any remaining tablets or poison out of reach, be non-judgemental, and inform the police/family; arrange psychiatric help afterwards.

VOMITING — WHEN NEVER TO INDUCE IT

- **Modern rule: a first aider should NOT induce vomiting at all**. Gastric lavage and activated charcoal are hospital procedures.
- It is **absolutely forbidden** when the poison is: (a) **a corrosive** (acid or alkali — it will burn the gullet again and may perforate it); (b) **a petroleum product** (kerosene, petrol, diesel, thinner, furniture polish — it will be inhaled and cause fatal **chemical pneumonitis**); (c) **phenol/carbolic acid**; (d) the casualty is **unconscious, drowsy or convulsing**; (e) the casualty is a **pregnant woman, a heart patient or a very young child**.
- **Never use salt water** to induce vomiting — it can itself kill by salt poisoning.
- If the casualty vomits **on his own**, turn the head to the side, keep the airway clear and **save a sample of the vomit** for the hospital.

14.4 Individual Poisons — Recognition and First Aid

Poison	Signs and symptoms	First aid / antidote
Strong acids (sulphuric, nitric, hydrochloric — <i>tezaab</i>)	Burning pain from mouth to stomach; stains and burns round the mouth (black/brown with sulphuric, yellow with nitric); vomiting of blood-stained material; difficulty in swallowing and speaking; shock	Never induce vomiting, never give sodium bicarbonate (gas may rupture the stomach). Wash the mouth; give small sips of cold water or milk ; wash contaminated skin with copious water for 20 min; airway care; urgent hospital
Strong alkalis (caustic soda/potash, lime, ammonia, drain and oven cleaner)	Soapy, slippery burns round the mouth; severe burning pain; vomiting; swelling of the throat with airway obstruction	Never induce vomiting, never give an acid (vinegar/lemon) . Sips of cold water or milk; watch the airway closely; urgent hospital
Carbolic acid / phenol / dettol-type disinfectants	Smell of phenol , white burns round the mouth, painless numbness of the mouth, vomiting, giddiness, dark (smoky) urine , collapse	No vomiting; wipe and wash the mouth and skin (with vegetable oil or plenty of water); airway care; hospital

Poison	Signs and symptoms	First aid / antidote
Kerosene, petrol, diesel, thinner, turpentine, furniture polish	Smell of the fluid, burning of the mouth and throat, vomiting, coughing and rapid breathing (aspiration into the lungs), fever, drowsiness, fits	Absolutely NO vomiting (aspiration is the killer). Keep the casualty upright and calm , give nothing by mouth, remove soaked clothing, oxygen if trained; urgent hospital
Organophosphate / carbamate insecticides (malathion, parathion, monocrotophos, dimethoate — the commonest serious poisoning in rural India)	Pin-point pupils, excessive salivation, sweating, tears and urination , vomiting, abdominal colic, diarrhoea, muscle twitching and weakness, slow pulse , frothing at the mouth, smell of the insecticide/garlic , difficulty in breathing, convulsions, coma (remember SLUDGE : Salivation, Lacrimation, Urination, Defaecation, Gastric cramps, Emesis)	Remove all contaminated clothing and wash the skin and hair thoroughly with soap and water (wear gloves — it is absorbed through skin); airway suction/positioning; do not do mouth-to-mouth ; rush to hospital — the antidote is atropine (+ pralidoxime)
Datura, belladonna, atropine, Datura seeds (often used criminally)	Widely dilated pupils and blurred vision , dry hot flushed skin, dry mouth , fast pulse, high temperature, delirium, hallucinations and excitement ('mad as a hatter, red as a beet, dry as a bone, blind as a bat, hot as a hare'), retention of urine, coma	Airway and cooling measures; protect the excited casualty from injury; hospital (antidote physostigmine ; sedation)
Opium, morphine, heroin, codeine (opioids)	Pin-point pupils , drowsiness — coma, slow shallow breathing , cyanosis, cold clammy skin, needle marks, slow pulse	Airway and artificial respiration (breathing stops before the heart); recovery position; hospital — antidote naloxone
Barbiturates, benzodiazepines and sleeping tablets	Drowsiness, slurred speech, unsteadiness, slow shallow breathing, coma; empty blister packs nearby	Airway, recovery position, artificial respiration if needed; keep the strips for the doctor; hospital (antidote for benzodiazepines is flumazenil)
Alcohol (ethyl)	Smell of drink, flushed face, unsteady gait, slurred speech, excitement then drowsiness, vomiting, dilated pupils , deep noisy breathing, coma with hypoglycaemia and hypothermia	Recovery position, keep warm (the casualty loses heat rapidly), never leave him alone or let him 'sleep it off' unattended; never assume it is only drink — look for head injury and low blood sugar; hospital if unconscious
Methyl alcohol (methanol, spurious liquor)	Vomiting, abdominal pain, blurring of vision and blindness , deep rapid breathing (acidosis), coma — often several victims together	Urgent hospital — the antidote is ethanol/fomepizole and dialysis; save the bottle; inform authorities
Paracetamol overdose	Very few symptoms for the first 24–48 hours, then nausea, vomiting and liver failure with jaundice — dangerous because it looks harmless at first	Hospital at once, even if the casualty feels well — the antidote N-acetylcysteine must be given within 8–10 hours
Aspirin (salicylate) overdose	ringing in the ears and deafness, nausea and vomiting, deep rapid breathing , sweating, fever, confusion, bleeding from the stomach	Hospital urgently; sips of water/milk if fully conscious; keep the packet
Iron tablets (common in children)	Vomiting of blood, diarrhoea, abdominal pain, shock, later liver damage	Hospital at once (antidote desferrioxamine); count the missing tablets
Carbon monoxide	Headache, giddiness, nausea, weakness, confusion, cherry-red skin , collapse; whole family affected ; source = charcoal <i>angeethi</i> , gas geyser, car exhaust	Ventilate and remove to fresh air (protect yourself), 100 % oxygen if available, CPR if needed; hospital — may need hyperbaric oxygen
Cyanide (industrial, silver polish, bitter almonds, some seeds)	Smell of bitter almonds , sudden giddiness, headache, gasping breathing, bright red skin , convulsions, instant collapse	Fresh air, do not do direct mouth-to-mouth (use a mask/bag), oxygen, immediate hospital — antidote kit (hydroxocobalamin, sodium thiosulphate/nitrite)
Aluminium phosphide (Celphos, 'rice tablet', wheat pill)	Garlic-like smell of the breath , vomiting, severe abdominal pain, intractable shock , difficulty in breathing — very high mortality; no antidote	Do not induce vomiting; nothing by mouth; airway and oxygen; immediate hospital; keep the tablet/wrapper; ventilate the room (phosphine gas is released and can affect the rescuer)
Rat poison (zinc/barium phosphide, warfarin-type)	Vomiting, abdominal pain, garlic smell (phosphide) or bleeding from the gums, nose and in urine (warfarin type)	Hospital; vitamin K is the antidote for warfarin-type poisons; keep the packet
Naphthalene balls, camphor, phenyl (household)	Vomiting, abdominal pain, excitement, convulsions (camphor), dark urine and anaemia (naphthalene)	Nothing by mouth if fitting; recovery position; hospital
Copper sulphate (neela thotha)	Metallic taste, blue-green vomit , burning abdominal pain, jaundice, kidney failure	Nothing by mouth; hospital urgently
Plant poisons — oleander (kaner), aconite, castor seed, abrus, yellow oleander seeds	Vomiting, abdominal pain, slow or irregular pulse and heart block (oleander/acnrite), tingling and numbness of the mouth (aconite), collapse	Nothing by mouth; complete rest lying down; cardiac monitoring needed — rush to hospital with a sample of the plant/seed
Lead, arsenic, mercury (chronic/occupational)	Colicky abdominal pain, constipation, anaemia, blue line on the gums and wrist drop (lead); vomiting, rice-water stools, garlic breath (arsenic); tremor, salivation, mental changes (mercury)	Remove from exposure, wash the skin, hospital for chelation (BAL, EDTA, penicillamine)

14.5 Common Antidotes — Quick Table

Poison	Antidote
Organophosphate insecticide	Atropine + pralidoxime (2-PAM)
Opium / morphine / heroin	Naloxone
Paracetamol	N-acetylcysteine (oral methionine)
Benzodiazepines	Flumazenil
Datura / atropine	Physostigmine / neostigmine
Cyanide	Hydroxocobalamin, sodium nitrite + sodium thiosulphate
Carbon monoxide	Oxygen (100 %, hyperbaric if severe)
Methanol / ethylene glycol	Ethanol or fomepizole
Iron	Desferrioxamine
Lead / arsenic / mercury	BAL (dimercaprol), EDTA, penicillamine
Warfarin / rat poison	Vitamin K (+ fresh frozen plasma)
Snake venom	Polyvalent anti-snake venom (ASV)
Aluminium phosphide	None — supportive treatment only
General adsorbent (many swallowed poisons)	Activated charcoal — hospital use only, not for corrosives, petroleum, iron, alcohol or cyanide

14.6 Food Poisoning

DEFINITION

- Food poisoning is an acute illness of the stomach and intestine caused by eating food or drinking water contaminated with bacteria, their toxins, viruses, parasites, chemicals or natural poisons.
- It usually causes vomiting, diarrhoea and abdominal pain within a few hours of the meal, and often affects several people who ate the same food — an outbreak.
- The chief danger is dehydration and shock, especially in children, the elderly and the sick.

Types and causes

Type	Organism / agent	Incubation	Typical source and features
Bacterial — toxin type	<i>Staphylococcus aureus</i>	1–6 hours (very short)	Cream cakes, custard, milk products, cooked meat handled by a person with a boil or septic finger. Severe vomiting and cramps, little or no fever
Bacterial — infective type	<i>Salmonella</i>	12–36 hours	Under-cooked egg, poultry, meat, unpasteurised milk. Fever, abdominal pain, diarrhoea, vomiting, headache
	<i>Clostridium perfringens</i>	8–22 hours	Meat and gravy kept warm for long. Cramps and diarrhoea; vomiting unusual
	<i>Bacillus cereus</i>	1–5 h (vomiting type) or 8–16 h (diarrhoeal type)	Re-heated/left-over rice , pasta
	<i>E. coli</i> (ETEC, O157)	12–72 hours	Contaminated water, salad, under-cooked minced meat. Watery or bloody diarrhoea
	<i>Vibrio cholerae</i>	Hours–5 days	Contaminated water — painless profuse ‘rice-water’ stools , rapid dehydration and collapse
	<i>Shigella</i> (bacillary dysentery)	1–3 days	Person-to-person, water, flies — blood and mucus in the stool , fever, tenesmus
	<i>Clostridium botulinum</i> (botulism)	12–36 hours	Improperly canned, tinned, bottled or fermented food ; the toxin attacks nerves — double vision, drooping eyelids, dry mouth, difficulty in speaking and swallowing, descending paralysis and paralysis of breathing . Little or no diarrhoea. Highly fatal — needs antitoxin and ventilation
Viral	Rotavirus, Norovirus, Hepatitis A and E	1–3 days (hepatitis 2–6 weeks)	Water, shellfish, food handled by an infected person; vomiting and watery diarrhoea; hepatitis A/E — jaundice
Parasitic	<i>Entamoeba histolytica</i> , <i>Giardia</i> , tapeworm, roundworm	Days–weeks	Contaminated water and raw vegetables — chronic loose stools, cramps, weight loss

Type	Organism / agent	Incubation	Typical source and features
Chemical	Insecticide residues, metal contamination (zinc, copper, tin from utensils) , food colours, adulterants, argemone oil in mustard oil (epidemic dropsy)	Minutes–hours	Metallic taste, vomiting, cramps; specific features by chemical
Natural toxins	Poisonous mushrooms , red kidney beans (raw), <i>khesari dal</i> (lathyrism), fish toxins (scombroid, puffer fish), sprouted potato (solanine), aflatoxin from mouldy groundnut/maize	Hours–days	Vomiting and cramps; mushroom may cause liver and kidney failure after 6–24 hours; aflatoxin causes liver damage

Signs and symptoms

- **Nausea and repeated vomiting, abdominal pain and cramps, diarrhoea** (watery, or with blood and mucus), flatulence.
- Fever, headache, malaise, weakness and muscle aches.
- **Signs of dehydration** — intense thirst, dry mouth and tongue, sunken eyes, loss of skin elasticity (pinched skin stays up), **sunken fontanelle in a baby**, little or no urine, irritability, drowsiness; in severe cases **shock** with a rapid weak pulse and cold clammy skin.
- **Several people affected after the same meal** — the hallmark of an outbreak.

First aid for food poisoning

1. **Rest** the casualty lying down, with a bowl and towel at hand; reassure him.
2. **Give plenty of fluid in small, frequent sips** — the mainstay of treatment. Best is **ORS (oral rehydration solution)**: one WHO sachet in **1 litre of clean water**; alternatively home-made — **1 teaspoon (a pinch, 2.6 g) of salt + 6 teaspoons (27 g) of sugar in 1 litre of boiled cooled water**, or rice water (*kanji*), lemon water, coconut water, buttermilk, dal water.
3. Give ORS **after every loose stool**: about 50–100 ml for a small child, 100–200 ml for an older child and **as much as an adult wants**.
4. **Do not give** aerated drinks, very sweet drinks, strong tea or coffee, alcohol, milk in large amounts, or any **anti-diarrhoeal medicine on your own** (they can be dangerous, especially in children and in dysentery).
5. Once vomiting stops, offer **light bland food** — rice, khichdi, banana, toast, curd, boiled potato (the 'BRAT' idea); continue breast-feeding a baby throughout.
6. **Keep a sample of the suspected food, the vomit and the stool** for examination, and note what was eaten, when and by whom.
7. **Maintain strict hygiene** — wash hands, disinfect the toilet, keep the casualty's utensils separate.
8. **Call a doctor / go to hospital** if: the casualty is a **baby, a small child, elderly, pregnant or already ill**; there is **blood in the stool or vomit**; **high fever**; **severe or continuous vomiting** so that fluids cannot be kept down; **signs of dehydration or shock**; symptoms last **more than 2–3 days**; there are **nervous symptoms (double vision, difficulty in swallowing or breathing, weakness)** suggesting **botulism**; or **mushroom or chemical poisoning** is suspected.
9. If the casualty becomes unconscious — **recovery position**, nothing by mouth, call an ambulance.
10. **Notify** the health authority if many people are affected (a public health emergency), e.g. after a marriage feast or in a hostel.

PREVENTION OF FOOD POISONING — THE FIVE KEYS (WHO)

- **1. Keep clean** — wash hands with soap before cooking and eating and after using the toilet; keep the kitchen, utensils and cloths clean; keep flies, rats and pets away; a food handler with diarrhoea, a boil or a septic finger **must not handle food**.
- **2. Separate raw and cooked food** — use different knives and boards; store raw meat below cooked food in the refrigerator.
- **3. Cook thoroughly** — especially meat, poultry, eggs and seafood; the centre should reach **70 °C**; re-heat leftovers until piping hot, and **never re-heat rice more than once**.
- **4. Keep food at safe temperatures** — do not leave cooked food at room temperature for more than **2 hours**; keep hot food above **60 °C** and cold food below **5 °C**; the **danger zone is 5–60 °C**; thaw frozen food in the refrigerator, not on the counter.
- **5. Use safe water and raw materials** — boiled or treated water, pasteurised milk, washed fruit and vegetables, no bulging or rusted tins, no food past its **expiry date**, no cut fruit or uncovered food from the roadside.
- Cover food, cover water containers, use a ladle (not hands) to serve, and never taste food that smells or looks

doubtful — **‘when in doubt, throw it out’.**

14.7 Prevention of Poisoning in General

- Keep **all medicines, insecticides, kerosene, acids, phenyl and cleaning agents locked away, out of the reach and sight of children**, and never in a cupboard with food.
- **Never store a poison in a soft-drink, water or milk bottle**, and never remove the original label.
- Read the label and dose **before** giving any medicine; never take medicine in the dark; never take someone else's prescription; discard expired medicines safely.
- **Never call medicine ‘sweets’** to persuade a child; use child-resistant caps.
- Follow the instructions when spraying pesticides — wear protective clothing, a mask and gloves, spray downwind, do not eat or smoke while spraying, and wash thoroughly afterwards.
- Ensure **ventilation** when using charcoal fires, gas geysers, paints and solvents; service gas appliances; fit a CO alarm if possible.
- Keep the numbers of the hospital, ambulance (**108**) and the **National Poisons Information Centre (1800-11-6117)** near the telephone.

14.8 Chapter Recap — One-Line Revision

- Poison = substance that **damages health or destroys life by chemical action**; routes = **swallowed, inhaled, absorbed, injected**.
- Classes = **corrosive, irritant, systemic/neurotic, cardiac, asphyxiant**.
- **Never induce vomiting** — above all with **corrosives, kerosene/petroleum, phenol**, or in an **unconscious or convulsing** casualty; **never use salt water**.
- **Always save the container, tablets, plant and vomit** — and treat every poisoning as a **medico-legal case**.
- **Pin-point pupils** = opium or organophosphate; **dilated pupils** = dhatura/atropine, alcohol; **cherry-red skin** = carbon monoxide; **garlic breath** = aluminium phosphide/phosphorus; **bitter almonds** = cyanide.
- Organophosphate poisoning → **SLUDGE signs**, wash the skin, antidote **atropine**; opioid → **naloxone**; paracetamol → **N-acetylcysteine**; CO → **oxygen**.
- **Aluminium phosphide (Celphos)** has **no antidote** and a very high mortality.
- Food poisoning is commonest from **Staphylococcus (1–6 h, vomiting)** and **Salmonella (12–36 h, fever)**; **botulism** (canned food) causes **paralysis and double vision**.
- Treatment of food poisoning = **rest + ORS in small frequent sips**; **ORS = 1 sachet in 1 litre** of clean water, or 1 tsp salt + 6 tsp sugar per litre.
- Prevention = **WHO five keys**: keep clean, separate raw and cooked, cook thoroughly, keep at safe temperature (danger zone **5–60 °C**), use safe water and raw material.

CHAPTER 15

Common Conditions

Foreign bodies in the ear, eye and nose; cramps; frost-bite and other cold injuries; heat disorders; bites and stings; epistaxis; snake bite and dog bite.

SYLLABUS COVERED IN THIS CHAPTER

- Common conditions — **foreign body in ear, eye and nose; cramps; frost-bite; bites and stings; epistaxis; snake bite; dog bite.**
- Related emergencies of heat and cold — heat cramps, heat exhaustion, heat stroke, hypothermia and chilblains.

15.1 Foreign Body in the Ear

Point	Detail
Common objects	Children: beads, seeds, grain, paper, cotton, small toys, pencil lead, pebbles. Adults: cotton-bud tips, ear-rings, insects (ants, mosquitoes, cockroaches), water, wax
Signs and symptoms	Pain, blocked feeling, deafness on that side, buzzing or noise in the ear (live insect), giddiness, discharge or bleeding if the canal or drum is injured; the child may keep touching the ear
First aid	Do NOT try to remove the object; never probe with a hair pin, matchstick, tweezers, ear bud or any instrument — you will push it deeper and may rupture the eardrum . • Reassure the casualty and keep the head tilted with the affected ear downwards — a loose object may fall out. • For an insect , sit the casualty with the head tilted, affected ear up, and gently pour lukewarm water, olive oil, glycerine or coconut oil into the ear to float or kill the insect; then tilt the ear down. (Never put liquid in if there is a discharge, a suspected perforation or a vegetable object.) • Never put water in for seeds, grains or peas — they swell. • Take the casualty to a doctor/ENT specialist
Prevention	Keep small objects away from children; teach children never to put anything in the ear; do not use cotton buds or hairpins to clean ears

15.2 Foreign Body in the Eye

- Common objects: **dust, grit, sand, eyelash, insect, husk, metal or glass fragment, chemical splash, coal dust.**
- Signs: **pain and grittiness, redness, watering, blinking and spasm of the lids (blepharospasm), inability to open the eye, blurred vision, sensitivity to light.**

First aid for a loose particle

- Tell the casualty not to rub the eye** — rubbing drives the particle in and scratches the cornea.
- Wash your hands; sit the casualty facing the light and gently separate the lids with the finger and thumb.
- Look for the particle: on the white of the eye, under the lower lid (pull the lower lid down) or under the upper lid (ask the casualty to look down while you lift the lid or draw the upper lid over a match stick/cotton bud).
- Irrigate the eye with clean water or sterile saline** poured from the **inner corner (nose side) outwards** so that the water does not run into the other eye; or use an **eye bath/eye cup**. Ask the casualty to **blink under water**.
- If the particle is visible and loose, **lift it off gently with the corner of a clean moist cloth, a cotton bud or the tip of a clean handkerchief**.
- If it does not come out, **cover the eye with an eye pad and refer to a doctor**.

NEVER — IN EYE INJURIES

- Never rub** the eye or let the casualty rub it.
- Never try to remove:** an object **sticking to or embedded in** the eyeball, an object on the **cornea (the coloured part)**, a **metal or glass splinter**, or anything that has **penetrated** the eye.
- Never apply pressure** on an injured eyeball, and never apply ointment, oil, kajal, breast milk or any home remedy.
- In a **penetrating injury:** lay the casualty on his back, ask him to keep both eyes still, **cover the injured eye lightly with a sterile pad or a paper/plastic cup taped over it (no pressure)** — cover the other eye too if possible — and take him **lying down** to an eye hospital. Nothing by mouth.
- For a **chemical splash:** **irrigate continuously with plenty of running water for at least 20 minutes**, holding the lids open, with the affected eye **lower**; then cover and refer. Do not try to neutralise the chemical.
- For **arc-eye/snow blindness/welder's flash** (burning, gritty, watering eyes hours after exposure): cover both eyes with soft pads, rest in a darkened room, and refer.

15.3 Foreign Body in the Nose

- Usually in **small children** — beads, buttons, seeds, paper, pieces of sponge, small batteries, peanuts; sometimes an insect in adults.
- Signs: **blockage and noisy breathing through one nostril, a blood-stained or foul-smelling discharge from that nostril (a classical sign of a long-standing foreign body)**, pain, swelling, sneezing.
- First aid: **Do not attempt to remove it and do not probe with tweezers, pins or fingers; keep the child calm**, tell him to **breathe through the mouth**, and ask an older child to **blow the nose gently once** while the other nostril is pressed closed. Do not let him sniff inwards. **Take the child to a doctor/ENT specialist — without delay if the object is a button battery** (it burns the tissue within hours) **or if breathing is affected**.
- Danger:** the object may be inhaled into the air passage and cause choking; a battery or a nut can cause severe damage and infection.

15.4 Cramps

DEFINITION

- A **cramp** is a **sudden, involuntary, painful and prolonged contraction (spasm) of a muscle or group of muscles**, which becomes hard and knotted and cannot be relaxed voluntarily.

Causes	Common sites
Loss of salt and water through profuse sweating (hot work, fever, heavy exercise) — heat cramps	Calf (commonest), foot, thigh, hamstring
Unaccustomed or prolonged exercise , poor warm-up, fatigue, swimming in cold water	Hand and finger (writer's cramp), abdominal muscles
Repeated occupational movements ; sitting or lying in an awkward position (night cramps in bed)	Neck, back
Loss of fluid by vomiting or diarrhoea (cholera cramps), dialysis	Whole limbs
Deficiency of calcium, magnesium, potassium or sodium ; pregnancy; some medicines (diuretics); poor circulation in the elderly	Any muscle

First aid for cramps

- Stretch the affected muscle gently** and hold it — this is the quickest relief. For a **calf cramp**: straighten the knee and **pull the foot and toes firmly upwards towards the shin** (or have the casualty stand and press the heel down). For a **thigh (hamstring) cramp**: straighten the knee and raise the leg. For a **foot cramp**: stand on the ball of the foot, or pull the toes up. For a **hand cramp**: straighten the fingers and press the palm flat on a hard surface.
- Massage the muscle firmly** towards the heart after the spasm eases, and apply gentle **warmth** (a warm compress) — or a cold compress if the muscle is strained.
- Rest** the muscle, then move it gently; do not resume hard work immediately.
- Give fluids** — **cool water or ORS/lightly salted water or lemon water with a pinch of salt** (about ½ teaspoon of salt per litre) if cramps are due to heat and sweating.
- Refer to a doctor** if cramps are severe, frequent, unexplained, accompanied by swelling/redness (may be **deep-vein thrombosis**), or occur with abdominal cramps and vomiting.

Prevention: warm up and stretch before exercise, drink enough water and take adequate salt in hot weather, keep a balanced diet rich in calcium, potassium and magnesium (bananas, milk, green vegetables), avoid sudden unaccustomed effort, and stretch the calves before sleeping.

15.5 Frost-Bite and Cold Injuries

DEFINITION

- Frost-bite** is the **freezing of the tissues of the exposed parts of the body** (fingers, toes, nose, ears, cheeks, chin) on exposure to extreme cold, in which ice crystals form in the tissues and the blood supply is cut off — it may end in **gangrene**.
- It is common in **high altitudes, snow-bound areas, mountaineers, soldiers (Siachen/Kargil)**, and in anyone **exposed to cold with wet clothing, wind, tight boots, alcohol or poor circulation**.

Stage	Features
1. Frost-nip (mildest, reversible)	Skin becomes white and numb ; no ice in the tissue; recovers completely on warming

Stage	Features
2. Superficial frost-bite (1st–2nd degree)	'Pins and needles' then complete numbness; skin hard, waxy white, cold and stiff on the surface but soft underneath ; on re-warming it becomes red, swollen, painful and blistered
3. Deep frost-bite (3rd–4th degree)	Whole part solid, hard like wood, white or blue-grey, completely numb, no movement ; later becomes black and mummified (gangrene) ; may need amputation

First aid for frost-bite

1. Move the casualty into shelter and out of the cold and wind.
2. Remove wet or tight clothing, boots, gloves, rings, bangles and watches from the affected part before it swells; handle the part **very gently**.
3. Warm the part with body heat first — put frost-bitten fingers in the casualty's own armpit or groin, or in your hands; cover the nose, ears or cheeks with warm dry hands.
4. Re-warm by immersing the part in warm water at 37–40 °C (body temperature — comfortably warm to your elbow, never hot) for about 20–40 minutes, until the skin becomes soft, red and sensation returns. Keep changing the water to maintain the temperature. **Do this only if there is no chance of the part re-freezing.**
5. Dry gently, apply a loose sterile dressing, separate the fingers and toes with dry gauze, raise the part on a pillow and keep it warm.
6. Give warm sweet drinks if the casualty is conscious; give paracetamol for pain (thawing is very painful).
7. Treat for hypothermia as well — it is usually present and is the greater danger.
8. Arrange hospital transfer; the casualty should **not walk** on frost-bitten feet.

FROST-BITE — THE 'NEVERS'

- **Never rub or massage** the frozen part, and **never rub it with snow** — the ice crystals tear the tissues.
- **Never use direct or dry heat** — no fire, stove, heater, hot-water bottle, radiator, exhaust or hot water above 40 °C: the numb skin is burnt without the casualty feeling it.
- **Never break blisters** and never apply ointment.
- **Never let the part thaw and re-freeze** — that causes far more damage than remaining frozen; if the casualty must still be moved through the cold, **keep the part frozen** and re-warm at the destination.
- **Never give alcohol or tobacco** — alcohol dilates skin vessels and increases heat loss; nicotine constricts vessels.
- **Never allow the casualty to walk** on thawed frost-bitten feet.

Other cold injuries

Condition	Features and first aid
Chilblains	Small, itchy, red-purple painful swellings on the fingers, toes, heels, nose or ears after exposure to cold and damp. Warm the part slowly, keep it dry, apply calamine or a soothing cream, do not scratch or apply direct heat; wear warm socks and gloves
Trench foot (immersion foot)	Prolonged standing in cold water or wet boots without freezing — the foot becomes white, numb, swollen and later painful and red. Dry and warm the feet, raise them, do not rub, change socks frequently, refer
Hypothermia	Fall of the body's core temperature below 35 °C (95 °F) . Causes: exposure to cold, wind and rain, immersion in cold water, inadequate clothing, alcohol, exhaustion; the elderly and infants are especially vulnerable. Signs: shivering (which stops as it becomes severe), cold pale dry skin, apathy, confusion, slurred speech, clumsiness and stumbling, slow shallow breathing, slow weak pulse , drowsiness — unconsciousness — cardiac arrest. First aid: move to shelter, remove wet clothing and dry the casualty , wrap in blankets/sleeping bag with a hat, insulate from the ground, re-warm gradually (body heat, warm room, covered warm water bottles on the trunk — not the limbs), give warm sweet drinks if fully conscious, handle very gently (rough handling can stop the heart), and transport horizontally to hospital. Never: rub the skin, give alcohol, use direct strong heat, put the casualty in a hot bath, or let him walk or exercise. Remember 'nobody is dead until warm and dead' — continue CPR for a long time

15.6 Heat Disorders (for comparison)

	Heat cramps	Heat exhaustion	Heat stroke (sun stroke)
Cause	Loss of salt and water in sweat	Loss of salt and water , with dehydration	Failure of the heat-regulating centre — sweating stops and the body cannot lose heat
Temperature	Normal	Normal or slightly raised (up to 40 °C)	Very high — above 40 °C (104 °F)
Skin	Normal, sweating	Pale, cold, clammy, sweating profusely	Hot, flushed, dry — sweating absent
Pulse	Normal	Rapid and weak	Rapid and full/bounding

	Heat cramps	Heat exhaustion	Heat stroke (sun stroke)
Other signs	Painful muscle spasms, especially in the legs and abdomen	Headache, giddiness, nausea, cramps, weakness, thirst, fainting, confusion	Severe headache, restlessness and confusion – fits – unconsciousness; noisy breathing
First aid	Rest in a cool place, stretch and massage the muscle, give ORS/salted water	Lie down in a cool shady place, raise the legs , loosen clothing, give ORS or lightly salted water in sips , fan and sponge; refer if no improvement	A medical emergency — call an ambulance; move to a cool place, remove outer clothing, cool rapidly by sponging with cold water, wrapping in a cold wet sheet, ice packs in the armpits, groin and neck, and fanning; stop cooling when the temperature falls to about 38 °C; recovery position if unconscious; nothing by mouth if not fully conscious ; monitor continuously

15.7 Bites and Stings (Insects and Other Creatures)

Insect stings — bee, wasp, hornet, ant

Point	Detail
Signs	Sharp pain, redness, local swelling and itching ; a sting may be left in the skin by a honey bee (wasps and hornets do not leave the sting)
First aid	Scrape the sting out sideways with the edge of a blunt knife, a credit card or a fingernail — do not use tweezers or squeeze it , which injects more venom. • Wash with soap and water. • Apply a cold compress/ice wrapped in cloth for 10–20 minutes to reduce pain and swelling; calamine lotion or a soothing antihistamine cream may be used. • Raise the part; remove rings if the hand is stung. • An oral antihistamine and paracetamol may be given for itching and pain. • Watch for 30 minutes for signs of allergy
When it is an emergency	(a) Anaphylaxis — swelling of the face, lips or tongue, wheeze, difficulty in breathing or swallowing, widespread rash, collapse → adrenaline auto-injector + call 112/108 (Chapter 7). (b) Sting inside the mouth or throat → swelling can block the airway: give the casualty ice to suck or cold water to sip and call an ambulance. (c) Multiple stings (bee/hornet swarm) → hospital
Prevention	Avoid bright clothes, perfumes and sweet foods outdoors; do not disturb hives or nests; keep food covered; use screens and repellents; shake out shoes and clothes in outdoor camps

Other bites and stings

Creature	Features	First aid
Scorpion	Intense burning pain and numbness at the site (often the hand or foot), local swelling, sweating, salivation, vomiting, restlessness; in children — rapid pulse, breathlessness, fits and collapse (Indian red scorpion is dangerous)	Reassure and keep the casualty still; wash the site; cold compress to relieve pain; immobilise and keep the part below heart level ; give paracetamol; no tourniquet, no cutting, no sucking; take to hospital at once, especially a child (pain relief and prazosin may be needed)
Spider / centipede	Local pain, redness, swelling; rarely cramps, sweating and fever	Wash, cold compress, elevate, analgesic; refer if there are general symptoms or the spider was dangerous
Mosquito, bug, flea, mite	Itchy red papules; danger is the disease transmitted — malaria, dengue, chikungunya, filaria, typhus	Calamine or antihistamine cream, avoid scratching; use repellents, nets and screens; watch for fever and see a doctor
Tick and leech	Attached to the skin, painless, engorged	Tick : grasp close to the skin with fine tweezers and pull straight out without twisting; do not burn it or apply kerosene/petrol; wash and disinfect; save the tick and watch for fever/rash. Leech : do not pull it off — apply salt, vinegar, lemon or a flame near it so that it drops off; then dress the bleeding point
Jellyfish / stingray / catfish (marine)	Severe burning pain, linear weals, swelling; collapse in severe cases	Get the casualty out of the water; pour sea water (not fresh water) over a jellyfish sting and remove tentacles with gloves/a stick; immerse a stingray/fish sting in hot water (45 °C) for 30–90 minutes ; treat pain and refer
Human bite	Punctured/lacerated wound, heavily infected	Wash thoroughly with soap and running water for 5 minutes, cover with a sterile dressing and always refer — antibiotics and tetanus prophylaxis are needed
Monkey, cat, bat, jackal, mongoose bite/scratch	Wound; all are rabies-prone	Treat exactly like a dog bite (below) — wash, dress, hospital for anti-rabies treatment

15.8 Epistaxis (Nose Bleed)

DEFINITION AND CAUSES

- **Epistaxis** is **bleeding from the nose**, usually from the rich network of vessels in the **front lower part of the nasal septum (Little's area / Kiesselbach's plexus)**.
- **Local causes: nose picking** (commonest in children), a blow on the nose, **fracture of the nose or of the base of the skull**, violent sneezing or nose blowing, foreign body, dryness of the air, infection (cold, sinusitis), polyp or tumour.
- **General causes: high blood pressure** (common in the elderly), heat and sun exposure, high altitude, **blood disorders (haemophilia, leukaemia, low platelets, dengue)**, liver disease, vitamin C or K deficiency, medicines — aspirin, warfarin and other blood thinners.

First aid

1. **Sit the casualty down with the head bent slightly FORWARD** over a bowl — never tilt the head back (blood then runs into the throat, causing swallowing of blood, vomiting and possible choking).
2. **Pinch the soft part of the nose (just below the bony bridge) firmly between the thumb and index finger and hold it continuously for 10 minutes**, without releasing to check.
3. Tell the casualty to **breathe through the mouth** and to **spit out** any blood in the mouth rather than swallow it.
4. Apply a **cold compress or ice pack over the bridge of the nose and the back of the neck**; loosen tight clothing at the neck.
5. After 10 minutes, release gently. **If the bleeding continues, pinch for two further periods of 10 minutes.**
6. When the bleeding stops, tell the casualty to **rest quietly, avoid blowing, picking or sniffing the nose, avoid hot drinks, alcohol, smoking and exertion for at least 4 hours (ideally 12–24 hours)**, and to sleep with the head slightly raised.
7. **Seek medical help if:** bleeding lasts **more than 20–30 minutes** or is very heavy; it recurs repeatedly; it follows a **head injury** (the fluid may be **CSF** — suspect a fractured base of skull: **do not** pinch or plug the nose, cover lightly and let it drain); the casualty is on **blood thinners** or has a bleeding disorder, high blood pressure, or becomes pale, faint or shocked; or the casualty is a small child or elderly.

EPISTAXIS — REMEMBER

- **Head FORWARD, not backward** — the single most-asked point.
- **Pinch the soft part for a full 10 minutes** — do not keep letting go.
- **Do not plug the nose with cotton wool** in first aid, and never plug it after a head injury.
- Do not let the casualty lie down flat or talk, laugh or cough.

15.9 Snake Bite

India has about 300 species of snakes, of which around **60 are venomous**. The '**big four**' responsible for almost all deaths are the **Indian cobra, common krait, Russell's viper and saw-scaled viper**. Most bites occur on the **legs and hands** of farmers and villagers, at night and in the monsoon.

Feature	Venomous (poisonous) snake bite	Non-venomous bite
Fang marks	Two distinct puncture marks (fang marks) , sometimes one	A row of small teeth marks (horse-shoe shape) , no fang punctures
Local reaction	Severe burning pain, rapid swelling, discoloration, bleeding and blistering (vipers); little local change with a krait/cobra neurotoxic bite	Slight pain, minimal swelling
General effects	Present — see below	Absent (only fear and anxiety)
Progress	Symptoms worsen with time	No progression

Type of venom	Snakes	Signs and symptoms
Neurotoxic (acts on nerves)	Cobra, krait, sea snake, coral snake	Drooping of the eyelids (ptosis), blurred/double vision, dribbling of saliva, difficulty in speaking and swallowing, weakness of the neck ('broken neck sign'), paralysis of limbs, and finally paralysis of the breathing muscles — death from respiratory failure. A krait bite may be almost painless and is often noticed only in the morning with abdominal pain and paralysis
Vasculotoxic / haemotoxic (acts on blood and vessels)	Russell's viper, saw-scaled viper, pit viper	Severe local pain, rapid swelling, blistering and tissue death; bleeding from the gums, nose, wound and in urine and vomit; non-clotting blood; shock and kidney failure
Myotoxic (acts on muscle)	Sea snakes	Muscle pain and stiffness, dark urine, kidney failure

First aid for snake bite — the modern protocol

1. **Reassure the casualty** — about 70 % of bites are by non-venomous snakes and even venomous bites are often 'dry'. **Fear itself can kill.** Keep him calm and still.
2. **Immobilise the bitten limb** with a **splint and a sling/bandage, exactly as for a fracture**, and keep it **at or slightly below the level of the heart**. Movement pumps venom into the circulation.
3. **Remove rings, bangles, watches, anklets and tight clothing or footwear** from the bitten limb before swelling starts.
4. **Wash/wipe the bite gently with soap and water** (do not scrub) and cover it with a **clean dry dressing**.
5. **Transport the casualty to hospital as quickly and as gently as possible** — preferably carried on a stretcher or a vehicle, **never walking or running**. **Anti-snake venom (ASV) is the only effective treatment** and is available only in hospital.
6. **Monitor breathing continuously** and be ready to give **artificial respiration/CPR** — in a cobra or krait bite, breathing may stop before the casualty reaches hospital.
7. If possible, **note the description of the snake** (or photograph it from a safe distance). **Never try to catch or kill the snake**, but if it has already been killed, take it along carefully.
8. **Nothing by mouth** — no food, no alcohol, no aspirin, no sedative, no stimulant.

Snake bite — the 'NEVERS' (ALL CLASSICAL EXAM POINTS)

- **Never apply a tight tourniquet** — it causes gangrene and does not stop venom spread. (A **broad crepe pressure-immobilisation bandage**, as for a sprain, may be applied over the whole limb with a splint in a confirmed neurotoxic bite where transport is long — but only if trained; it must never be a narrow tight band.)
- **Never cut, slash, incise or scarify** the wound.
- **Never suck the venom out** with the mouth or with any device.
- **Never apply potassium permanganate, chillies, cow dung, herbs, kerosene, ice or a 'snake stone'**, and never cauterise the wound.
- **Never give alcohol, aspirin or sedatives**, and never let the casualty walk.
- **Never rely on a snake charmer, mantra, tantrik or traditional healer** — the delay kills. Go straight to a hospital that has ASV.
- **Never wash away all traces vigorously** — gentle cleaning only (a swab may help identify the venom).

Prevention: wear **shoes and long trousers** in fields and at night, carry a **torch**, do not put hands into holes, grain stores, wood piles or under stones, **sleep on a cot under a mosquito net** (kraits bite sleepers on the floor), clear rubbish and rat holes round the house, and never handle or tease a snake.

15.10 Dog Bite and Rabies

Point	Detail
The danger	Rabies (hydrophobia) — a viral disease of the brain that is almost 100 % fatal once symptoms appear, but is completely preventable by prompt wound care and vaccination. India accounts for a large share of the world's rabies deaths, and dogs cause about 95 % of them
Virus and spread	Rabies (Lyssa) virus , present in the saliva of an infected animal; enters through a bite, scratch, or a lick on broken skin or mucous membrane . Animals involved: dog, cat, monkey, jackal, fox, mongoose, bat, cattle, horse (rodents like rats and squirrels, and birds, do not transmit rabies)
Incubation period	Usually 1–3 months (range 5 days to 1 year or more) — shorter when the bite is on the face, head, neck or hands or is deep/multiple
Features in the animal	Change in behaviour, restlessness, biting without provocation, drooling of saliva, hoarse bark, refusal of food and water, paralysis, death within 10 days
Features in man	Pain, tingling or itching at the healed bite site; fever, headache, anxiety; then hydrophobia (painful spasm of the throat at the sight or thought of water) , excessive salivation, spasms, terror, delirium, paralysis, coma and death

First aid for an animal bite — what you do decides the outcome

1. **Wash the wound immediately and thoroughly with plenty of soap and running water for at least 15 minutes** — this is the **single most important step** and removes/kills much of the virus. Wash all bite and scratch sites.
2. **Apply an antiseptic** — **povidone-iodine (Betadine)**, or 70 % alcohol/spirit, after washing.
3. **Control bleeding** with direct pressure and cover with a **clean, loose, sterile dressing**.

4. **Do NOT:** suture or tightly close the wound at once, apply chillies, lime, turmeric, oil, herbs, cow dung, kerosene, plaster or any irritant; do not cauterise; do not massage the wound; do not cover it with an air-tight dressing.
5. **Take the casualty to a hospital/health centre at once** for:
 - **Anti-rabies vaccine (ARV)** — modern cell-culture vaccine given **intramuscularly on days 0, 3, 7, 14 (and 28)** (Essen schedule), or intradermally as per the centre's protocol;
 - **Rabies immunoglobulin (RIG)** infiltrated **around the wound** in **Category III** exposures (deep/multiple bites, bites on the head, face, neck, hands or genitals, licks on broken skin, bat exposure);
 - **Tetanus prophylaxis** and **antibiotics** as required.
6. **Observe the animal for 10 days** if it can be safely confined; if it remains healthy, the course may be modified by the doctor. If the animal is stray, unknown, dead, or behaving abnormally, **complete the full course**.
7. **Never delay** vaccination to 'see what happens' — once symptoms begin, rabies cannot be cured.

WHO category	Type of exposure	Treatment
Category I	Touching or feeding an animal; licks on intact skin	No treatment (wash the area)
Category II	Nibbling of uncovered skin; minor scratches or abrasions without bleeding	Wash + vaccine
Category III	Single or multiple transdermal bites, scratches with bleeding; licks on broken skin or mucous membrane; contact with bats	Wash + vaccine + rabies immunoglobulin (RIG)

Prevention: vaccinate pet dogs and cats yearly, control and sterilise stray dogs, do not tease or feed stray animals, teach children not to approach strange dogs, never disturb a feeding or sleeping dog or a bitch with pups, and report bites promptly. **World Rabies Day — 28 September.**

15.11 Chapter Recap — One-Line Revision

- **Foreign body in ear/nose:** never probe — refer to a doctor; for an **insect in the ear** float it out with warm oil or water; **never use water for seeds** (they swell).
- **Foreign body in the eye:** never rub; irrigate from the **inner to the outer corner**; never remove an embedded or corneal object; **chemical splash — 20 minutes of running water.**
- **Cramps:** stretch the muscle (calf → straighten the knee and pull the toes up), massage, warmth and **ORS/salted water.**
- **Frost-bite:** re-warm with **water at 37–40 °C**; **never rub, never use dry heat, never break blisters, never let it re-freeze.**
- **Hypothermia** = core temperature **below 35 °C**; re-warm gradually, handle gently, warm sweet drinks, no alcohol.
- **Heat stroke — hot dry skin, no sweating, temperature above 40 °C, confusion —** cool rapidly, emergency; **heat exhaustion —** pale, clammy, sweating → ORS and rest.
- **Bee sting — scrape the sting out sideways,** cold compress; watch for **anaphylaxis**; sting in the mouth → ice to suck + ambulance.
- **Scorpion sting —** pain relief, cold compress, immobilise, hospital (especially a child); **no tourniquet, no cutting.**
- **Epistaxis — sit up, head forward, pinch the soft part for 10 minutes,** cold compress, no blowing for 4 hours.
- **Snake bite —** reassure, immobilise like a fracture, keep the limb at/below heart level, rush to hospital for ASV; **no tourniquet, no cutting, no sucking, no alcohol, no traditional healers.**
- **Dog bite —** wash with soap and running water for 15 minutes, antiseptic, loose dressing, **anti-rabies vaccine (days 0, 3, 7, 14, 28) + RIG for Category III + tetanus cover.**

CHAPTER 16

Transport of Injured Persons

Principles and rules of transport, manual methods of lifting and carrying by one, two, three and four bearers, drags, stretchers and their improvisation, loading and carrying, positions for different injuries and ambulance transport.

SYLLABUS COVERED IN THIS CHAPTER

- Transport of injured persons — general rules and principles, methods of carrying a casualty by one and more bearers, drags and emergency moves.
- Stretchers — types, testing, blanketing, loading and unloading, carrying over obstacles and stairs; positions of the casualty during transport; ambulance and hand-over.

16.1 Principles and General Rules of Transport

THE GOLDEN PRINCIPLE

- **A casualty must not be moved until his injuries have been attended to and he is fit to be moved** — except when there is **immediate danger to life** (fire, explosion, gas, drowning, falling masonry, traffic, rising water or the need to reach the airway).
- **'Treat first, transport next'** — bleeding is controlled, fractures are immobilised and shock is treated **before** the journey.
- The object of transport is to move the casualty **to shelter or hospital without increasing the injury, without causing pain, and without risk to the bearers.**

1. **Decide whether the casualty needs to be moved at all**, and by what method — the method depends on the nature and site of the injury, the weight and size of the casualty, the number of helpers, the distance and the nature of the ground.
2. **Explain the plan to the casualty** and to the helpers; appoint **one person as leader** who gives all the commands.
3. **Immobilise fractures and dress wounds** before moving; support the injured part throughout.
4. **Move the casualty feet first** on a level surface and **head first when going up** stairs, a slope or into an ambulance (the head stays higher) — except for a fracture of the lower limb going downhill, when the injured limb is kept uppermost.
5. **Keep the casualty's body in line**; support the head and neck in any suspected spinal injury.
6. Bearers must **step off with the same foot and break step** (not march in step) so that the stretcher does not sway; move **smoothly and slowly, without jolting**.
7. **Watch the casualty's face continuously** for pain, vomiting or loss of consciousness; keep the airway clear.
8. **Keep the casualty warm and covered**, and protect him from sun, rain, dust and onlookers.
9. **Never leave the casualty alone**, and never allow him to walk or sit up if he has a serious injury or shock.
10. **Lift with your legs, not your back** — feet apart, back straight, knees bent, casualty held close to the body, and lift on the word of command by straightening the knees.
11. **Send a written report** with the casualty and hand him over personally to the doctor or nurse.

16.2 Emergency Moves and Drags (used only in immediate danger)

Method	How it is done	When used
Blanket (drag) lift	Roll the casualty gently onto a blanket (log-roll), gather the blanket at the head end and drag him head-first along the floor	Unconscious or heavy casualty, smoke-filled room, narrow passage — the safest drag for a suspected spinal injury
Shoulder / armpit drag	Support the casualty's head on your forearms, pass your hands under his armpits, grasp his crossed forearms and drag him backwards	Short distance over a smooth floor; no lifting equipment
Ankle drag	Grasp both ankles and pull the casualty in a straight line	Very short emergency distance on a smooth surface only — never with a head, neck or back injury
Clothes drag	Grasp the clothing at the shoulders/collar, supporting the head with your forearms, and pull	Rapid removal from fire, water's edge or traffic
Fireman's crawl	Tie the casualty's wrists together, put the loop round your neck and crawl on hands and knees	Escape from a low, smoke-filled space

16.3 Methods of Carrying by ONE Bearer

Method	Technique	Suitable for / not for
Human crutch (assisted walking)	The casualty's arm is placed round your neck and you hold his wrist; your other arm goes round his waist; you support him on the injured side and walk in step with him (using a stick on the sound side)	Conscious , able to walk, minor injury of one leg. Not for serious injury, fracture of the lower limb, shock or giddiness
Cradle carry	Lift the casualty across your arms — one arm under the knees, the other round the back	Children and light adults
Pick-a-back (piggy-back)	The casualty, sitting on a chair or table, puts his arms round your neck; you hold his thighs and carry him on your back	Conscious, light casualty who can hold on. Not for unconscious or arm-injured casualties
Fireman's lift (shoulder carry)	The casualty is drawn across your shoulder , his body hanging down your back, one arm held down in front of you — leaving one of your hands free to open doors or hold a rail	Light, unconscious or conscious casualty; useful on a ladder or in a narrow passage. Not for chest, abdominal, spinal or multiple injuries, fractured ribs, breathlessness, or a heavy casualty
Drag methods	As above	Immediate danger only

16.4 Methods of Carrying by TWO or More Bearers

Method	How the bearers hold	Use
Two-handed seat	The two bearers face each other, squat on either side of the casualty, pass their near arms under his back and their far arms under his thighs, and grasp each other's wrists (or clothing)	A casualty who cannot use his arms or is unable to help; short distance
Three-handed seat	Each bearer grasps his own wrist with one hand and the other bearer's wrist with the other; the fourth hand is used as a back-rest	A casualty with one injured leg — the free hand supports the injured limb
Four-handed seat	Each bearer grasps his own left wrist with his right hand and the other bearer's right wrist with his left hand , forming a square seat; the casualty sits on it and holds the bearers' shoulders	A conscious casualty who can use both arms — no leg injury; for carrying over a distance
Fore-and-aft carry	One bearer lifts under the armpits from behind, the other holds under the knees — the casualty is carried lengthwise	Through narrow passages, doorways and staircases
Chair carry	The casualty is seated on a strong chair, tied to it, and the chair is tilted back and carried by two bearers (one at the back holding the chair back, one in front holding the front legs)	Conscious casualty; best method for stairs and narrow turns . Not for spinal, pelvic or lower-limb fractures or an unconscious casualty
Three- or four-bearer lift (blanket/scoop lift)	Bearers kneel on one knee on the same side (or on both sides), slide their hands under the casualty's head/shoulders, back, hips and legs and, on the word of command, lift together and rest him on their knees , then stand and place him on the stretcher	Unconscious casualty, suspected fracture of the spine, pelvis or thigh — the method of choice for placing a casualty on a stretcher
Log-roll	Four or five bearers ; one holds the head in line with the body and gives the commands while the others turn the casualty as a single unit, keeping the head, neck, trunk and legs in a straight line	Suspected spinal injury , to place a spine board or blanket under the casualty, or to clear the airway of vomit

CHOOSING THE METHOD — A QUICK GUIDE

- **Conscious, can walk, minor leg injury** → human crutch.
- **Conscious, can use both arms, must be carried some distance** → four-handed seat.
- **Conscious, one leg injured** → three-handed seat.
- **Cannot use his arms** → two-handed seat.
- **Narrow passage, stairs** → fore-and-aft or chair carry.
- **Unconscious, light casualty, single rescuer, immediate danger** → fireman's lift or a drag.
- **Spinal injury, pelvic or thigh fracture, unconscious casualty** → stretcher with a 3-4 bearer lift or log-roll onto a spine board — never a seat carry.

16.5 Stretchers

Type	Description and use
Standard (Furley) stretcher	The ordinary ambulance stretcher — two poles with runners, traverse bars, a canvas bed and a pillow sack; about 1.9–2.4 m long ; folds lengthwise
Trolley (wheeled) stretcher	Wheeled cot used in ambulances and hospitals; height adjustable, with straps and rails
Scoop stretcher	Splits into two halves that are slid under the casualty from either side and clipped together — ideal for a spinal injury with minimum movement
Spinal (long) board / short board / vacuum mattress	Rigid board on to which the casualty is log-rolled and strapped, with head blocks and a cervical collar — the standard for suspected spinal injury
Neil Robertson stretcher	A canvas-and-bamboo/slatted stretcher that wraps round the casualty and is lifted vertically — used to remove a casualty from a ship's hold, mine, tank, well, lift-shaft or any confined vertical space
Paraguard / basket (Jordan frame) stretcher	Rigid basket stretcher for mountain and helicopter rescue
Pole-and-canvas / Utila stretcher	Light folding stretcher used by the armed forces and civil defence
Improvised stretcher	Made from: two strong poles/bamboos and a blanket (fold the blanket round the poles so that the casualty's weight holds the ends), two poles and two or three coats/jackets (sleeves turned inside out and the poles passed through them), a door, shutter, bench, table top, ladder (padded), plank, or a strong sheet/tarpaulin/sack — anything rigid, long and strong enough

STRETCHER RULES

- **Always test an improvised stretcher** with a person of similar weight **before** putting the casualty on it.
- **Prepare (blanket) the stretcher** before the casualty is lifted: lay a blanket diagonally/lengthwise so that half can be wrapped over the casualty; add a pillow for the head.
- **Bring the stretcher to the casualty**, not the casualty to the stretcher.
- **Load and unload from the head or foot end** wherever possible, and always lift on the **word of command** of one leader.
- **Secure the casualty** with straps or broad bandages (chest, hips, knees and ankles) for a journey, on a slope, or in a helicopter.
- **Carry feet first on the level, head first up** an incline, stairs or into the ambulance, and **feet first down** — except that a casualty with a **fractured lower limb** is carried so that the **injured limb is uppermost**, i.e. **head first downhill**.
- **Four bearers** are ideal; **two can manage** on level ground; bearers should be of nearly equal height, and must **break step**.
- Keep the stretcher **horizontal at all times** — over obstacles, walls and ditches use the drill of passing the stretcher over while supporting it, never tilting it.

16.6 Position of the Casualty during Transport

Condition	Position
Unconscious but breathing	Recovery (lateral) position — or supine with the head turned and the airway maintained, strapped to the stretcher
Shock, haemorrhage, fainting	Flat on the back with the legs raised 20–30 cm , head low and turned to the side
Head injury / stroke	Flat, head and shoulders slightly raised , head supported in line; recovery position if unconscious
Suspected spinal injury	Flat on the back on a rigid spine board , head immobilised with blocks and a collar, body strapped — moved only by log-roll or scoop
Chest injury / breathlessness / heart attack / asthma	Half-sitting (Fowler's) position , leaning towards the injured side in a chest injury
Abdominal injury or pain	On the back with the knees drawn up and supported by a pillow
Fracture of the lower limb	Flat, limb immobilised and supported; carried with the injured limb uppermost on a slope
Fracture of the pelvis	Flat on the back, knees and ankles tied together with padding between
Pregnant casualty	Left lateral position
Burns	Flat, burnt part raised, covered with a clean sheet
Snake bite	Lying still, bitten limb splinted and kept at or below heart level
Vomiting / bleeding from the mouth	Head turned to the side, or recovery position

16.7 Ambulance Transport and Hand-Over

- **Call 108 (free ambulance) or 112**; give the exact location, number and condition of casualties and the hazards present (Chapter 1).
- Use an **ambulance** for every serious casualty. A private vehicle should be used only if no ambulance is available — and then the casualty must be **laid flat with an attendant beside him**, never propped up in a seat or carried on a two-wheeler.
- **Load the casualty head first** into the ambulance, secure the stretcher, keep him warm and keep the airway under observation throughout.
- The driver must travel **smoothly and steadily, not fast and jerky** — ‘there is no hurry like a slow hurry’; sirens and speed frighten the casualty and worsen shock.
- **Continue monitoring and recording** the level of response, breathing, pulse and bleeding during the journey; carry the first-aid kit within reach.
- **Hand over personally** to the doctor or nurse with a clear verbal and written report: **what happened, when, what you found (signs and symptoms with times), what treatment you gave, and any poison container, medicine, amputated part or belongings**.
- In **mass casualty and disaster** situations, transport in the order of **triage priority (red → yellow → green)**, not in the order in which casualties were found.
- **Air ambulance / helicopter** is used for mountains, floods and remote areas; the casualty must be **firmly strapped**, and the rescuer must approach the helicopter only **from the front, when signalled by the crew**.

PROTECTING THE BEARERS — CORRECT LIFTING TECHNIQUE

- Stand with the **feet apart**, one foot slightly forward; **bend the knees, keep the back straight** and the head up.
- Get a **firm grip** and hold the load **close to the body**; do not twist while lifting.
- **Lift by straightening the legs**, smoothly, on the command ‘ready — lift’.
- **Never lift more than you can manage**; get more help, or use mechanical aids.
- Bearers should be of **similar height**, know their orders, and **break step** while carrying.

16.8 Chapter Recap — One-Line Revision

- **Never move a casualty until his injuries are attended to** — except in immediate danger.
- Choice of method depends on the **injury, weight of the casualty, number of helpers, distance and ground**.
- **Four-handed seat** → conscious casualty who can use both arms; **three-handed seat** → one injured leg; **two-handed seat** → cannot use the arms; **human crutch** → can walk.
- **Fireman's lift** leaves one hand free and is for a light casualty — never for chest, abdominal or spinal injuries.
- **Fore-and-aft and chair carry** are best for **stairs and narrow passages**.
- **Spinal injury** — **log-roll onto a spine board/scoop stretcher** with 4–5 bearers, the head held in line.
- **Blanket drag** is the safest drag; **ankle drag** must never be used with a spinal injury.
- Carry **feet first on the level, head first going up** — but **injured limb uppermost** on a slope; bearers must **break step**.
- **Neil Robertson stretcher** — vertical removal from a confined space (well, ship's hold, mine).
- Always **test an improvised stretcher** first, keep it **horizontal**, and **hand over with a written report**.

CHAPTER 17

Use of Common Medicines

Drugs and dosage forms, routes of administration, the rules of giving medicine, groups of common medicines with their uses, doses, cautions and side effects, storage, and the dangers of self-medication.

SYLLABUS COVERED IN THIS CHAPTER

- Use of common medicines — definitions, dosage forms, routes of administration, rules and precautions for giving medicines, weights and measures, storage and expiry.
- Common groups — analgesics and antipyretics, antiseptics and disinfectants, antacids, antihistamines, ORS, antibiotics, antidiarrhoeals, ointments, drops, inhalers and emergency drugs used in first aid.

17.1 Basic Terms

Term	Meaning
Drug	Any substance used in the diagnosis, prevention, treatment or cure of disease, or to alter a body function
Medicine	A drug prepared in a form fit for administration to man (tablet, syrup, injection)
Dose	The measured quantity of a medicine to be taken at one time; daily dose = total in 24 hours
Prescription	A written order of a registered medical practitioner for a medicine (Rx)
Over-the-counter (OTC)	Medicines that may be sold without a prescription — e.g. paracetamol, antacids, ORS, calamine
Schedule H / H1 drug	Indian drug law: medicines (most antibiotics, sedatives) that may be sold only on a doctor's prescription ; marked with Rx and a red line on the pack
Generic and brand name	Generic = the drug's own name (paracetamol); brand = a company's name (Crocin, Calpol). Generic medicines are cheaper and equally effective
Side effect / adverse reaction	An unwanted effect of a medicine taken in the correct dose
Allergy to a medicine	Rash, itching, swelling, wheeze or collapse after a drug — must be recorded and never given again
Antidote	A substance that counteracts a poison
Expiry date	The date after which the medicine must not be used — it may be ineffective or harmful
Placebo	An inactive preparation given for its psychological effect

17.2 Dosage Forms and Routes of Administration

Route	Forms used	Notes
Oral (by mouth)	Tablet, capsule, syrup, suspension, powder, granules, drops, lozenge, ORS	Commonest, safest and cheapest; never give anything by mouth to an unconscious, drowsy, convulsing or vomiting casualty
Sub-lingual (under the tongue)	Nitroglycerine/sorbitrate tablet or spray	Very rapid absorption — used in angina/heart attack
Topical (on the skin)	Ointment, cream, lotion, gel, powder, paint, spray	For local action — antiseptics, calamine, burn creams; wash the hands after applying
Inhalation	Metered-dose inhaler, rotahaler, nebuliser, oxygen	Asthma reliever (salbutamol) ; acts within minutes
Instillation (drops)	Eye, ear and nose drops	Check the label — eye drops must never be an ear preparation ; use a separate dropper; discard 4 weeks after opening
Rectal	Suppository, enema	Paracetamol suppository in a vomiting child; diazepam for a prolonged fit (medical direction)
Injection (parenteral)	Intramuscular (IM), intravenous (IV), subcutaneous (SC), intradermal (ID)	Not for a first aider — the only exception is the casualty's own adrenaline auto-injector (IM in the outer thigh) in anaphylaxis

Weights, measures and common abbreviations

Measure / abbreviation	Equivalent / meaning
1 teaspoon (tsp)	5 ml

Measure / abbreviation	Equivalent / meaning
1 dessertspoon	10 ml
1 tablespoon (tbsp)	15 ml
1 cup / glass	About 200–250 ml
1 gram (g)	1,000 milligram (mg)
1 litre	1,000 ml
15–16 drops	About 1 ml (with a standard dropper)
OD / BD / TDS / QID	Once / twice / three times / four times a day
SOS / PRN	'If necessary' — only when required
HS	At bedtime
AC / PC	Before food / after food
STAT	At once, immediately
IM / IV / SC / PO	Intramuscular / intravenous / subcutaneous / by mouth

17.3 Rules and Precautions for Giving Medicines

THE 'FIVE RIGHTS' OF GIVING MEDICINE

- **Right patient** — confirm the name; never give one person's medicine to another.
- **Right drug** — read the **label three times**: when you take it out, before you pour, and when you put it back. Never give a medicine from an unlabelled or doubtful container.
- **Right dose** — measure exactly with a marked spoon, cup or syringe; a child's dose is by **age/weight**, never a guess.
- **Right route** — oral, topical, inhaled, drops; an ear drop is never put in the eye.
- **Right time** — with or without food, at the correct intervals, for the full course.
- Add a sixth: **Right documentation** — record what was given, how much and when.

1. A first aider gives **only simple, well-known medicines**, and only to a **fully conscious** casualty who can swallow. He **never prescribes, never injects, and never gives a prescription-only (Schedule H) drug**.
2. **Ask about allergies** (especially to penicillin, sulpha and aspirin) and about medicines already being taken, **before** giving anything.
3. **Check the expiry date** and the appearance — discard discoloured, cracked, damp, sticky or foul-smelling medicines and cloudy solutions.
4. **Give with plenty of water** (a full glass), sitting or standing — never lying flat; do not crush or open a capsule or an enteric-coated tablet unless told to.
5. **Never give anything by mouth** to an unconscious, semi-conscious, convulsing casualty, to one with an abdominal injury, or when an operation is likely.
6. Be extra careful with **children, pregnant and breast-feeding women, the elderly, and casualties with liver, kidney, heart, asthma or ulcer disease**.
7. **Do not mix medicines with alcohol**; warn about drowsiness (do not drive) with antihistamines and sedatives.
8. **Complete the full course** of an antibiotic even if the casualty feels better, and never share or reuse left-over antibiotics.
9. **Record and report any side effect** and stop the medicine if a rash, swelling or breathlessness appears — it may be the start of **anaphylaxis**.
10. **Store** medicines in a **cool, dry, dark place, locked and out of the reach of children**; keep the ones needing refrigeration (insulin, some eye drops, vaccines) at **2–8 °C**; keep them in their original containers with the labels intact.
11. **Dispose** of expired medicines safely — do not throw them where children, animals or water sources can reach them; return them to a pharmacy if possible.

17.4 Common Medicines Used in First Aid

A. Analgesics (pain relievers) and antipyretics (fever reducers)

Medicine	Use and adult dose	Cautions and side effects
Paracetamol (acetaminophen) — Crocin, Dolo, Calpol	The safest first-aid analgesic and antipyretic — fever, headache, body ache, toothache, mild pain. Adult 500 mg–1 g every 6–8 hours; maximum 3–4 g in 24 hours . Child 10–15 mg/kg per dose	Overdose causes fatal liver damage with very few early symptoms — never exceed the dose, never combine two brands containing paracetamol; avoid in liver disease and with alcohol
Aspirin (acetyl-salicylic acid) — Disprin, Ecosprin	Pain, fever, inflammation; 300 mg chewed at once in a suspected heart attack ; low dose (75–150 mg) daily for heart protection (on medical advice)	Never give to a child under 16 (risk of Reye's syndrome), nor in peptic ulcer, bleeding disorders, dengue, asthma or aspirin allergy ; causes gastric irritation and bleeding; not to be given in stroke or in a bleeding casualty
Ibuprofen / diclofenac (NSAIDs) — Brufen, Voveran	Pain and inflammation of sprains, strains, backache, toothache, joint pain, period pain; ibuprofen 200–400 mg 8-hourly after food	Take after food ; avoid in ulcer, asthma, kidney disease, pregnancy, dehydration and in those on blood thinners; may cause acidity and bleeding
Antispasmodic — dicyclomine, hyoscine (Cyclopan, Buscopan)	Colicky abdominal pain, period pain	Never give in undiagnosed abdominal pain (it masks appendicitis and other surgical emergencies); causes dry mouth and blurred vision

B. Antiseptics and disinfectants

Preparation	Use	Points to remember
Povidone-iodine (Betadine) 5–10 %	Cleaning skin and wounds, dog bites, minor cuts, before dressing	The most useful all-round antiseptic; avoid in iodine allergy, thyroid disease, large raw areas and in newborns
Chlorhexidine (Savlon) / cetrimide	Cleaning wounds and skin, washing hands	Mild and non-irritant; dilute as directed; do not use in the eye or ear
Spirit / 70 % alcohol, alcohol swabs	Skin disinfection before injection; cleaning instruments	Never pour on an open wound — intensely painful and damages tissue; inflammable
Hydrogen peroxide 3 %	Cleaning dirty or infected wounds, loosening crusts, mouth wash (diluted)	Foams on contact; do not use on healing tissue or in deep closed wounds
Potassium permanganate (KMnO₄, 'pink lotion')	Weak solution (1:10,000, a pale pink colour) for soaking infected wounds, ulcers and fungal infections of the feet	Crystals are corrosive and poisonous — must be fully dissolved; stains skin and clothes
Tincture iodine / gentian violet	Old-fashioned wound paints	Painful on raw surfaces; largely replaced by povidone-iodine
Soap and clean running water	The best and most important 'antiseptic' for any wound or bite	15 minutes of washing for an animal bite or chemical contamination
Bleaching powder / sodium hypochlorite	Disinfectant for floors, toilets, utensils, water and spills of blood or vomit	A disinfectant kills germs on objects; an antiseptic is used on living tissue — do not confuse the two

C. Medicines for allergy, stomach and bowel

Medicine	Use and dose	Cautions
Antihistamines — chlorpheniramine (Avil) 4 mg, cetirizine 10 mg, loratadine 10 mg	Itching, urticaria/hives, insect stings, allergic rash, hay fever, mild allergic reaction; one tablet at night (cetirizine) or 8-hourly (chlorpheniramine)	Cause drowsiness (do not drive or operate machinery); antihistamines are NOT a substitute for adrenaline in anaphylaxis ; avoid with alcohol
Antacids — magnesium/aluminium hydroxide gel, calcium carbonate; ranitidine, omeprazole/pantoprazole	Heartburn, acidity, gastritis; antacid gel 2 teaspoons after meals and at bedtime	Chest pain is not always acidity — think of a heart attack; antacids interfere with the absorption of other medicines (give 2 hours apart)
ORS (oral rehydration salts)	The most important medicine in diarrhoea, vomiting, heat exhaustion, cholera and burns . One sachet in 1 litre of clean water; give frequent small sips after every loose stool (50–200 ml in children, freely in adults)	Use within 24 hours of mixing; never make it stronger or weaker; never use hot water or add sugar/salt to a ready sachet; continue breast-feeding
Zinc tablets (20 mg)	With ORS in childhood diarrhoea for 14 days — reduces severity and recurrence	Give as advised by a doctor

Medicine	Use and dose	Cautions
ORS alternatives / home fluids	Rice water (<i>kanji</i>), lemon water with salt and sugar, coconut water, buttermilk, dal water, curd	Avoid aerated drinks and very sweet juices (they worsen diarrhoea)
Anti-emetic — domperidone, ondansetron, promethazine	Vomiting, motion sickness (promethazine/dimenhydrinate 30–60 minutes before travel)	Only on medical advice in children; drowsiness
Anti-diarrhoeal — loperamide	Adult travellers' diarrhoea only	Never give to children , and never in dysentery (blood/mucus in the stool) or high fever — it can be dangerous. Fluid replacement, not a 'stopping' medicine, is the treatment of diarrhoea
Laxative — isabgol (psyllium husk), lactulose, bisacodyl	Occasional constipation, with plenty of water and fibre	Never give a laxative in abdominal pain, vomiting or suspected appendicitis or obstruction ; avoid habitual use
Oral glucose / sugar / glucose gel / honey	Hypoglycaemia , exhaustion, heat illness — 3–4 teaspoons of glucose or sugar in water	Only to a fully conscious casualty who can swallow

D. Skin preparations, drops and emergency medicines

Preparation	Use	Cautions
Silver sulphadiazine cream (Silverex)	Burns and scalds — after cooling, on medical advice	Not on the face; do not apply to a fresh burn before cooling; avoid in sulpha allergy
Calamine lotion	Itching, sunburn, prickly heat, insect bites, chickenpox	Soothing and safe; shake well; do not apply to broken skin
Antibiotic ointment — framycetin, neomycin-bacitracin, mupirocin	Minor infected cuts, grazes and boils	Thin layer; stop if a rash develops; never put antibiotic ointment in the eye unless it is an eye preparation
Antifungal cream/powder — clotrimazole	Ringworm, athlete's foot, intertrigo	Apply for 2–4 weeks and keep the area dry
Analgesic balm / spray, ice/hot packs	Muscle ache, sprains, backache	Not on broken skin, burns or the eyes; use cold for the first 48 hours
Eye drops/ointment — lubricant drops, antibiotic drops	Dry, irritated eye; minor conjunctivitis (on medical advice)	Never use steroid drops without a doctor ; discard 4 weeks after opening; do not touch the dropper to the eye
Ear drops — olive oil, wax-softening drops	Softening wax, an insect in the ear	Not if there is a discharge or a suspected perforation
Nasal drops/spray — saline, xylometazoline	Blocked nose	Decongestant sprays must not be used for more than 3–5 days (rebound congestion); not for infants
Salbutamol inhaler (blue reliever)	Asthma attack — help the casualty take his own inhaler: 2 puffs, repeated every 2 minutes up to 10 puffs ; use a spacer if available	Causes tremor and palpitation; if there is no relief in 5–10 minutes — ambulance
Nitroglycerine / sorbitrate (sub-lingual tablet or spray)	Angina/heart attack — help the casualty take his own tablet under the tongue	Causes headache and a fall in blood pressure; the casualty must be sitting; not if the BP is low
Adrenaline auto-injector (EpiPen)	Anaphylaxis — 0.3–0.5 mg IM into the outer thigh (child 0.15 mg)	The life-saving drug of anaphylaxis ; a second dose may be given after 5–15 minutes; hospital always
Smelling salts (sal volatile)	Traditionally used for faintness in a conscious casualty	Never hold under the nose of an unconscious casualty; fresh air and lying down are better
Oral rehydration, glucose, salt	See above	The commonest and most useful 'medicines' in the first-aid box

E. Antibiotics, anti-malarials and vaccines — what a first aider should know

- **Antibiotics (amoxicillin, azithromycin, cefixime, ciprofloxacin, metronidazole, doxycycline)** are **prescription-only** medicines. A first aider **does not start antibiotics**; his duty is to clean and cover the wound and refer.
- Rules when a doctor has prescribed them: **take the full course, at the correct intervals, with water, and do not stop when you feel better; never share or re-use them; never take a left-over antibiotic for a new illness.** Misuse causes **antibiotic resistance**, the greatest threat in modern medicine.
- Watch for **allergy (rash, swelling, breathlessness — especially with penicillins)**, and for diarrhoea or fungal infection after a course.
- **Metronidazole must never be taken with alcohol**; **tetracycline/doxycycline** must be avoided in children and pregnancy and not taken with milk; **ciprofloxacin** is avoided in growing children.
- **Anti-malarials (chloroquine, artemisinin combinations, primaquine)** are given only after a **blood test** — fever with chills in a malarious area must be tested, not treated blindly.

- **Tetanus toxoid (TT) and anti-rabies vaccine** are the two immunisations a first aider must remember to arrange — for wounds and for animal bites respectively.
- **Vitamins and tonics** are not a treatment for illness; iron and folic acid for anaemia and vitamin D/calcium for bone health are used on medical advice. Iron is taken **after food** and stains the stools black.

17.5 Dangers of Self-Medication

WHY SELF-MEDICATION IS DANGEROUS

- It **masks the real disease** — a painkiller hides appendicitis, an antacid hides a heart attack, an anti-diarrhoeal hides dysentery.
- **Wrong drug, wrong dose, wrong duration** — leading to failure of treatment or to poisoning (paracetamol and liver failure being the classic example).
- **Allergic and adverse reactions**, including fatal anaphylaxis.
- **Drug interactions** with medicines already being taken.
- **Antibiotic resistance** and **habit formation** (sedatives, painkillers, cough syrups).
- **Delay in seeking proper medical help** — often the real cause of death.
- **Golden rule for the first aider:** *give only simple, familiar remedies to a fully conscious casualty, in the correct dose — and when in doubt, give nothing and refer.*

17.6 Medicines for the First-Aid Box — Summary List

Purpose	Medicine
Pain and fever	Paracetamol tablets; aspirin (adults, for suspected heart attack); ibuprofen
Wound care	Povidone-iodine solution/ointment, chlorhexidine, hydrogen peroxide, spirit swabs, antibiotic ointment
Allergy and itching	Antihistamine tablets, calamine lotion, the casualty's own adrenaline auto-injector
Dehydration	ORS sachets , glucose powder, zinc tablets (children)
Acidity	Antacid gel or tablets
Burns	Silver sulphadiazine cream, non-adherent dressings
Eye	Sterile eye wash/saline, eye pads
Faintness	Glucose/sugar; smelling salts
Fungal/skin	Clotrimazole cream, antiseptic powder
Others	Thermometer, gloves, torch, scissors, notebook, list of emergency numbers, first-aid manual

17.7 Chapter Recap — One-Line Revision

- A first aider gives **only simple medicines to a fully conscious casualty** — he never prescribes, never injects and never gives Schedule H drugs.
- **Five rights:** right **patient, drug, dose, route and time** (+ record).
- **Paracetamol** is the safest analgesic/antipyretic: **500 mg–1 g 6–8 hourly, maximum 3–4 g/day**; overdose destroys the **liver**.
- **Aspirin:** **300 mg chewed in a suspected heart attack; never under 16 years (Reye's syndrome)**, never in ulcer, dengue or stroke.
- **ORS = 1 sachet in 1 litre** of clean water, used within **24 hours**; it is the treatment of diarrhoea — **loperamide is never given to children**.
- **Adrenaline (0.3–0.5 mg IM in the outer thigh)** is the drug of choice in **anaphylaxis**; antihistamines are not a substitute.
- **Asthma** → the casualty's own **blue salbutamol inhaler, 2 puffs up to 10**; **angina** → his own sub-lingual nitroglycerine.
- **Antiseptic** is used on living tissue, a **disinfectant** on objects; **soap and running water** is the best wound cleanser.
- **Never** give an antispasmodic or laxative in undiagnosed abdominal pain; **never** use medicines past their **expiry date**.
- Store medicines **cool, dry, dark, labelled, locked and out of children's reach**; complete every antibiotic course.

APPENDIX

Quick Reference and Rapid Revision

Glossary and abbreviations, all normal values, position chart, master list of do's and don'ts, all mnemonics, antidote chart, records and reports, fire and disaster basics, and a chapter-wise list of one-line facts for the last day before the examination.

18.1 Glossary of First-Aid and Medical Terms

Term	Meaning	Term	Meaning
Abrasion	Graze; scraping off of the skin surface	Hypoxia	Lack of oxygen in the tissues
Amnesia	Loss of memory	Hypothermia	Core body temperature below 35 °C
Anaphylaxis	Severe generalised allergic reaction	Incision	Clean cut by a sharp object
Angina	Chest pain from reduced blood flow to the heart muscle	Ischaemia	Reduced blood supply to a part
Anoxia	Complete absence of oxygen	Laceration	Tear with ragged edges
Antidote	Substance that counteracts a poison	Lumbar	Region of the lower back
Antiseptic	Substance that stops germs growing on living tissue	Malaena	Black tarry stool from digested blood
Apnoea	Absence of breathing	Oedema	Swelling from fluid in the tissues
Asphyxia	Oxygen lack with CO ₂ retention from interference with breathing	Oliguria	Reduced urine output
Aspiration	Inhalation of vomit, blood or fluid into the lungs	Opisthotonos	Arching of the back in tetanus/convulsion
Asystole	Complete absence of heart activity	Pallor	Paleness of the skin
Aura	Warning sensation before a fit	Paraesthesia	Tingling, 'pins and needles'
Avulsion	Tearing away of a flap of tissue	Perfusion	Flow of blood through the tissues
Bradycardia	Pulse below 60/min	Petechiae	Pin-point bleeding spots in the skin
Capillary refill	Time for colour to return after pressing a nail bed (normal < 2 s)	Prognosis	Expected outcome of an illness or injury
Cardiac arrest	Stoppage of effective heart action	Prone	Lying face downwards
Casualty	A person who is injured or suddenly taken ill	Pyrexia	Fever
Cerebral	Relating to the brain	Reduction	Setting of a fracture or dislocation (not for a first aider)
Compression	Pressure on the brain from bleeding or swelling	Rigor	Shivering attack with fever
Concussion	Temporary shaking-up of the brain	ROSC	Return of spontaneous circulation after CPR
Contusion	Bruise — bleeding into tissues with unbroken skin	Sign	What the first aider observes
Crepitus	Grating of broken bone ends	Splint	Rigid support for a fracture
Cyanosis	Blue discoloration from lack of oxygen	Sprain	Injury to a ligament round a joint
Defibrillation	Electric shock to restore a normal heart rhythm	Strain	Over-stretching of a muscle or tendon
Dislocation	Displacement of bones at a joint	Stridor	Harsh noisy breathing from upper-airway narrowing
Dyspnoea	Difficult or laboured breathing	Supine	Lying on the back
Epistaxis	Bleeding from the nose	Symptom	What the casualty complains of
Fracture	Break in the continuity of a bone	Syncope	Fainting
Gangrene	Death of tissue from loss of blood supply	Tachycardia	Pulse above 100/min
Haematemesis	Vomiting of blood	Tetany	Muscle spasms from low calcium or over-breathing
Haematuria	Blood in the urine	Tourniquet	Constricting band to stop limb bleeding (last resort)
Haemoptysis	Coughing up of blood from the lungs	Triage	Sorting of casualties by priority
Haemorrhage	Escape of blood from a vessel	Trismus	Lock-jaw — spasm of the jaw muscles
Hyperglycaemia	High blood sugar	Unconsciousness	State of not responding to stimuli
Hypoglycaemia	Low blood sugar (below 70 mg/dl)	Ventricular fibrillation	Chaotic, ineffective quivering of the heart

18.2 Abbreviations Used in First Aid

Abbr.	Full form	Abbr.	Full form
ABC	Airway, Breathing, Circulation	GCS	Glasgow Coma Scale
AED	Automated External Defibrillator	IM / IV	Intramuscular / Intravenous
ACLS	Advanced Cardiac Life Support	MI	Myocardial Infarction (heart attack)
ARV / RIG	Anti-Rabies Vaccine / Rabies Immunoglobulin	ORS	Oral Rehydration Solution
ASV	Anti-Snake Venom	PEP	Post-Exposure Prophylaxis
AVPU	Alert, Voice, Pain, Unresponsive	RICE	Rest, Ice, Compression, Elevation
BLS	Basic Life Support	ROSC	Return of Spontaneous Circulation
BP	Blood Pressure	RR	Respiratory Rate
CPR	Cardio-Pulmonary Resuscitation	SpO ₂	Oxygen saturation of the blood
CSF	Cerebro-Spinal Fluid	START	Simple Triage And Rapid Treatment
DRABC	Danger, Response, Airway, Breathing, Circulation	TT	Tetanus Toxoid
EMS	Emergency Medical Services	VF	Ventricular Fibrillation
FAST	Face, Arms, Speech, Time (stroke test)	WHO	World Health Organization

18.3 All Normal Values on One Page

Parameter	Adult	Child	Infant
Pulse (beats/min)	60–100 (avg 72)	80–120	100–160
Respiration (breaths/min)	12–20	20–30	30–60
Blood pressure (mm Hg)	120/80	~100/65	~80/50
Temperature	37 °C / 98.4 °F	Same	Same
SpO ₂	95–100 %	95–100 %	95–100 %
Blood volume	5–6 litres	2–2.5 litres	250–350 ml
Capillary refill	< 2 seconds	< 2 s	< 2 s
Urine output	1–1.5 l/day (≥ 30 ml/h)	1–2 ml/kg/h	2 ml/kg/h
CPR compression depth	5–6 cm	≈ 5 cm (1/3 chest)	≈ 4 cm (1/3 chest)
CPR rate / ratio	100–120/min; 30 : 2	100–120/min; 30 : 2 (15 : 2 for 2 professionals)	100–120/min; 30 : 2 (15 : 2)
Rescue breaths alone	10–12/min	12–20/min	12–20/min
Fasting blood sugar	70–110 mg/dl	Same	Same
Haemoglobin	M 13–18, F 12–16 g/dl	11–16 g/dl	14–20 g/dl (newborn)
Blood pH	7.35–7.45	Same	Same
Tidal volume	500 ml	—	—
Number of bones	206	—	270–300 at birth

18.4 Position of the Casualty — Master Chart

Condition	Correct position
Unconscious, breathing	Recovery (lateral) position
Cardiac arrest / CPR	Flat on the back on a hard surface
Shock, bleeding, fainting	Flat, legs raised 20–30 cm, head low and turned to one side
Heart attack, angina	Half-sitting (W position), knees bent
Asthma, breathlessness	Sitting up, leaning slightly forward
Chest injury	Half-sitting, leaning towards the injured side
Abdominal injury / pain	On the back, knees drawn up and supported
Head injury, stroke	Flat with head and shoulders slightly raised; recovery position if unconscious
Spinal injury	Do not move — flat, head held in line, spine board
Fracture of the lower limb	Flat, limb immobilised and supported
Fracture of the pelvis	Flat, knees and ankles tied with padding between
Nose bleed	Sitting up, head bent forward

Condition	Correct position
Snake bite	Lying still, limb splinted at or below heart level
Pregnant casualty	Left lateral position
Burns	Flat, burnt part raised and covered
Vomiting or bleeding into the mouth	Head turned to the side / recovery position
Heat stroke	Flat in a cool place, clothing removed, actively cooled
Hypothermia	Flat and horizontal, insulated, handled gently

18.5 Master List of 'Never Do' Points

THE FIRST AIDER'S ABSOLUTE PROHIBITIONS

- **Never** rush into danger — electricity, gas, fire, water, traffic, a confined space.
- **Never** give anything by mouth to an unconscious, drowsy, convulsing or vomiting casualty, or to one likely to need an operation.
- **Never** remove a blood-soaked dressing — add another pad over it.
- **Never** remove a deeply embedded foreign body, and never probe a wound.
- **Never** apply a tourniquet except for uncontrollable, life-threatening limb bleeding — and never loosen it once applied.
- **Never** attempt to reduce (set) a fracture or dislocation, and never test for crepitus.
- **Never** apply direct heat, hot-water bottles or alcohol to a shocked casualty.
- **Never** put anything in the mouth of a person having a fit, and never restrain him.
- **Never** induce vomiting in poisoning — above all with corrosives and petroleum products.
- **Never** apply ice, oil, butter, toothpaste or turmeric to a burn, and never prick blisters.
- **Never** rub a frost-bitten part or use dry heat on it, and never let it re-freeze.
- **Never** cut, suck or apply a tourniquet to a snake bite, and never let the casualty walk.
- **Never** plug the ear or nose that is bleeding after a head injury.
- **Never** move a suspected spinal-injury casualty without full support and helpers.
- **Never** exceed your training, never accept a fee, and never discuss the casualty with onlookers or the press.

18.6 All the Mnemonics in One Place

Mnemonic	Stands for	Used for
3 P's	Preserve life, Prevent worsening, Promote recovery	Aims of first aid
DRSABCD	Danger, Response, Send for help, Airway, Breathing, Circulation/CPR, Defibrillation	Action plan at every emergency
AVPU	Alert, Voice, Pain, Unresponsive	Level of consciousness
SAMPLE	Signs/symptoms, Allergies, Medication, Past history, Last meal, Events	History taking
AMPLE	Allergies, Medicines, Past illness, Last meal, Events	Hand-over to hospital
RPM	Respiration, Perfusion, Mental status	START triage assessment
FISH SHAPED	Fainting, Infantile convulsions, Shock, Head injury, Stroke, Heart attack, Asphyxia, Poisoning, Epilepsy, Diabetes	Causes of unconsciousness
AEIOU-TIPS	Alcohol, Epilepsy, Insulin, Overdose, Uraemia, Trauma, Infection, Poisoning, Stroke	Causes of coma
FAST	Face, Arms, Speech, Time	Recognising a stroke
RICE / PRICE	(Protect), Rest, Ice, Compression, Elevation	Sprains, strains, bruises
HARM	Heat, Alcohol, Running, Massage — to be avoided for 48–72 hours	After a sprain or strain
5 P's	Pain, Pallor, Pulselessness, Paraesthesia, Paralysis	Signs of a too-tight bandage / circulation loss
SLUDGE	Salivation, Lacrimation, Urination, Defaecation, Gastric cramps, Emesis	Organophosphate (insecticide) poisoning
Reach-Throw-Row-Go	Order of water rescue	Drowning
Stop-Drop-Wrap-Roll	Action when clothes catch fire	Burns
5 Rights	Right patient, drug, dose, route, time	Giving medicines
WHO 5 Keys	Keep clean, separate raw/cooked, cook thoroughly, safe temperature, safe water	Preventing food poisoning

Mnemonic	Stands for	Used for
30/30 rule	Shelter if thunder < 30 s after the flash; wait 30 min after the last thunder	Lightning safety

18.7 Antidotes and Specific Treatments — Ready Chart

Emergency	Specific treatment / antidote
Anaphylaxis	Adrenaline 0.5 mg IM (outer thigh)
Ventricular fibrillation / cardiac arrest	CPR + defibrillation (AED)
Heart attack	Aspirin 300 mg chewed + rest + hospital
Hypoglycaemia (conscious)	Oral sugar/glucose
Asthma attack	Salbutamol (blue) inhaler, 2–10 puffs
Angina	Sub-lingual nitroglycerine
Snake bite	Anti-snake venom (ASV) in hospital
Dog bite	Wash 15 min + anti-rabies vaccine (± RIG)
Dirty/punctured wound	Tetanus toxoid (± immunoglobulin)
Organophosphate poisoning	Atropine + pralidoxime
Opioid overdose	Naloxone
Paracetamol overdose	N-acetylcysteine
Carbon monoxide poisoning	100 % oxygen (hyperbaric if severe)
Cyanide poisoning	Hydroxocobalamin / nitrite–thiosulphate kit
Iron poisoning	Desferrioxamine
Warfarin / rat poison	Vitamin K
Diarrhoea and dehydration	ORS (+ zinc in children)
Heat stroke	Rapid external cooling
Hypothermia	Gradual re-warming, gentle handling
Frost-bite	Re-warming in water at 37–40 °C

18.8 First-Aid Records and Hand-Over Report

Every serious case must be recorded. A simple format:

Item	What to write
Identification	Name, age, sex, address/employee number, contact person
Date and time	Time of the incident, time you arrived, time the ambulance was called and arrived
History	What happened (mechanism of injury), where, how; what the casualty and witnesses said; SAMPLE history
Findings	Level of response (AVPU), airway, breathing rate, pulse rate and character, skin colour and temperature, pupils, bleeding, injuries found in the head-to-toe examination
Observations chart	Pulse, breathing and response repeated every 10 minutes with the time of each reading
Treatment given	Dressings, bandages, splints, position, CPR (with times), medicines given (name, dose, time), oxygen
Property	Valuables, documents, dentures, spectacles handed over — to whom and when (with a witness)
Disposal	Sent home / to hospital / to a doctor; by what transport; accompanied by whom
Signature	Name, designation and signature of the first aider

Keep the record **factual** — no opinions or blame. It may be needed for insurance, compensation or a court of law, and the information is **confidential**.

18.9 Fire, Rescue and Disaster — Ready Reference

Situation	Action
Clothes on fire	STOP – DROP – WRAP – ROLL: stop the casualty running, lay him down with the burning side uppermost, wrap him in a blanket/coat/rug (not nylon) and roll him on the ground; then treat the burns
Fire in a building	Raise the alarm and call 101 ; close doors behind you; do not use the lift; crawl low under the smoke keeping close to the floor; test doors with the back of the hand before opening; if trapped, block gaps with wet cloth and signal from a window

Situation	Action
Types of extinguisher	Water — wood, paper, cloth (Class A); foam — flammable liquids (B); CO₂ / dry powder — electrical (C) and liquids; wet chemical/fire blanket — cooking oil (K/F) . Never use water on an electrical or oil fire . Remember PASS : Pull, Aim, Squeeze, Sweep
Gas leak	Do not operate any switch or flame; close the valve; open doors and windows; evacuate; call the gas agency/fire brigade
Road accident	Park safely with hazard lights on, wear a reflective jacket, switch off the ignition and apply the handbrake of the crashed vehicle, warn oncoming traffic, do not smoke, call 112/108 , do not remove a helmet unless the airway is blocked (two-rescuer removal), treat the casualties where they lie and remember the Good Samaritan protection
Mass casualty / disaster	Take charge, ensure scene safety, triage (red–yellow–green–black) , set up a collection point, treat only airway and severe bleeding while triaging, record and transport by priority, inform the district control room (1070/1077)
Collapsed structure / confined space	Never enter without breathing apparatus, a harness and helpers; beware of crush syndrome on release; note the time of the crush

18.10 Chapter-wise One-Line Facts for the Last Day

Outline of first aid

- First aid = immediate, temporary, skilled help with available material.
- Aims = **preserve life (first), prevent worsening, promote recovery**.
- Term coined by **Esmarch**; St John Ambulance **1877**; Red Cross **1863 (Henri Dunant)**; Indian Red Cross **1920**.
- First-aid symbol = **white cross on green**; World First Aid Day = **2nd Saturday of September**; World Red Cross Day = **8 May**.
- Ambulance **108**, universal emergency **112**, fire **101**, police **100**, childline **1098**, poison information **1800-11-6117**.
- Triage: **red = immediate, yellow = urgent, green = delayed, black = dead/expectant**; START assesses **RPM**.
- Golden hour = **60 minutes**; brain damage in **3–4 minutes**; breathing checked for **≤ 10 seconds**.
- Factories Act: first-aid box per **150 workers**; ambulance room above **500 workers**.

Structure and functions of the body

- 206 bones** (axial 80 + appendicular 126); hand 27, foot 26; vertebrae **7-12-5-5-4**; ribs **12 pairs (7 true, 3 false, 2 floating)**.
- Femur** largest, **stapes** smallest, **clavicle** most often fractured, **hyoid** has no joint, **patella** is the largest sesamoid.
- Heart: 4 chambers, 4 valves, pacemaker **SA node**, thickest wall **left ventricle**, cardiac output **5 l/min**.
- Pulmonary artery** carries deoxygenated and **pulmonary vein** oxygenated blood.
- Blood = plasma **55 %** + cells; RBC life **120 days**; **O– universal donor, AB+ universal recipient**.
- Right lung **3 lobes**, left **2**; inhaled objects go to the **right** bronchus; expired air = **16 % O₂, 4 % CO₂**.
- Vital centres in the **medulla**; balance in the **cerebellum**; temperature in the **hypothalamus**.
- 12 cranial nerves** (vagus longest, trigeminal largest) and **31 pairs of spinal nerves**.
- Liver** largest gland, **thyroid** largest endocrine gland, **pituitary** master gland, **skin** largest organ.

Dressings and bandages

- Dressing touches the wound; bandage holds the dressing.
- Triangular bandage = **1 m square cut diagonally**; forms = open, broad, narrow.
- Arm sling** — forearm/ribs; **elevation sling** — bleeding hand/collar bone; **collar-and-cuff** — humerus.
- Roller widths: finger 2.5, hand 5, arm 5–6, leg 7.5–9, trunk 10–15 cm.
- Turns: circular, spiral, **reverse spiral (forearm/leg)**, **figure-of-eight (ankle)**, spica (shoulder/hip/thumb), recurrent (stump).
- Overlap **two-thirds**, bandage **below upwards**, finish with a **reef knot**, keep finger tips visible.

CPR, artificial respiration and asphyxia

- Adult CPR: **100–120/min, 5–6 cm, 30 : 2**, change rescuer every **2 minutes**.
- Infant: **two fingers, 4 cm, 5 breaths first, brachial pulse**; two professionals use **15 : 2**; newborn **3 : 1**.
- AED pads: **below the right collar bone + left mid-axillary line**.
- Choking: **5 back blows + 5 abdominal thrusts**; infants get **chest thrusts, never abdominal**; unconscious → CPR.
- Mouth-to-mouth **10–12/min**; Schafer **12–15**; Holger Nielsen **12** (best manual); **Sylvester 12**; **Eve's rocking 10–12**.
- Asphyxia stages: **dyspnoea → convulsions → apnoea**; CO poisoning gives **cherry-red skin**.

- Drowning: **reach-throw-row-go**, 5 rescue breaths, never drain the lungs, always hospitalise (secondary drowning).

Wounds, bleeding, shock and electric shock

- Incised bleeds most; punctured risks **tetanus**; contusion is a **closed** wound.
- Tetanus: *Clostridium tetani*, incubation **7–10 days**, toxin **tetanospasmin**, sign **trismus**.
- Bleeding control = **rest + elevation + direct pressure for 10 minutes**; pressure points **brachial** and **femoral**.
- Blood loss of **1–1.5 litres** causes definite shock; one-third loss may be fatal.
- Shock position = **flat, legs raised 20–30 cm**, warm, **nothing by mouth**.
- Anaphylaxis → **adrenaline IM outer thigh**; heart attack → **aspirin 300 mg chewed**.
- Burns: cool **10–20 minutes**; **rule of nines** (head 9, arm 9, leg 18, trunk 18 + 18, perineum 1); child head 18, leg 14.
- Electric shock: **switch off first**; **AC more dangerous than DC**; **50–100 mA** → **VF**; high voltage — keep **18 m** away.

Fractures, unconsciousness, fits and poisons

- Fracture types: simple, **compound (most dangerous)**, comminuted, **greenstick (children)**, complicated, impacted, depressed.
- Immobilise the **joints above and below**; bandage **above and below, never over** the fracture; **never reduce**.
- **Shoulder** is the most commonly dislocated joint; dislocation = **fixed joint, no crepitus**.
- **Sprain = ligament, strain = muscle/tendon**; treat by **RICE**, avoid **HARM**.
- Concussion = temporary with **equal pupils and rapid pulse**; compression = progressive with **unequal pupils and slow full pulse**.
- Fit: **never put anything in the mouth, never restrain**; status epilepticus = fit **> 5 minutes** → ambulance.
- Hysterical fit: **only before an audience, no tongue bite, no incontinence, no cyanosis, pupils normal**.
- Poisoning: **never induce vomiting** (especially corrosives and kerosene); save the container; **pin-point pupils** = opium/organophosphate, **dilated** = datura.
- Food poisoning: **Staphylococcus 1–6 h (vomiting)**, **Salmonella 12–36 h (fever)**, **botulism 12–36 h (paralysis)**; treat with **ORS**.

Common conditions, transport and medicines

- Epistaxis → **sit up, head forward, pinch the soft part 10 minutes**.
- Snake bite → **immobilise like a fracture, limb at/below heart level, hospital for ASV**; no tourniquet, no cutting, no sucking.
- Dog bite → **wash with soap and water 15 minutes** + vaccine on days **0, 3, 7, 14, 28** (+ RIG in Category III).
- Frost-bite → re-warm at **37–40 °C**; never rub or use dry heat; hypothermia = core temperature **< 35 °C**.
- Heat stroke = **hot dry skin, no sweating, > 40 °C** → cool rapidly; heat exhaustion = **pale, clammy, sweating** → ORS.
- Bee sting → **scrape out sideways**; sting in the mouth → ice and ambulance.
- Carries: **four-handed seat** (both arms usable), **three-handed** (one leg injured), **two-handed** (arms unusable), **fireman's lift** (one hand free), **chair/fore-and-aft** (stairs).
- Carry **feet first on the level, head first up**; bearers **break step**; **Neil Robertson stretcher** for vertical rescue.
- Paracetamol **500 mg–1 g 6-hourly (max 3–4 g/day)**; aspirin **never under 16** (Reye's syndrome).
- **ORS = 1 sachet per litre**, used within 24 hours; loperamide never in children; **5 rights** of medicine administration.

18.11 Last Word

HOW TO USE THIS BOOK IN THE FINAL WEEK

- **Day 1–3:** read one Part a day, marking the coloured boxes only.
- **Day 4–5:** revise all the **tables** — most objective questions come straight from comparison tables (arterial vs venous, concussion vs compression, epilepsy vs hysteria, fracture vs dislocation, heat stroke vs heat exhaustion, hypoglycaemia vs hyperglycaemia).
- **Day 6:** revise every **Chapter Recap** and the **normal values, doses, rates and timings** — numbers are the easiest marks in the paper.
- **Day 7:** read this Appendix from beginning to end; it is the whole syllabus in miniature.
- In the examination, remember the three constants: **airway before everything, bleeding next, shock always** — and when two options look correct, choose the one that **protects life first and does the least to the casualty**.